

Series Estimation of Cointegrated System with Time Varying Coefficients^{*}

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This paper proposes a series estimation approach for cointegrated system with coefficients that evolve smoothly over time. We first establish the limit theory for the estimator. The framework is then extended to incorporate stationary regressors entering the system with time-invariant coefficients. While the associated estimators retain standard asymptotic properties, the presence of stationary components alters the distribution of the estimators for the time-varying coefficients and motivates us to consider a series-type dynamic OLS estimator. Under suitable conditions, the estimator admits a tractable mixed normal limiting distribution. A bootstrap procedure is also proposed to construct pointwise confidence intervals. Monte Carlo simulations indicate that the series-type dynamic OLS estimator can improve finite-sample performance. An empirical application to the consumption–wealth ratio shows that allowing for time variation in the cointegrating relationship can enhance equity premium forecasts relative to conventional fixed-coefficient specifications.

Keywords: series estimation, time-varying cointegration, dynamic OLS estimator, consumption–wealth ratio

JEL-Classification: C22, C32, C14, G17

1 – Introduction

There has been growing interest in the limit theory of nonparametric time-varying coefficient cointegration regressions. This literature dates back to Park and Hahn 1999, who develop a series estimator for time-varying cointegrating vectors in a single-equation framework. Bierens and Martins 2010 propose a time-varying vector error correction model in which the cointegration relationship varies smoothly over time and is modelled using orthonormal series expansion. They also propose a likelihood ratio test for time-invariant cointegration.

Within a kernel estimation framework, Phillips, Li, and Gao 2017 show that conventional asymptotic theory developed under stationarity breaks down when regressors are integrated, owing to a kernel-induced degeneracy in the weighted signal matrix. To address this issue, they introduce a novel rotational decomposition and establish the corresponding limit theory. Building on this insight, Li, Phillips, and Gao 2020 extend the framework to accommodate cointegrated regressors as well as mixtures of stochastic and deterministic trend components. More recently, Kapetanios et al. 2020 develop a test for time-varying cointegration and apply it to the UK “great ratios.” While the associated limit theory is now well understood, statistical inference in the kernel estimation framework is formulated in terms of rotated transformations of the estimator rather than the original parameters. This makes economic interpretation less transparent and may complicate empirical implementation in practice.

In this paper, we develop a series estimation approach for cointegrated systems with time-varying cointegrating coefficients. Our starting point is the triangular representation of cointegrated system in Phillips 1991, which we extend to allow for smooth variation in the long-run parameters. By approximating the evolving cointegrating relationship using a finite series expansion based on Chebyshev time polynomials as in Bierens 1997 and Bierens and Martins 2010, we show that, in contrast to kernel-based methods, the resulting estimator does not suffer from degeneracy in the weighted signal matrix when regressors are integrated. We establish the associated large-sample theory and derive the asymptotic distribution of the estimator.

We further extend the framework to allow for stationary regressors entering the model with time-invariant coefficients. In this setting, the asymptotic properties of the estimators associated with the time-invariant parameters remain unchanged and continue to follow the standard limit

theory developed for cointegrated systems with mixed integrated and stationary regressors, as in Phillips 1995. In contrast, the asymptotic distribution of the estimators for the time-varying cointegrating coefficients is altered by the presence of stationary components. The resulting limit theory nonetheless provides useful guidance for obtaining more tractable inference, which motivates us to consider a specification that augments the regression with leads and lags of first differences of the integrated regressors. Under suitable regularity conditions, we show that the resulting series estimator admits a mixed normal limiting distribution. This extends the dynamic OLS framework of Saikkonen 1991 and Stock and Watson 1993 to a setting with time-varying cointegrating coefficients implemented through orthonormal series approximations. For empirical implementation, we also propose a bootstrap procedure to construct pointwise confidence intervals for the time-varying coefficient estimates.

The finite-sample performance of the proposed estimator is evaluated using an extensive Monte Carlo study. The results suggest that augmenting the regression with a modest number of leads and lags of first differences of the integrated regressors can improve estimation accuracy in many designs, particularly when correlation between the regression disturbances and regressor innovations is more pronounced. The qualitative performance patterns are broadly robust across alternative coefficient dynamics and sample sizes.

We illustrate the proposed approach in an empirical application to the consumption–wealth ratio (*cay*) and its implications for forecasting the U.S. quarterly equity premium. We find substantial time variation in the estimated cointegrating coefficients, consistent with evidence of instability in the long-run relationship among consumption, wealth, and labour income documented in the previous literature. We show that allowing for coefficient variation can improve out-of-sample forecasting performance relative to the historical mean benchmark in the pre-pandemic period. Although predictive gains weaken once post-COVID observations are included, the time-varying *cay* continues to deliver lower forecast errors than the fixed-coefficient measure, indicating that modelling time variation in the cointegrating relationship remains empirically useful.

The rest of the paper is organized as follows. Section 2 presents the model, the series estimation methodology, and the baseline asymptotic theory. Section 3 extends the framework to allow for stationary regressors and derives the corresponding limit results. Section 4 examines finite-sample performance of the proposed estimator using Monte Carlo simulations. Section 5 illustrates the

usefulness of the method in an application to the consumption–wealth ratio and equity premium forecasting. Section 6 concludes. Technical proofs, additional empirical results, and detailed data descriptions are reported in the Appendix.

2 – Model and estimation theory

2.1. Model setup

We consider the time-varying coefficients cointegrated system (cf. Phillips 1991)

$$\mathbf{y}_t = \mathbf{A}_t \mathbf{X}_t + \mathbf{u}_{0t}, \quad (1)$$

$$\mathbf{X}_t = \mathbf{X}_{t-1} + \mathbf{u}_{xt}, \quad (2)$$

where $t = 1, 2, \dots, n$. $\mathbf{y}_t$ denotes a d -dimensional vector, and $\mathbf{X}_t$ is a $K_1 \times 1$ vector of integrated regressors. The matrix of time-varying cointegrating coefficients is given by $\mathbf{A}_t := \mathbf{A}(\tau_t)$, where $\mathbf{A}_t$ is $d \times K_1$ with elements modeled as functions of scaled time point $\tau_t := t/n$. The process $(\mathbf{X}_t)_t$ is initialized at $\mathbf{X}_0 = O_p(1)$.

The innovations $\mathbf{u}_t = (\mathbf{u}_{0t}^\top, \mathbf{u}_{xt}^\top)^\top$ are determined according to the linear process

$$\mathbf{u}_t = \Phi(\mathcal{L})\boldsymbol{\varepsilon}_t = \sum_{j=0}^{\infty} \Phi_j \boldsymbol{\varepsilon}_{t-j}, \quad (3)$$

where $\Phi(\mathcal{L}) = \sum_{j=0}^{\infty} \Phi_j \mathcal{L}^j$, $(\Phi_j)_j$ is a sequence of $(d + K_1) \times (d + K_1)$ matrices, $\mathcal{L}$ is the lag operator, and $(\boldsymbol{\varepsilon}_t)_t$ is a sequence of independent and identically distributed (*i.i.d.*) random vectors with dimension $d + K_1$. We impose the following conditions.

Assumption 1. (i) $(\boldsymbol{\varepsilon}_t)_t$ is an *i.i.d.* $(d + K_1)$ -dimensional random vector with $\mathbb{E}(\boldsymbol{\varepsilon}_t) = \mathbf{0}_{(d+K_1) \times 1}$, $\Sigma_\varepsilon := \mathbb{E}(\boldsymbol{\varepsilon}_t \boldsymbol{\varepsilon}_t^\top)$, where $\lambda_{\min}(\Sigma_\varepsilon) > 0$, and $\mathbb{E}\|\boldsymbol{\varepsilon}_t\|^p < \infty$, for some $p > 4$. (ii) The coefficient matrices in (3) satisfy $\sum_{j=0}^{\infty} j \|\Phi_j\| < \infty$ and $|\Phi(1)| \neq 0$.

Under Assumption 1, $\mathbf{u}_t$ has covariance matrix $\Sigma = \sum_{j=0}^{\infty} \Phi_j \Sigma_\varepsilon \Phi_j^\top$, with $\mathbb{E}\|\mathbf{u}_t\|^p < \infty$, for some $p > 4$. By functional central limit theory (FCLT) for linear process (cf. Phillips and Solo

1992), we have

$$\frac{1}{\sqrt{n}} \sum_{t=1}^{\lfloor nr \rfloor} \mathbf{u}_t \Rightarrow \mathbf{B}(r) \text{ on } D([0, 1], \mathbb{R}^{d+K_1}), \quad (4)$$

where $\mathbf{B}(r)$ denotes a vector Brownian motion with long-run covariance matrix $\boldsymbol{\Omega} = \boldsymbol{\Phi}(1)\boldsymbol{\Sigma}_\varepsilon\boldsymbol{\Phi}(1)^\top > 0$ and $\boldsymbol{\Phi}(1) = \sum_{j=0}^{\infty} \boldsymbol{\Phi}_j$. For the ease of theoretical development, we partition $\boldsymbol{\Sigma}$, $\boldsymbol{\Phi}(1)$, and $\boldsymbol{\Omega}$ comfortably with $\mathbf{u}_t$ as

$$\boldsymbol{\Sigma} = \begin{bmatrix} \boldsymbol{\Sigma}_{00} & \boldsymbol{\Sigma}_{0x} \\ \boldsymbol{\Sigma}_{x0} & \boldsymbol{\Sigma}_{xx} \end{bmatrix}, \quad \boldsymbol{\Phi}(1) = \begin{bmatrix} \boldsymbol{\Phi}_0(1) \\ \boldsymbol{\Phi}_x(1) \end{bmatrix}, \quad \boldsymbol{\Omega} = \begin{bmatrix} \boldsymbol{\Omega}_{00} & \boldsymbol{\Omega}_{0x} \\ \boldsymbol{\Omega}_{x0} & \boldsymbol{\Omega}_{xx} \end{bmatrix}, \quad (5)$$

where $\boldsymbol{\Phi}_0(1)$ and $\boldsymbol{\Phi}_x(1)$ are of dimension $d \times (d + K_1)$ and $K_1 \times (d + K_1)$, respectively. Using the Beveridge–Nelson (BN) decomposition, we have the following decomposition for $\mathbf{u}_t$:

$$\mathbf{u}_t = \boldsymbol{\Phi}(1)\boldsymbol{\varepsilon}_t - \Delta \tilde{\mathbf{u}}_t, \text{ for } \tilde{\mathbf{u}}_t = \sum_{j=0}^{\infty} \tilde{\boldsymbol{\Phi}}_j \boldsymbol{\varepsilon}_{t-j}, \quad \tilde{\boldsymbol{\Phi}}_j = \sum_{k=j+1}^{\infty} \boldsymbol{\Phi}_k, \quad (6)$$

where Assumption 1(ii) ensures $\sum_{j=0}^{\infty} \|\tilde{\boldsymbol{\Phi}}_j\| < \infty$ (Phillips and Solo 1992). Corresponding to the partition of $\mathbf{u}_t$, we write

$$\mathbf{u}_{0t} = \sum_{j=0}^{\infty} \boldsymbol{\Phi}_{0j} \boldsymbol{\varepsilon}_{t-j}, \quad \tilde{\mathbf{u}}_{0t} = \sum_{j=0}^{\infty} \tilde{\boldsymbol{\Phi}}_{0j} \boldsymbol{\varepsilon}_{t-j}, \quad (7)$$

and also

$$\mathbf{u}_{xt} = \sum_{j=0}^{\infty} \boldsymbol{\Phi}_{xj} \boldsymbol{\varepsilon}_{t-j}, \quad \tilde{\mathbf{u}}_{xt} = \sum_{j=0}^{\infty} \tilde{\boldsymbol{\Phi}}_{xj} \boldsymbol{\varepsilon}_{t-j}, \quad (8)$$

where $\boldsymbol{\Phi}_{0j}$, $\tilde{\boldsymbol{\Phi}}_{0j}$ are of dimension $d \times (d + K_1)$ and $\boldsymbol{\Phi}_{xj}$, $\tilde{\boldsymbol{\Phi}}_{xj}$ are of dimension $K_1 \times (d + K_1)$, respectively. In addition, the limit theory also involves the partitioned components of the one-sided long-run covariance matrix

$$\boldsymbol{\Lambda} := \begin{bmatrix} \boldsymbol{\Lambda}_{00} & \boldsymbol{\Lambda}_{0x} \\ \boldsymbol{\Lambda}_{x0} & \boldsymbol{\Lambda}_{xx} \end{bmatrix} = \sum_{j=1}^{\infty} \mathbb{E}(\mathbf{u}_t \mathbf{u}_{t-j}^\top).$$

As usual, we have $\Omega = \Sigma + \Lambda + \Lambda^\top$ and

$$\Lambda = \sum_{j=1}^{\infty} \sum_{k=0}^{\infty} \Phi_{k+j} \Sigma_\varepsilon \Phi_k^\top = \sum_{k=0}^{\infty} \left(\sum_{j=1}^{\infty} \Phi_{k+j} \right) \Sigma_\varepsilon \Phi_k^\top = \sum_{k=0}^{\infty} \tilde{\Phi}_k \Sigma_\varepsilon \Phi_k^\top = \mathbb{E}(\tilde{\mathbf{u}}_t \mathbf{u}_t^\top).$$

2.2. Estimation procedure

Our objective is to develop estimators for the time-varying cointegrating coefficients $\mathbf{A}(\tau_t)$ and the time-invariant coefficients Γ . We first state the assumption imposed on $\mathbf{A}(\tau_t)$:

Assumption 2. Let $\mathbf{A}^{(i,j)}(\tau_t)$ be the (i, j) th elements in $\mathbf{A}(\tau_t)$, where $i = 1, 2, \dots, d$, $j = 1, 2, \dots, K$. $\mathbf{A}^{(i,j)}(\cdot)$ is a squared integrable real function: $\mathbf{A}^{(i,j)}(\cdot) \in L^2(0, 1) = \left\{ a(r) : \int_0^1 a^2(r) dr < \infty \right\}$ and is $q + 1$ times continuously differentiable on $[0, 1]$, where $q > 3$.

Under Assumption 2, $\mathbf{A}(\tau_t)$ admits an orthogonal expansion

$$\mathbf{A}(\tau_t) = \sum_{i=0}^{\infty} \mathbf{B}_i \psi_i(\tau_t),$$

where $\mathbf{B}_i$ is of dimension $d \times K_1$ and

$$\psi_0(\tau_t) = 1, \quad \psi_j(\tau_t) = \sqrt{2} \cos(j\pi\tau_t), \quad (9)$$

where $j = 1, 2, 3, \dots$. Then, $\{\psi_j(r)\}_j$ is an orthonormal basis in $L^2[0, 1]$. (9) defines the Chebyshev time polynomials, first used in econometrics by Bierens 1997 in the context of testing a unit root against a nonlinear trend–stationary alternative. They are also employed by Dong and Linton 2018 for estimating deterministic time trends in additive nonparametric models. Some basic properties of the Chebyshev time polynomials are summarized in Section D.

Let k be a positive integer. We then define a truncated series

$$\mathbf{A}_k(\tau_t) = \sum_{i=0}^{k-1} \mathbf{B}_i^{(k)} \psi_i(\tau_t) = \mathbf{B}^{(k)} (\boldsymbol{\psi}_{(k)}(\tau_t) \otimes \mathbf{I}_{K_1}), \quad (10)$$

where $\boldsymbol{\psi}_{(k)}(\tau_t) = (\psi_0(\tau_t), \psi_1(\tau_t), \dots, \psi_{k-1}(\tau_t))^\top$. Under Assumption 2, by directly applying

Theorem D.1 in Appendix D, we have

$$\|\mathbf{A}(\tau_t) - \mathbf{A}_k(\tau_t)\| = O\left(\frac{1}{k^q}\right), \quad (11)$$

for any $\tau_t \in [0, 1]$. This shows that $\mathbf{A}(\tau_t)$ can be well approximated by (10), and the approximation error goes to zero as $k \rightarrow \infty$.

By replacing $\mathbf{A}(\tau_t)$ in (1) with (10), we have the approximated model:

$$\mathbf{y}_t = \sum_{i=0}^{k-1} \mathbf{B}_i \psi_i(\tau_t) \mathbf{X}_t + \mathbf{u}_{0t}^{(k)}, \quad (12)$$

where $\mathbf{u}_{0t}^{(k)} = \mathbf{u}_{0t} + (\mathbf{A}_t - \mathbf{A}_k(\tau_t)) \mathbf{X}_t$. Define

$$\begin{aligned} \mathbf{X}_t^{(k)} &= (\mathbf{X}_t^\top, \psi_1(\tau_t) \mathbf{X}_t^\top, \dots, \psi_{k-1}(\tau_t) \mathbf{X}_t^\top)^\top \\ \mathbf{B}^{(k)} &= (\mathbf{B}_0, \mathbf{B}_1, \dots, \mathbf{B}_{k-1}). \end{aligned} \quad (13)$$

We can write (12) more conveniently as

$$\mathbf{y}_t = \mathbf{B}^{(k)} \mathbf{X}_t^{(k)} + \mathbf{u}_{0t}^{(k)}. \quad (14)$$

$\mathbf{B}^{(k)}$ can thus be estimated by LS (least squares):

$$\widehat{\mathbf{B}}_n^{(k)} = \left(\sum_{t=1}^n \mathbf{y}_t \mathbf{X}_t^{(k)\top} \right) \left(\sum_{t=1}^n \mathbf{X}_t^{(k)} \mathbf{X}_t^{(k)\top} \right)^{-1}$$

Then, for $\tau_t \in [0, 1]$, the series estimator of $\mathbf{A}(\tau_t)$ is

$$\widehat{\mathbf{A}}_n^{(k)}(\tau_t) = \widehat{\mathbf{B}}_n^{(k)} \mathbf{T}_{(k)}(\tau_t), \quad (15)$$

based on (10), where $\mathbf{T}_{(k)}(\tau_t) = \boldsymbol{\psi}_{(k)}(\tau_t) \otimes \mathbf{I}_{K_1}$.

2.3. Asymptotic theory

We now introduce an additional assumption on the series expansion order k . As expected, it is inversely related to q : the smoother the function $\mathbf{A}(\tau_t)$, the lower the required order of the series expansion.

Assumption 3. $k = \lfloor c \cdot n^\kappa \rfloor$ is a sequence of integers for some $c > 0$, where $\frac{1}{q} < \kappa < \min \left\{ \frac{1}{3}, \frac{1}{2} - \frac{1}{p} \right\}$.

Theorem 1 derives the first-order asymptotic expansion of the series estimator (15) and establishes its leading term.

Theorem 1. Suppose that Assumptions 1-3 are satisfied. For any given $\tau_t \in [0, 1]$, we have, as $n \rightarrow \infty$

$$n \cdot \left(\widehat{\mathbf{A}}_n^{(k)}(\tau_t) - \mathbf{A}(\tau_t) + \mathbf{\Gamma}(\tau_t) \right) = \mathbf{\Gamma}_{ux}^{(k)} \left(\mathbf{\Lambda}_{xx}^{(k)} \right)^{-1} \mathbf{T}_{(k)}(\tau_t) + o_p(\sqrt{k}), \quad (16)$$

where $\mathbf{\Gamma}(\tau_t) := \mathbf{A}(\tau_t) - \mathbf{A}_k(\tau_t)$ and

$$\begin{aligned} \mathbf{\Lambda}_{xx}^{(k)} &:= \int_0^1 \left\{ \boldsymbol{\psi}_{(k)}(r) \boldsymbol{\psi}_{(k)}^\top(r) \otimes (\mathbf{B}_X(r) \mathbf{B}_X^\top(r)) \right\} dr, \\ \mathbf{\Gamma}_{ux}^{(k)} &:= \int_0^1 d\mathbf{B}_0(r) \left(\boldsymbol{\psi}_{(k)}(r) \otimes \mathbf{B}_X(r) \right)^\top + \boldsymbol{\ell}_{1(k)} \otimes (\mathbf{\Lambda}_{0x} + \mathbf{\Sigma}_{0x}). \end{aligned}$$

Note that (16) is only a stochastic expansion and does not by itself imply a mixed normal limit, since $\mathbf{B}_X(r)$ and $\mathbf{B}_0(r)$ are correlated. Define

$$\widehat{\mathbf{V}}_{n(k)}(\tau_t) = \left\{ \mathbf{T}_{(k)}^\top(\tau_t) \mathbf{\Lambda}_{n,xx}^{-1} \mathbf{T}_{(k)}(\tau_t) \right\} \otimes \widehat{\mathbf{\Omega}}_{n,00},$$

where $\widehat{\mathbf{\Omega}}_{n,00}$ is a consistent estimator of $\mathbf{\Omega}_{00}$. In the next corollary, we show that, mixed Normal asymptotic distribution is achieved in the pure cointegration case when $\mathbf{X}_t$ is strictly exogenous.

Corollary 1. Suppose that conditions in Theorem 1 are satisfied and $\mathbb{E}(\mathbf{u}_{0t} \otimes \mathbf{u}_{xs}) = \mathbf{0}_{dK_1 \times 1}$ for any t, s , we have

$$n \left(\widehat{\mathbf{V}}_{n(k)}(\tau_t) \right)^{-1/2} \text{vec} \left(\widehat{\mathbf{A}}_n^{(k)}(\tau_t) - \mathbf{A}(\tau_t) + \mathbf{\Gamma}(\tau_t) \right) \xrightarrow{d} \mathcal{N}(\mathbf{0}_{dK_1 \times 1}, \mathbf{I}_{dK_1}), \quad (17)$$

for any fixed $\tau_t \in [0, 1]$.

In view of the fact that $\left\| \widehat{\mathbb{V}}_{n(k)}(\tau_t) \right\| = O(1)k$, the convergence rate in (17) is $O_p(1) \frac{n}{\sqrt{k}}$. The convergence rate is comparable with the kernel estimate in the literature. Theorem 3.1 in Phillips, Li, and Gao 2017 shows that the *type I* super convergence rate is given by $nh^{1/2}$, where h is the bandwidth parameter. Thinking of k^{-1} as equivalent to the bandwidth h , the convergence rate in the two situations are quite comparable. However, due to kernel estimation degeneracy issues and rotation matrix introduced in their approach, the kernel estimate also involves a slower *type II* convergence rate, which is given by nh .

3 – Including stationary regressors

In this section, we consider a refinement of the model (1) in which stationary regressors are added to model the "short run" effects:

$$\mathbf{y}_t = \mathbf{A}_t \mathbf{X}_t + \mathbf{\Gamma} \mathbf{Z}_t + \mathbf{u}_{0t}, \quad (18)$$

where $\mathbf{Z}_t$ is a K_2 -dimensional vector of stationary variables. It is worth mentioning that $\mathbf{Z}_t$ enter the system (18)-(2) with time-invariant coefficients $\mathbf{\Gamma}$. Allowing these coefficients to vary over time would undermine the interpretation of $\mathbf{A}_t$ as the evolving long-run relation between $\mathbf{y}_t$ and $\mathbf{X}_t$, since a time-varying $\mathbf{\Gamma}_t$ interacting with stationary regressors would introduce additional low-frequency components in $\mathbf{y}_t$ that are incompatible with the cointegration structure. Fixing $\mathbf{\Gamma}$ ensures that all systematic time variation in the long-run component is captured solely by $\mathbf{A}_t$, while $\mathbf{Z}_t$ accounts only for transitory fluctuations. This separation guarantees that the contribution of stationary regressors remains purely short-run, without contaminating the low-frequency variation driven by the integrated process $\mathbf{X}_t$ and prevents spurious drift that would arise from time-varying coefficients on $\mathbf{Z}_t$.

We assume that $(\mathbf{u}_t^\top, \mathbf{Z}_t^\top)^\top$ satisfies the following invariance principle:

$$\frac{1}{\sqrt{n}} \sum_{t=1}^{\lfloor nr \rfloor} (\mathbf{u}_t^\top, \mathbf{Z}_t^\top)^\top \Rightarrow \mathbf{B}(r) \text{ on } D([0, 1], \mathbb{R}^{d+K_1+K_2}), \quad (19)$$

where $\mathbf{B}(r) = (\mathbf{B}_0^\top(r), \mathbf{B}_X^\top(r), \mathbf{B}_Z^\top(r))^\top$ is a vector Brownian motion with long-run covariance

matrix Ω . We have $\Omega = \Sigma + \Lambda + \Lambda^\top$ and the following three-way partition

$$\Omega := \begin{bmatrix} \Omega_{00} & \Omega_{0x} & \Omega_{0z} \\ \Omega_{x0} & \Omega_{xx} & \Omega_{xz} \\ \Omega_{z0} & \Omega_{zx} & \Omega_{zz} \end{bmatrix}, \quad \Lambda := \begin{bmatrix} \Lambda_{00} & \Lambda_{0x} & \Lambda_{0z} \\ \Lambda_{x0} & \Lambda_{xx} & \Lambda_{xz} \\ \Lambda_{z0} & \Lambda_{zx} & \Lambda_{zz} \end{bmatrix}.$$

With the estimation procedure introduced in Section 2.2, we now have an extended model:

$$\mathbf{y}_t = \mathbf{B}^{(k)} \mathbf{X}_t^{(k)} + \Gamma \mathbf{Z}_t + \mathbf{u}_{0t}^{(k)}, \quad (20)$$

where $\mathbf{u}_{0t}^{(k)} = \mathbf{u}_{0t} + (\mathbf{A}_t - \mathbf{A}_k(\tau_t)) \mathbf{X}_t$ and $\mathbf{B}^{(k)}, \mathbf{X}_t^{(k)}$ are defined as in (13). $\mathbf{B}^{(k)}$ and Γ can thus be estimated by LS (least squares):

$$\begin{bmatrix} \widehat{\mathbf{B}}_n^{(k)} & \widehat{\Gamma}_n \end{bmatrix} = \begin{bmatrix} \sum_{t=1}^n \mathbf{y}_t \mathbf{X}_t^{(k)\top} \\ \sum_{t=1}^n \mathbf{y}_t \mathbf{Z}_t^\top \end{bmatrix} \begin{bmatrix} \sum_{t=1}^n \mathbf{X}_t^{(k)} \mathbf{X}_t^{(k)\top} & \sum_{t=1}^n \mathbf{X}_t^{(k)} \mathbf{Z}_t^\top \\ \sum_{t=1}^n \mathbf{Z}_t \mathbf{X}_t^{(k)\top} & \sum_{t=1}^n \mathbf{Z}_t \mathbf{Z}_t^\top \end{bmatrix}^{-1}.$$

Finally, the series estimator for $\mathbf{A}(\cdot)$ is obtained in the same manner as in (15).

Our next theorem characterises the limit theory of the estimator for Γ , the coefficient matrix of the stationary regressors.

Theorem 2. Suppose that Assumptions 1-3 are satisfied. If, in addition, (19) holds and $\mathbb{E}(\mathbf{u}_{0t} \otimes \mathbf{Z}_t) = \mathbf{0}_{dK_2 \times 1}$, then

$$\sqrt{n} \cdot \text{vec}(\widehat{\Gamma}_n - \Gamma) \xrightarrow{d} \mathcal{N}\left(\mathbf{0}_{dK_2 \times 1}, (\Sigma_{zz}^{-1} \otimes \mathbf{I}_d) \Omega^{(0z)} (\Sigma_{zz}^{-1} \otimes \mathbf{I}_d)\right),$$

where $\Omega^{(0z)} = \sum_{j=-\infty}^{\infty} \mathbb{E}(\mathbf{u}_{0t} \mathbf{u}_{0t+j}^\top \otimes \mathbf{Z}_t \mathbf{Z}_{t+j}^\top)$.

Of course, we need the condition $\mathbb{E}(\mathbf{u}_{0t} \otimes \mathbf{Z}_t) = \mathbf{0}_{dK_2 \times 1}$, so that $\mathbf{Z}_t$ is weakly exogenous and Γ can be consistently estimated. Interestingly, the asymptotic distribution of $\widehat{\Gamma}_n$ coincides with the case when the cointegrating coefficients $\mathbf{A}_t$ are constant (e.g., Park and Phillips 1989, Phillips 1995), provided that series expansion order k satisfies Assumption 3. The reason is that, when k grows sufficiently fast, the series approximation error becomes asymptotically negligible, so

that it does not contribute to the first order asymptotic expansion of $\sqrt{n} \cdot \text{vec} \left(\widehat{\Gamma}_n - \Gamma \right)$, and the standard $\sqrt{n}$ -asymptotic theory for regressions with stationary regressors remains valid.

Theorem 3 below establishes the first-order asymptotic expansion for the series estimator (15) for the time-varying coefficients $\mathbf{A}(\cdot)$.

Theorem 3. Suppose that Assumptions 1-3 are satisfied. If, in addition, (19) holds. For any given $\tau_t \in [0, 1]$, we have, as $n \rightarrow \infty$

$$n \cdot \left(\widehat{\mathbf{A}}_n^{(k)}(\tau_t) - \mathbf{A}(\tau_t) + \Gamma(\tau_t) \right) = \widetilde{\Gamma}_{ux}^{(k)} \left(\Lambda_{xx}^{(k)} \right)^{-1} \mathbf{T}_{(k)}(\tau_t) + o_p \left(\sqrt{k} \right), \quad (21)$$

where $\Gamma(u) := \mathbf{A}(u) - \mathbf{A}_k(u)$, $\Lambda_{xx}^{(k)}$ is defined as in Theorem 1, and

$$\begin{aligned} \widetilde{\Gamma}_{ux}^{(k)} := & \int_0^1 \left(d\mathbf{B}_0(r) - \Sigma_{0z} \Sigma_{zz}^{-1} d\mathbf{B}_Z(r) \right) \left(\psi_{(k)}(r) \otimes \mathbf{B}_X(r) \right)^\top + \ell_{1(k)} \otimes (\Lambda_{0x} + \Sigma_{0x}) \\ & - \ell_{1(k)} \otimes \left\{ \Sigma_{0z} \Sigma_{zz}^{-1} (\Lambda_{zx} + \Sigma_{zx}) \right\}. \end{aligned}$$

Compared to Theorem 1, we see that the leading term is altered once stationary regressors are included, unless $\mathbb{E}(\mathbf{u}_{0t} \otimes \mathbf{Z}_t) = \mathbf{0}_{dK_2 \times 1}$. Nevertheless, if the primary interest lies in estimating the time-varying long-run relationship $\mathbf{A}_t$, the representation in (17) provides guidance on how stationary variables may be incorporated to obtain a more tractable asymptotic distribution. Define the orthogonalized Brownian motion $\mathbf{B}_{0,z}(r) = \mathbf{B}_0(r) - \Sigma_{0z} \Sigma_{zz}^{-1} \mathbf{B}_Z(r)$. Then

$$\begin{aligned} \mathbb{E} \left[\mathbf{B}_{0,z}(r) \mathbf{B}_X^\top(r) \right] &= \mathbb{E} \left[\mathbf{B}_0(r) \mathbf{B}_X^\top(r) \right] - \Sigma_{0z} \Sigma_{zz}^{-1} \mathbb{E} \left[\mathbf{B}_Z(r) \mathbf{B}_X^\top(r) \right] \\ &= \Omega_{0x} - \Sigma_{0z} \Sigma_{zz}^{-1} \Omega_{zx} \\ &= \Sigma_{0x} + \Lambda_{0x} + \Lambda_{x0}^\top - \Sigma_{0z} \Sigma_{zz}^{-1} (\Sigma_{zx} + \Lambda_{zx} + \Lambda_{xz}^\top) \\ &= \Sigma_{0x} - \Sigma_{0z} \Sigma_{zz}^{-1} \Sigma_{zx} + \Lambda_{0x} - \Sigma_{0z} \Sigma_{zz}^{-1} \Lambda_{zx} + \Lambda_{x0}^\top - \Sigma_{0z} \Sigma_{zz}^{-1} \Lambda_{xz}^\top. \end{aligned}$$

A sufficient condition for the asymptotic distribution to be mixed normal is therefore $\mathbb{E} \left[\mathbf{B}_{0,z}(r) \mathbf{B}_X^\top(r) \right] = \mathbf{0}_{d \times K_1}$. To interpret this condition in terms of the underlying innovations, define the orthogonalized residual $\bar{\mathbf{u}}_{0,t} = \mathbf{u}_{0t} - \Sigma_{0z} \Sigma_{zz}^{-1} \mathbf{Z}_t$. Then, the preceding covariance components admit the

representations

$$\Sigma_{0x} - \Sigma_{0z}\Sigma_{zz}^{-1}\Sigma_{zx} + \Lambda_{0x} - \Sigma_{0z}\Sigma_{zz}^{-1}\Lambda_{zx} = \sum_{j=0}^{\infty} \mathbb{E}(\bar{\mathbf{u}}_{0,t}\mathbf{u}_{x,t-j}^{\top}), \quad (22)$$

$$\Lambda_{x0}^{\top} - \Sigma_{0z}\Sigma_{zz}^{-1}\Lambda_{xz}^{\top} = \sum_{j=0}^{\infty} \mathbb{E}(\bar{\mathbf{u}}_{0,t}\mathbf{u}_{x,t+j}^{\top}). \quad (23)$$

Achieving a mixed normal limiting distribution requires both (22) and (23) to vanish. Consider a simple case where $\Lambda_{0x} = \Lambda_{zx} = \mathbf{0}$. In this case, by simply choosing $\mathbf{Z}_t = \Delta\mathbf{X}_t$ would make (22) vanish, but still not enough to kill (23). The structure of (22) and (23), nevertheless, motivates us to consider the following model specification by including both leads and lags of $\Delta\mathbf{X}_t = \mathbf{u}_{xt}$

$$\mathbf{y}_t = \mathbf{B}^{(k)}\mathbf{X}_t^{(k)} + \sum_{j=-K}^K \Gamma_j\mathbf{X}_{t-j} + \tilde{\mathbf{v}}_t^{(k)}, \quad (24)$$

where K is the number of leads (lags), $\tilde{\mathbf{v}}_t^{(k)} = \mathbf{v}_t + \sum_{|j|>K} \Gamma_j\mathbf{X}_{t-j} + (\mathbf{A}_t - \mathbf{A}_k(\tau_t))\mathbf{X}_t$, and $(\mathbf{v}_t)_t$ is a stationary process satisfying $\mathbb{E}[\mathbf{u}_{xt}\mathbf{v}_{t+j}^{\top}] = \mathbf{0}_{K_1 \times d}$ for all j . The detailed justification of the specification (24) is provided in Section B.

We now present the following theorem for the series estimator $\hat{\mathbf{A}}_n^{(k)}(\tau_t) = \tilde{\mathbf{B}}_n^{(k)}\mathbf{T}_{(k)}(\tau_t)$, where $\tilde{\mathbf{B}}_n^{(k)}$ is obtained from the LS estimation of (24).

Theorem 4. Suppose that Assumptions 1-3 are satisfied. If, in addition, $\frac{K^2}{n^{1-\gamma}} \rightarrow 0$ and $\sqrt{n} \sum_{|j|>K} \|\Gamma_j\| \rightarrow 0$, where γ is defined as in Assumption 3. For any given $\tau_t \in [0, 1]$, we have, as $n \rightarrow \infty$

$$n \left(\tilde{\mathbb{V}}_{n(k)}(\tau_t) \right)^{-1/2} \text{vec} \left(\hat{\mathbf{A}}_n^{(k)}(\tau_t) - \mathbf{A}(\tau_t) + \mathbf{\Gamma}(\tau_t) \right) \xrightarrow{d} \mathcal{N}(\mathbf{0}_{dK_1 \times 1}, \mathbf{I}_{dK_1}), \quad (25)$$

where

$$\tilde{\mathbb{V}}_{n(k)}(\tau_t) := \left\{ \mathbf{T}_{(k)}^{\top}(\tau_t) \left(\Lambda_{n,xx}^{(k)} \right)^{-1} \mathbf{T}_{(k)}(\tau_t) \right\} \otimes \hat{\Omega}_{n,0 \cdot x},$$

and $\hat{\Omega}_{n,0 \cdot x}$ is a consistent estimator of $\Omega_{0 \cdot x} = \Omega_{00} - \Omega_{0x}\Omega_{xx}^{-1}\Omega_{x0}$.

Theorem 4 shows that asymptotic distribution of the estimator is mixed Normal in the general model setup (1)-(2), provided that enough number of leads and lags of $\Delta\mathbf{X}_t$ are augmented into the model. In this sense, the specification may be viewed as a time-varying extension of the

dynamic OLS framework of Stock and Watson 1993, which builds on the approach of Saikkonen 1991 by augmenting the cointegration regression with leads and lags of first differences to address endogeneity and serial correlation. We shall name the series estimator based on the specification (24) as series-type dynamic OLS estimator.

The requirement $\sqrt{n} \sum_{|j|>K} \|\Gamma_j\| \rightarrow 0$ implies that K has to be large, so the approximation error becomes asymptotically negligible. In contrast, $\frac{K^2}{n^{1-\gamma}} \rightarrow 0$ restricts the rate at which K may diverge. In practice, some experiments show that a small value of K may be advisable, as detailed in Section 4. It may be noted that a similar tuning issue arises in the canonical time-varying cointegration framework of Park and Hahn 1999, where efficient inference relies on HAC-type estimation of long-run covariance matrices and hence requires the choice of a truncation lag (or bandwidth) parameter. By contrast, in the present setting endogeneity and serial correlation are addressed directly through finite lead-lag augmentation. As a result, implementation is operationally more straightforward and less sensitive to an additional layer of smoothing choices.

In practice, pointwise confidence intervals for $\hat{\mathbf{A}}_n^{(k)}(\tau_t)$ can be constructed using a bootstrap procedure similar to the ones outlined in Chang, Park, and Song 2006. We summarize the implementation steps below.

Step 1. Fit the regression (1) to obtain $\hat{\mathbf{u}}_{0t} = \mathbf{y}_t - \mathbf{A}_{nt}\mathbf{X}_t$ and define $\hat{\mathbf{u}}_t = (\hat{\mathbf{u}}_{0t}^\top, \mathbf{u}_{xt}^\top)^\top$, where $\mathbf{u}_{xt} = \Delta\mathbf{X}_t$.

Step 2. Fit a VAR(q) model to $(\hat{\mathbf{u}}_t)_t$, where q is a suitably chosen finite lag order. Obtain $(\varepsilon_t^*)_t$ by randomly resampling the centered fitted residuals $(\hat{\varepsilon}_{qt})_t$ and construct the bootstrap observations $(\mathbf{u}_t^*)_t$ recursively from the estimated VAR(q) model given the initial values.

Step 3. Partition $\mathbf{u}_t^* = (\mathbf{u}_{0t}^{*\top}, \mathbf{u}_{xt}^{*\top})^\top$ analogously as $\mathbf{u}_t = (\mathbf{u}_{0t}^\top, \mathbf{u}_{xt}^\top)^\top$. Obtain the bootstrap observations $(\mathbf{X}_t^*)_t$ by $\mathbf{X}_t^* = \mathbf{X}_0^* + \sum_{j=1}^t \mathbf{u}_{jt}^*$ with $\mathbf{X}_0^* = \mathbf{X}_0$ and generate bootstrap observations $(\mathbf{y}_t^*)_t$ from $\mathbf{y}_t^* = \mathbf{A}_{nt}\mathbf{X}_t^* + \mathbf{u}_{0t}^*$. The estimate $\mathbf{A}_{nt}$ of $\mathbf{A}_t$ here is not necessarily the same as the one used in Step 1 and will be employed to construct pointwise confidence intervals.

4 – Monte Carlo simulations

We use the Data Generating Process (D.G.P.):

$$y_t = A_t X_t + u_{0t},$$

where A_t has the following functional forms

- M1: $A_t = 1 + \frac{t}{n}$;
- M2: $A_t = e^{-\frac{t}{n}}$.

The regressor X_t is generated according to $X_t = X_{t-1} + u_{xt}$. We generate $\mathbf{u}_t = (u_{0t}, u_{xt})^\top$ from a bivariate MA(1) process

$$\mathbf{u}_t = \boldsymbol{\varepsilon}_t + \boldsymbol{\Theta} \boldsymbol{\varepsilon}_{t-1}, \quad \boldsymbol{\varepsilon}_t \stackrel{iid}{\sim} \mathcal{N}(0, \boldsymbol{\Sigma}_\varepsilon)$$

with

$$\boldsymbol{\Theta} = \begin{pmatrix} 0.3 & -0.4 \\ \theta_{21} & 0.6 \end{pmatrix}, \quad \boldsymbol{\Sigma}_\varepsilon = \begin{pmatrix} 1 & \sigma_{21} \\ \sigma_{21} & 1 \end{pmatrix},$$

and θ_{21}, σ_{21} allowed to vary. We consider values of $\{0.8, 0.4, 0, -0.8\}$ for θ_{21} and $\{-0.85, -0.5, 0.5\}$ for σ_{21} , as in Phillips and Loretan 1991.

To compute the series estimator, we consider both the baseline specification

$$y_t = A_t X_t + u_{0t} \tag{26}$$

and a slightly more flexible specification in which number of lags (K_a) and leads (K_e) may be different:

$$y_t = A_t X_t + \sum_{j=-K_e}^{K_a} \gamma_j \Delta X_{t-j} + u_{0t}. \tag{27}$$

Following Phillips and Loretan 1991, we consider several combinations of the numbers of lags (K_a) and leads (K_e): (i) $K_a = K_e = 0$; (ii) $K_a = 2, K_e = 0$; (iii) $K_a = 4, K_e = 0$; (iv) $K_a = 2, K_e = 1$; and (v) $K_a = 4, K_e = 2$.¹

1. Note that, in all these specifications based on (27), ΔX_t is always included in the model.

Regarding to the truncation parameter k , we use the generalized cross validation (GCV) method as in Gao, Tong, and Wolff 2002 to select an optima value k :

$$\hat{k} = \arg \min_{k \in \mathcal{K}_n} \left(1 - \frac{df(k, K_1, K_2)}{n} \right)^{-2} \frac{1}{n} \sum_{t=1}^n \left(\hat{u}_{0t}^{(k)} \right)^2,$$

where $df(k, K_1, K_2)$ is the total number of parameters to be estimated, $\mathcal{K}_n = \{1, 2, \dots, 10, 11, 12\}$, and $\hat{u}_{0t}^{(m)}$ is the fitted residuals, either from (26) or (27).

The performance of the estimators is evaluated by the mean squared error (MSE): $\frac{1}{n} \sum_{t=1}^n \left(\hat{A}_t - A_t \right)^2$. The objective is to assess the accuracy of estimating the time-varying parameters A_t and, in particular, to examine whether augmenting the regression with leads and lags improves estimation accuracy.

Tables 1-2 report the average MSE of the series estimator of A_t under designs M1 and M2, respectively. For each combination of $(\theta_{21}, \sigma_{21})$ and sample size n , we compare the baseline specification (26) with the augmented specification (27) across a range of choices for the numbers of lags (K_a) and leads (K_e). Under (26), the reported MSEs are in levels, whereas all other entries are ratios relative to the baseline performance. Values below 1 indicate an improvement from augmentation, and boxed entries mark the best-performing specification.

Several patterns emerge. First, the gains from (27) are substantially larger when $\sigma_{21} < 0$. This is consistent with our DGP, under which $\Omega_{0x} = \theta_{21}(1.3 - 0.4\sigma_{21}) + (2.08\sigma_{21} - 0.64)$, which is larger in magnitude when σ_{21} is negative. Second, augmenting the model with leads of ΔX_t generally provides additional improvements: specifications including leads deliver the best performance in 31 out of 36 cases under M1 and 29 out of 36 cases under M2. Finally, the qualitative conclusions are highly stable across M1 and M2, indicating robustness to the functional form of the time-varying coefficient. Overall, in line with the simulation evidence, these results suggest that modest lead-lag augmentation is a robust way to improve estimation accuracy, especially in situations with stronger endogeneity.

Table 1 – Small sample properties of the series estimator of A_t under M1: $A_t = 1 + \frac{t}{n}$

θ_{21}	$n = 200$				$n = 500$				$n = 1000$			
	0.8	0.4	0.0	−0.8	0.8	0.4	0.0	−0.8	0.8	0.4	0.0	−0.8
$\sigma_{21} = -0.85$												
OLS	0.030	0.014	0.007	0.002	0.007	0.003	0.001	0.000	0.002	0.001	0.000	0.000
(0,0)	0.418	0.242	0.158	0.172	0.398	0.235	0.168	0.204	0.405	0.246	0.170	0.290
(2,0)	0.246	0.194	0.155	0.147	0.217	0.186	0.158	0.168	0.209	0.181	0.171	0.252
(4,0)	0.240	0.202	0.150	0.131	0.218	0.175	0.159	0.165	0.210	0.181	0.170	0.237
(2,1)	0.246	0.204	0.155	0.115	0.204	0.188	0.161	0.137	0.189	0.176	0.167	0.215
(4,2)	0.242	0.195	0.155	0.106	0.209	0.184	0.151	0.116	0.194	0.173	0.171	0.185
(1,1)	0.231	0.199	0.149	0.122	0.215	0.186	0.167	0.135	0.206	0.188	0.172	0.215
(4,4)	0.248	0.192	0.143	0.095	0.205	0.184	0.159	0.109	0.200	0.174	0.163	0.175
(8,8)	0.269	0.220	0.152	0.097	0.211	0.185	0.152	0.098	0.198	0.163	0.157	0.159
$\sigma_{21} = -0.5$												
OLS	0.010	0.009	0.005	0.002	0.002	0.002	0.001	0.000	0.001	0.000	0.000	0.000
(0,0)	0.729	0.598	0.473	0.370	0.741	0.588	0.480	0.451	0.715	0.571	0.492	0.495
(2,0)	0.585	0.515	0.454	0.387	0.567	0.537	0.450	0.443	0.535	0.503	0.482	0.520
(4,0)	0.620	0.510	0.458	0.374	0.557	0.520	0.436	0.457	0.536	0.491	0.455	0.498
(2,1)	0.589	0.511	0.482	0.342	0.525	0.514	0.432	0.396	0.491	0.497	0.448	0.450
(4,2)	0.565	0.550	0.454	0.312	0.549	0.503	0.456	0.369	0.494	0.453	0.482	0.414
(1,1)	0.554	0.517	0.455	0.334	0.554	0.523	0.442	0.389	0.496	0.524	0.483	0.432
(4,4)	0.643	0.535	0.463	0.331	0.539	0.504	0.442	0.344	0.466	0.463	0.449	0.399
(8,8)	0.690	0.599	0.525	0.323	0.528	0.520	0.432	0.327	0.495	0.467	0.454	0.374
$\sigma_{21} = 0.5$												
OLS	0.001	0.002	0.002	0.003	0.000	0.000	0.000	0.001	0.000	0.000	0.000	0.000
(0,0)	1.297	1.204	1.200	1.075	1.187	1.285	1.174	0.897	1.167	1.125	1.237	0.925
(2,0)	1.035	1.010	1.114	1.003	0.906	1.021	1.053	0.937	0.909	0.995	1.049	1.058
(4,0)	1.046	1.011	1.088	1.064	0.883	1.001	1.008	0.964	0.893	0.920	1.003	1.029
(2,1)	0.965	1.036	1.208	0.886	0.746	0.914	0.999	0.801	0.728	0.851	0.986	0.856
(4,2)	0.972	1.030	1.147	0.917	0.735	0.944	0.997	0.791	0.690	0.825	1.001	0.821
(1,1)	0.965	1.022	1.113	0.893	0.777	0.975	1.089	0.797	0.786	0.927	1.081	0.806
(4,4)	0.986	1.040	1.176	0.892	0.740	0.943	0.992	0.802	0.695	0.792	0.970	0.847
(8,8)	1.118	1.165	1.204	0.941	0.753	0.942	0.978	0.785	0.625	0.819	0.985	0.786

Notes. OLS is based on specification (26). All other rows are based on specification (27), with different combinations of the numbers of leads K_e and lags K_a . Entries in the row “OLS” report MSE in levels, while the remaining entries are ratios relative to OLS performance. For each combination of $(\theta_{21}, \sigma_{21})$ and sample size n , boxed entries indicate the best performance.

Table 2 – Small sample properties of the series estimator of A_t under M2: $A_t = e^{-\frac{t}{n}}$

θ_{21}	$n = 200$				$n = 500$				$n = 1000$			
	0.8	0.4	0.0	−0.8	0.8	0.4	0.0	−0.8	0.8	0.4	0.0	−0.8
$\sigma_{21} = -0.85$												
OLS	0.028	0.013	0.006	0.002	0.006	0.003	0.001	0.000	0.002	0.001	0.000	0.000
(0,0)	0.424	0.230	0.152	0.129	0.416	0.250	0.163	0.173	0.412	0.260	0.171	0.196
(2,0)	0.258	0.199	0.145	0.110	0.228	0.192	0.152	0.143	0.204	0.182	0.156	0.173
(4,0)	0.269	0.190	0.145	0.100	0.222	0.178	0.158	0.137	0.204	0.178	0.160	0.163
(2,1)	0.262	0.201	0.138	0.093	0.220	0.180	0.168	0.123	0.190	0.167	0.159	0.148
(4,2)	0.260	0.217	0.146	0.084	0.213	0.182	0.150	0.109	0.195	0.177	0.153	0.124
(1,1)	0.253	0.203	0.150	0.102	0.217	0.184	0.156	0.127	0.189	0.176	0.150	0.153
(4,4)	0.266	0.217	0.152	0.081	0.214	0.186	0.155	0.107	0.194	0.171	0.153	0.118
(8,8)	0.313	0.243	0.164	0.089	0.223	0.190	0.148	0.090	0.197	0.168	0.149	0.107
$\sigma_{21} = -0.5$												
OLS	0.010	0.008	0.005	0.002	0.002	0.002	0.001	0.000	0.001	0.000	0.000	0.000
(0,0)	0.764	0.602	0.444	0.311	0.735	0.528	0.407	0.372	0.761	0.589	0.480	0.409
(2,0)	0.640	0.561	0.405	0.318	0.556	0.455	0.395	0.372	0.548	0.506	0.444	0.398
(4,0)	0.692	0.531	0.418	0.292	0.582	0.446	0.406	0.348	0.556	0.471	0.415	0.383
(2,1)	0.614	0.566	0.439	0.291	0.524	0.461	0.406	0.340	0.508	0.465	0.437	0.363
(4,2)	0.608	0.594	0.428	0.282	0.506	0.446	0.405	0.307	0.487	0.461	0.432	0.327
(1,1)	0.619	0.562	0.407	0.293	0.534	0.470	0.421	0.321	0.527	0.486	0.443	0.341
(4,4)	0.627	0.599	0.451	0.294	0.530	0.462	0.424	0.307	0.513	0.463	0.414	0.322
(8,8)	0.715	0.645	0.483	0.313	0.565	0.479	0.393	0.302	0.517	0.487	0.430	0.305
$\sigma_{21} = 0.5$												
OLS	0.001	0.001	0.002	0.003	0.000	0.000	0.000	0.001	0.000	0.000	0.000	0.000
(0,0)	1.204	1.146	1.281	0.975	1.262	1.207	1.158	0.991	1.259	1.218	1.170	0.992
(2,0)	0.956	0.995	1.233	0.991	0.972	0.997	1.012	1.027	0.967	0.953	1.028	1.022
(4,0)	0.953	0.986	1.230	0.991	0.959	0.937	1.068	0.986	0.874	0.928	1.014	0.992
(2,1)	1.008	1.098	1.274	0.858	0.840	0.938	1.047	0.860	0.751	0.854	0.994	0.875
(4,2)	1.039	1.105	1.258	0.883	0.831	0.965	1.053	0.888	0.718	0.883	0.962	0.858
(1,1)	0.973	1.035	1.255	0.864	0.881	0.942	1.104	0.871	0.768	0.918	1.044	0.886
(4,4)	1.083	1.119	1.297	0.906	0.783	0.933	1.069	0.834	0.705	0.842	0.999	0.831
(8,8)	1.130	1.197	1.407	1.033	0.811	0.944	1.041	0.844	0.661	0.837	0.960	0.840

Notes. See Notes in Table 1.

5 – An empirical application

We illustrate the proposed method in an empirical application to the consumption–wealth ratio (*cay*) and its ability to predict U.S. quarterly equity premium. The *cay* variable, originally proposed by Lettau and Ludvigson 2001, measures the deviation in the common trend between consumption (*c*), aggregate wealth (*a*), and labor income (*y*):

$$cay_t = c_t - \omega a_t - (1 - \omega)y_t, \quad (28)$$

where ω equals the average share of asset holdings in total wealth. Under certain conditions, it can be shown that

$$cay_t = \mathbb{E}_t \sum_{i=1}^{\infty} \rho_{\omega}^i \{ [\omega r_{a,t+i} + (1 - \omega)r_{h,t+i}] - \Delta c_{t+i} \} + (1 - \omega)z_t, \quad (29)$$

where $\mathbb{E}_t$ is the conditional expectation on information available at time t , ρ_{ω} is the steady-state ratio of new investment to total wealth, $r_{a,t}$ is the return on asset holdings, $r_{h,t}$ is the return on human capital, and z_t is a mean zero stationary random variable. (29) suggests that the consumption-wealth ratio *cay*_{*t*} may contain information about future asset returns $r_{a,t+i}$. In particular, its predictive content is stronger when future returns on human capital $r_{h,t+i}$ and consumption growth Δc_{t+i} exhibit limited volatility, or these variables are closely related to expected returns on assets.

An important first step in using *cay*_{*t*} to forecast asset returns is the estimation of cointegration relationship between consumption, aggregate wealth and labor income in (28). Lettau and Ludvigson 2001 employs Stock and Watson 1993’s dynamic least squares (DLS) technique which involves estimating the following single equation model

$$c_t = \alpha + \beta_a a_t + \beta_y y_t + \sum_{i=-K}^K b_{a,i} \Delta a_{t-i} + \sum_{i=-K}^K b_{y,i} \Delta y_{t-i} + \epsilon_t.$$

With the estimated parameters $\hat{\beta}_a$ and $\hat{\beta}_y$, we then define $\widehat{cay}_t = c_t - \hat{\beta}_a a_t - \hat{\beta}_y y_t$. This can be thought of as the “residuals” from the cointegration regression and therefore represents the

deviations from the shared trend in consumption, assets, and labor income.

While Lettau and Ludvigson 2001 argue that the parameters in *cay* are steady-state wealth shares and therefore unlikely to vary over time, empirical evidence has generally questioned the stability of this equilibrium relationship. In particular, Dauber and Lawrenz 2025 conduct an extensive analysis by estimating $(\beta_a, \beta_y)^\top$ over different sample periods and find that the coefficients are not stable over time. Motivated by these findings, we use the series estimator to obtain a time-varying estimate of the cointegrating vector and construct a corresponding time-varying measure of *cay*. We then assess whether this measure improves the forecasting performance for the U.S. equity risk premium relative to the conventional *cay* based on a fixed-coefficient cointegration regression.

We use quarterly data spanning the period of 1952Q1-2024Q4. Data on consumption² and income are taken from the Bureau of Economic Analysis (BEA). The wealth series are obtained from the Board of Governors of the Federal Reserve System (FED). The aggregated wealth and labour income series are all deflated using the PCE-type deflator and then transformed into per capita terms (in logarithm). Finally, we use CRSP value-weighted index return provided by the Center for Research in Security Prices (CRSP). The risk-free rate is the Treasury-bill rate. More detailed description of how each variables are constructed can be found in section F.

Our estimation is based on the series-type dynamic OLS method proposed in section 3. As in section 4, GCV method is used to select the truncation parameter k over the grid $\{1, 2, 3, \dots, 11, 12\}$. Motivated by our simulation findings and parsimonious considerations, we include 1 lead and 1 lag of Δa_t and Δy_t .

To avoid complications arising from the pandemic period, we first report estimation results using the pre-pandemic sample ending in 2019Q4. Figure 1 presents the empirical estimates of $\beta_{a,t}$ and $\beta_{y,t}$ together with their 95% confidence intervals obtained from the bootstrap procedure described in Section 3. Overall, there is substantial time variation in both parameters. In particular, there are brief periods in the 1960s and around 2000 during which $\beta_{a,t}$ becomes statistically insignificant. Since theory in Lettau and Ludvigson 2001 implies that $\beta_{a,t} + \beta_{y,t} = 1$, the two coefficients should move in opposite directions. Consistent with this implication, the figure shows

2. The results in this section are based on the nondurables plus services minus clothing and footwear (NDS). As demonstrated in Martin Lettau's website (<https://sites.google.com/view/martinlettau/data>), we also consider the measure based on personal consumption expenditures (PCE). These results are reported in section E.

that when $\beta_{a,t}$ declines, $\beta_{y,t}$ tends to increase, and vice versa. However, the estimation results change markedly once post-pandemic observations are included. As shown in Figure 2, both parameters exhibit substantially greater volatility. Although we observe an increase in $\beta_{a,t}$ and a decline in $\beta_{y,t}$ since the 1990s, $\beta_{y,t}$ is also statistically insignificant throughout this period.

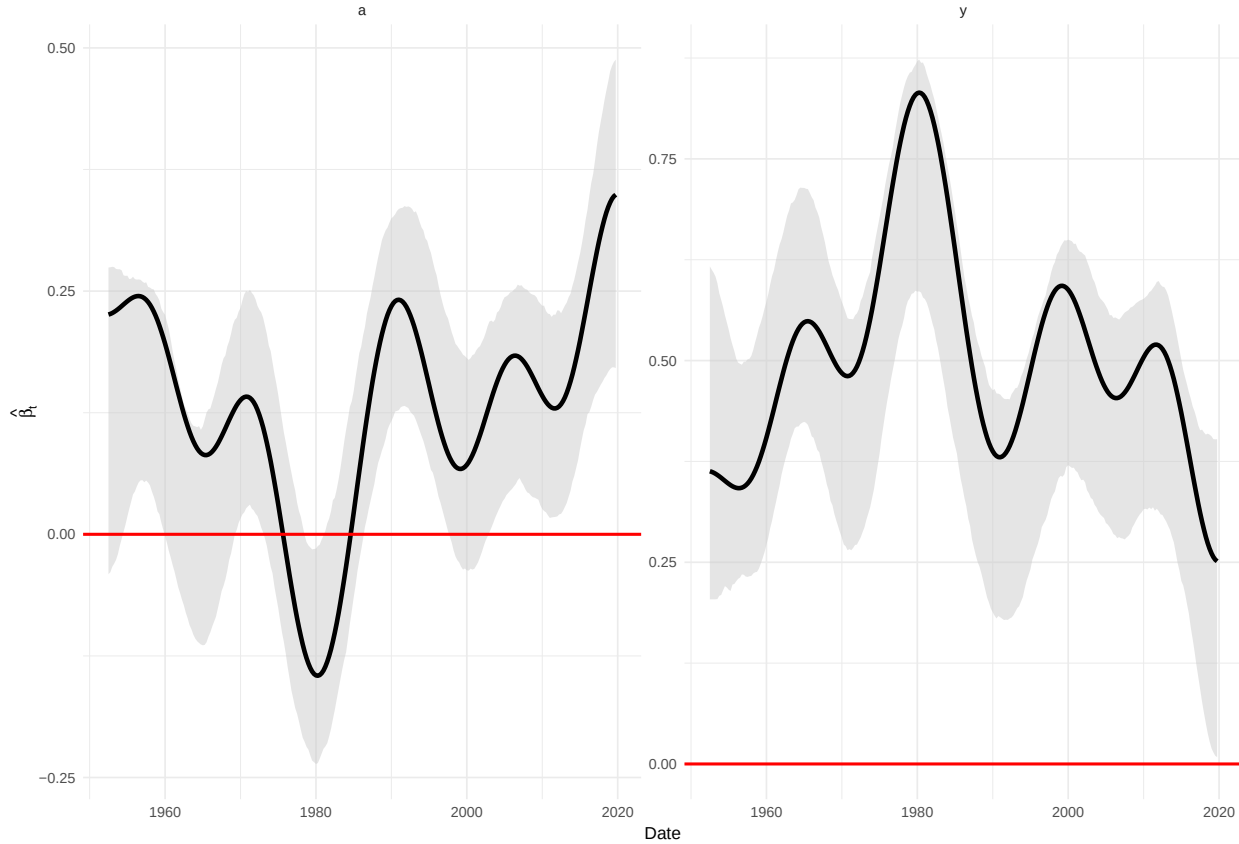


Figure 1 – Empirical estimates for $\beta_{a,t}$ and $\beta_{y,t}$.

Notes. The figure plots the estimated time-varying cointegrating coefficients $\beta_{a,t}$ and $\beta_{y,t}$ over the sample period 1951Q4–2019Q4. Point estimates are obtained using the series-type dynamic OLS estimator with one lead and one lag of the first differences of the regressors. Shaded areas represent 95% bootstrap confidence intervals.

We now assess whether the *cay* measure constructed using time-varying coefficients improves the out-of-sample forecasting performance of the quarterly U.S. equity premium (defined as the CRSP value-weighted index return minus the lagged Treasury-bill rate) relative to the conventional *cay* measure based on a fixed-coefficient cointegration regression. The out-of-sample (OOS) evaluation begins in 1995Q1. Forecasts are generated recursively using an expanding estimation window. Forecast performance is evaluated using the out-of-sample R^2 statistic (in percent),

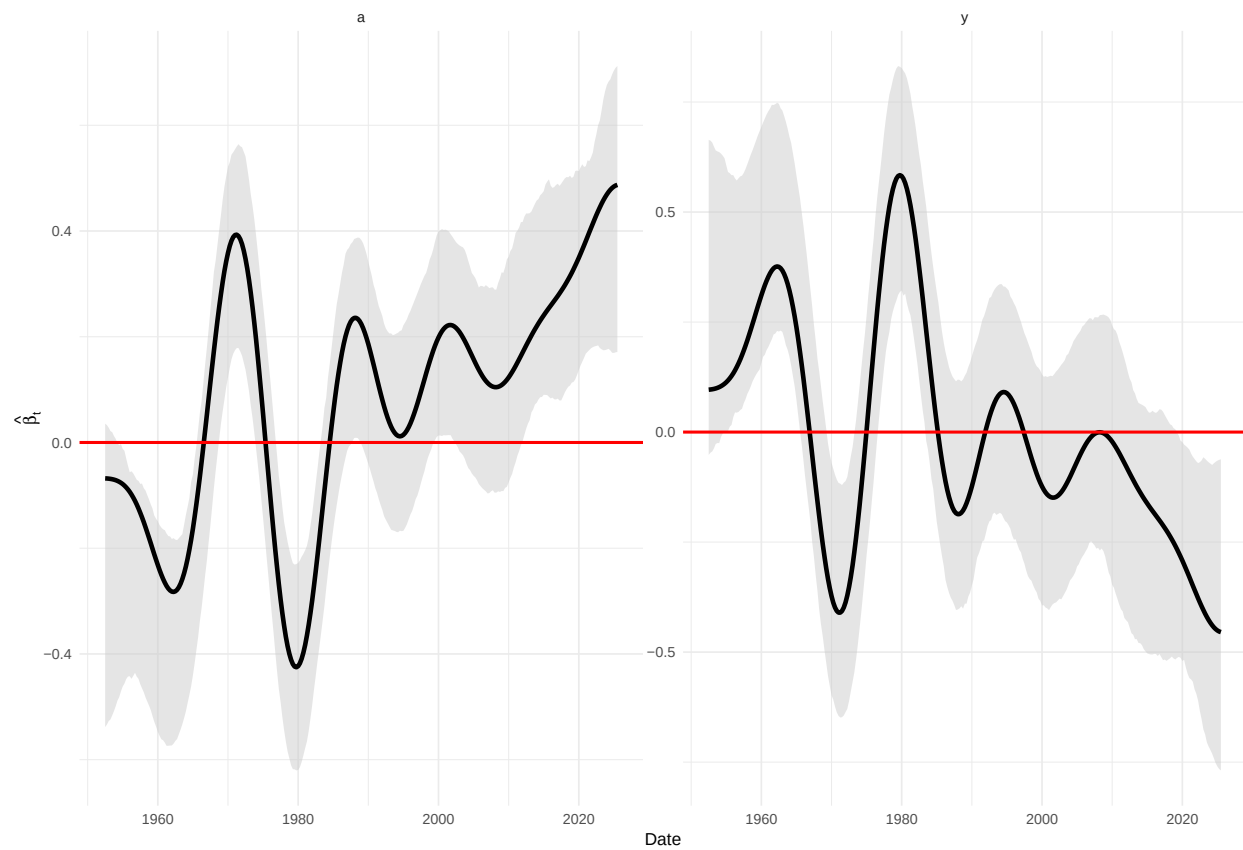


Figure 2 — Empirical estimates for $\beta_{a,t}$ and $\beta_{y,t}$.

Notes. See Note in Figure 1.

computed relative to the historical mean benchmark:

$$R_{\text{OOS}}^2 = 1 - \frac{\sum_{t=T_0}^{T-1} (r_{t+1} - \hat{r}_{t+1|t})^2}{\sum_{t=T_0}^{T-1} (r_{t+1} - \bar{r}_{t+1|t})^2}. \quad (30)$$

Here r_{t+1} denotes the realized equity premium, $\hat{r}_{t+1|t}$ is the one-step-ahead forecast from the predictive regression using the *cay* measure constructed from either the fixed-coefficient or the time-varying cointegration regression, and $\bar{r}_{t+1|t}$ denotes the historical mean forecast. To assess the statistical significance of differences in mean squared forecast errors (MSE) between the *cay*-based forecasts and the historical mean benchmark, we report p -values based on the Clark and West 2007's test.

Table 3 reports the results for two evaluation samples: 1995Q1–2019Q4 and 1995Q1–2024Q4, and are further decomposed into overall, recession, and expansion periods based on National Bureau of Economic Research (NBER) recession indicator. Several patterns emerge. First, in the pre-pandemic evaluation sample (Panel A), the *cay* measure constructed using time-varying coefficients delivers positive out-of-sample R^2 overall, whereas the conventional fixed-coefficient measure performs poorly with negative R^2 . The gains from the time-varying specification are primarily driven by the expansion periods, where the time-varying *cay* produces a positive R^2 , while the fixed-coefficient measure yields substantially negative forecasting performance. Second, once the post-COVID observations are included (Panel B), the overall predictability of the equity premium weakens substantially for both predictors. Nevertheless, the time-varying *cay* continues to outperform the fixed-coefficient measure, delivering uniformly smaller forecast errors and less negative R^2 across all subsamples.

To gain further insight into the source of the forecasting gains produced by the time-varying *cay*, we examine how forecasting performance evolves over time. Figure 3 plots the cumulative difference in mean squared forecast errors relative to the historical mean benchmark. The results show that the time-varying *cay* generally outperforms the benchmark over most of the evaluation period up to 2019Q4. In contrast, the fixed-coefficient *cay* measure only briefly outperforms the time-varying specification in the early 2000s, around the recession following the dot-com bubble collapse. After this period, and particularly following the global financial crisis of 2007–2009, the time-varying *cay* consistently delivers lower forecast errors than the fixed-coefficient measure.

Table 3 — Quarterly U.S. equity premium out-of-sample forecasting results

Predictor	Overall		Recession		Expansion	
	$R^2\%$	p -value	$R^2\%$	p -value	$R^2\%$	p -value
Panel A: 1995Q1–2019Q4						
$\widehat{cay}_f$	-8.760	0.396	5.835	0.067	-13.688	0.516
$\widehat{cay}_{tvp}$	0.478	0.266	-0.860	0.605	0.930	0.147
Panel B: 1995Q1–2024Q4						
$\widehat{cay}_f$	-8.983	0.506	2.721	0.167	-13.935	0.598
$\widehat{cay}_{tvp}$	-6.168	0.920	-0.237	0.527	-8.677	0.919

Notes. The table reports out-of-sample forecasting performance for the quarterly U.S. equity premium using *cay* measures constructed from fixed ($\widehat{cay}_f$) and time-varying ($\widehat{cay}_{tvp}$) coefficient cointegration regressions. Forecast performance is evaluated using the OOS R^2 statistic (in percent) defined in (30) relative to the historical mean benchmark. p -values are based on the Clark and West 2007’s test for equal predictive accuracy. The forecast evaluation begins in 1995Q1 and forecasts are generated recursively using an expanding estimation window. Results are reported for two evaluation samples, 1995Q1–2019Q4 and 1995Q1–2024Q4, and are further decomposed into recession and expansion periods according to the NBER recession indicator.

Once post-COVID observations are included, the historical mean benchmark begins to dominate both predictors. Nevertheless, the time-varying *cay* continues to perform better than the fixed-coefficient measure, indicating that allowing for time variation in the cointegrating relationship still provides improvements relative to the conventional specification.

6 — Conclusion

This paper develops a series estimation framework for cointegrated system with smoothly time-varying cointegrating coefficients. The proposed approach avoids the degeneracy that arises in kernel-based estimation with integrated regressors and yields a tractable mixed normal asymptotic distribution under suitable dynamic augmentation. A practical bootstrap device is also presented for constructing pointwise confidence intervals for the estimated coefficients.

Simulation evidence indicates that moderate lead–lag augmentation can improve finite-sample performance in a range of settings. An application to the consumption–wealth ratio highlights substantial time variation in the long-run relationship and shows that incorporating this variation can enhance equity premium forecasts relative to conventional fixed-coefficient specifications. These results underscore the empirical relevance of modelling parameter instability in cointegrated

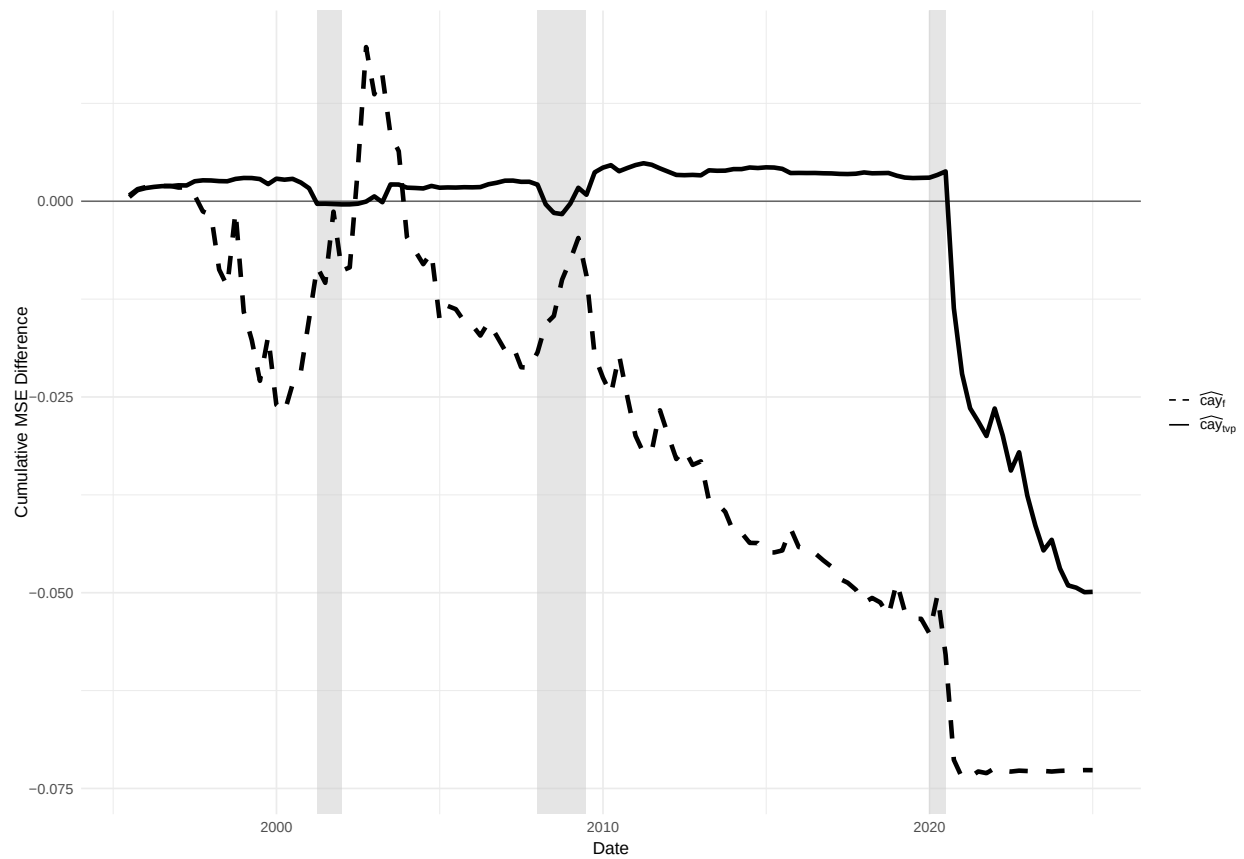


Figure 3 — Cumulative difference in MSE difference relative to the historical mean benchmark

Notes. The figure plots the cumulative difference in MSE relative to the historical mean benchmark for forecasts of the quarterly U.S. equity premium. Positive values indicate that the corresponding predictor outperforms the historical mean benchmark. The series $\widehat{cay}_f$ and $\widehat{cay}_{tvp}$ denote the *cay* measures constructed from the fixed-coefficient and time-varying cointegration regressions, respectively. Shaded areas correspond to NBER recession periods.

systems.

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A — Proofs of the main theorems

A.1. Proof of Theorem 1

Recall that the estimator for $\mathbf{B}^{(k)}$ is given by

$$\widehat{\mathbf{B}}_n^{(k)} = \left(\sum_{t=1}^n \mathbf{y}_t \mathbf{X}_t^{(k)} \right) \left(\sum_{t=1}^n \mathbf{X}_t^{(k)} \mathbf{X}_t^{(k)\top} \right)^{-1}.$$

Simple algebra gives

$$\begin{aligned} n \cdot \left(\widehat{\mathbf{B}}_n^{(k)} - \mathbf{B}^{(k)} \right) &= \left(\frac{1}{n} \sum_{t=1}^n \mathbf{u}_{0t}^{(k)} \mathbf{X}_t^{(k)\top} \right) \left(\frac{1}{n^2} \sum_{t=1}^n \mathbf{X}_t^{(k)} \mathbf{X}_t^{(k)\top} \right)^{-1} \\ &= \left(\frac{1}{n} \sum_{t=1}^n \mathbf{u}_{0t} \mathbf{X}_t^{(k)\top} \right) \left(\frac{1}{n^2} \sum_{t=1}^n \mathbf{X}_t^{(k)} \mathbf{X}_t^{(k)\top} \right)^{-1} \\ &\quad + \left(\frac{1}{n} \sum_{t=1}^n (\mathbf{A}_t - \mathbf{A}_k(t/n)) \mathbf{X}_t \mathbf{X}_t^{(k)\top} \right) \left(\frac{1}{n^2} \sum_{t=1}^n \mathbf{X}_t^{(k)} \mathbf{X}_t^{(k)\top} \right)^{-1} \\ &:= \boldsymbol{\Gamma}_{n,ux}^{(k)} \left(\boldsymbol{\Lambda}_{n,xx}^{(k)} \right)^{-1} + \boldsymbol{\Delta}_{n,xx}^{(k)} \left(\boldsymbol{\Lambda}_{n,xx}^{(k)} \right)^{-1}. \end{aligned}$$

By Lemma C.1, we have

$$\boldsymbol{\Lambda}_{n,xx}^{(k)} = \boldsymbol{\Lambda}_{xx}^{(k)} + o_p(1),$$

where $\boldsymbol{\Lambda}_{xx}^{(k)} = \int_0^1 \left(\boldsymbol{\psi}_{(k)}(r) \boldsymbol{\psi}_{(k)}^\top(r) \right) \otimes (\mathbf{B}_X(r) \mathbf{B}_X^\top(r)) dr$ and $\left\| \left(\boldsymbol{\Lambda}_{n,xx}^{(k)} \right)^{-1} \right\|_{sp} = O_p(1)$. In addition, Lemma C.3 leads to

$$\frac{1}{n} \boldsymbol{\Delta}_{n,xx}^{(k)} = \boldsymbol{\Delta}_{xx}^{(k)} + o_p(1),$$

where $\left\| \boldsymbol{\Delta}_{xx}^{(k)} \right\|_{sp} = O_p\left(\frac{1}{k^q}\right)$. This implies that $\left\| \boldsymbol{\Delta}_{n,xx}^{(k)} \left(\boldsymbol{\Lambda}_{n,xx}^{(k)} \right)^{-1} \right\|_{sp} \leq \left\| \boldsymbol{\Delta}_{xx}^{(k)} \right\|_{sp} \left\| \left(\boldsymbol{\Lambda}_{n,xx}^{(k)} \right)^{-1} \right\|_{sp} = O_p\left(\frac{n}{k^q}\right) = o_p(1)$.

Define $\boldsymbol{\Gamma}(\tau_t) := \mathbf{A}(\tau_t) - \mathbf{A}_k(\tau_t)$ and $\mathbf{T}_{(k)}(\tau_t) = \boldsymbol{\psi}_{(k)}(\tau_t) \otimes \mathbf{I}_{K_1}$ for any given $\tau_t = t/n$. Then,

we have

$$\begin{aligned} n \cdot \left(\widehat{\mathbf{A}}_n^{(k)}(\tau_t) - \mathbf{A}(\tau_t) + \mathbf{\Gamma}(\tau_t) \right) &= n \cdot \left[\left(\widehat{\mathbf{B}}_n^{(k)} - \mathbf{B}^{(k)} \right) \left(\boldsymbol{\psi}_{(k)}(\tau_t) \otimes \mathbf{I}_{K_1} \right) \right] \\ &= \mathbf{\Gamma}_{ux}^{(k)} \mathbf{\Lambda}_{xx}^{(k)-1} \mathbf{T}_{(k)}(\tau_t) + o_p(\sqrt{k}), \end{aligned}$$

where $\mathbf{\Gamma}_{ux}^{(k)} = \int_0^1 d\mathbf{B}_0(r) \left(\boldsymbol{\psi}_{(k)}(r) \otimes \mathbf{B}_X(r) \right)^\top + \boldsymbol{\ell}_{1(k)} \otimes (\mathbf{\Lambda}_{0x} + \mathbf{\Sigma}_{0x})$, which follows from Lemma C.1 and Lemma C.2.

A.2. Proof of Corollary 1

Let

$$\widehat{\mathbb{V}}_{n(k)}(\tau_t) = \left\{ \mathbf{T}_{(k)}^\top(\tau_t) \left(\mathbf{\Lambda}_{n,xx}^{(k)} \right)^{-1} \mathbf{T}_{(k)}(\tau_t) \right\} \otimes \widehat{\mathbf{\Omega}}_{n,00},$$

where $\widehat{\mathbf{\Omega}}_{n,00}$ is a consistent estimator of $\mathbf{\Omega}_{00}$. Then, we have

$$\begin{aligned} &n \left(\widehat{\mathbb{V}}_{n(k)}(\tau_t) \right)^{-1/2} \text{vec} \left(\widehat{\mathbf{A}}_n^{(k)}(\tau_t) - \mathbf{A}(\tau_t) + \mathbf{\Gamma}(\tau_t) \right) \\ &= \left(\left\{ \mathbf{T}_{(k)}^\top(u) \left(\mathbf{\Lambda}_{n,xx}^{(k)} \right)^{-1} \mathbf{T}_{(k)}(u) \right\} \otimes \widehat{\mathbf{\Omega}}_{n,00} \right)^{-1/2} \left(\mathbf{T}_{(k)}^\top(\tau_t) \otimes \mathbf{I}_d \right) \text{vec} \left\{ \mathbf{\Gamma}_{n,ux}^{(k)} \left(\mathbf{\Lambda}_{n,xx}^{(k)} \right)^{-1} \right\} + o_p(1). \end{aligned}$$

Using the joint convergence result of

$$\begin{pmatrix} \frac{1}{\sqrt{n}} \sum_{t=1}^{\lfloor nr \rfloor} \mathbf{u}_{0t} \\ \frac{1}{\sqrt{n}} \sum_{t=1}^{\lfloor nr \rfloor} \mathbf{u}_{xt} \end{pmatrix} \Rightarrow \begin{pmatrix} \mathbf{B}_0(r) \\ \mathbf{B}_X(r) \end{pmatrix},$$

on $D([0, 1], \mathbb{R}^{d+K_1})$, together with Lemma C.1 and Lemma C.2, we have the joint stochastic expansion

$$\begin{pmatrix} \mathbf{\Lambda}_{n,xx}^{(k)} \\ \mathbf{\Gamma}_{n,ux}^{(k)} \end{pmatrix} = \begin{pmatrix} \mathbf{\Lambda}_{xx}^{(k)} \\ \mathbf{\Gamma}_{ux}^{(k)} \end{pmatrix} + o_p(1),$$

where $\mathbf{\Lambda}_{xx}^{(k)}$ and $\mathbf{\Gamma}_{ux}^{(k)}$ are defined as in Theorem 1. Since $\mathbb{E}(\mathbf{u}_{0t} \otimes \mathbf{u}_{xs}) = \mathbf{0}_{dK_1 \times 1}$ for any t, s , we have $\mathbf{\Omega}_{0x} = \mathbf{0}_{dK_1}$, so that Brownian motions $\mathbf{B}_0(r)$ and $\mathbf{B}_X(r)$ are uncorrelated and hence independently distributed. Then, as in Phillips and Hansen 1990, together with the continuous

mapping theorem, we can show that

$$\{\mathbf{T}_{(k)}^\top(\tau_t) \otimes \mathbf{I}_d\} \left\{ \left(\boldsymbol{\Lambda}_{xx}^{(k)} \right)^{-1} \otimes \mathbf{I}_d \right\} \int_0^1 \left((\boldsymbol{\psi}_{(k)}(r) \otimes \mathbf{B}_X(r)) \otimes \mathbf{I}_d \right) d\mathbf{B}_r(\boldsymbol{\Omega}_{00})$$

has a mixed Normal distribution with stochastic covariance matrix $\mathbb{V}_{(k)}(\tau_t) := \left\{ \mathbf{T}_{(k)}^\top(\tau_t) \left(\boldsymbol{\Lambda}_{xx}^{(k)} \right)^{-1} \mathbf{T}_{(k)}(\tau_t) \right\} \otimes \boldsymbol{\Omega}_{00}$. Clearly, we have

$$\|\mathbb{V}_{(k)}(\tau)\|_{sp} \leq \|\boldsymbol{\Omega}_{00}\|_{sp} \left\| \left(\boldsymbol{\Lambda}_{xx}^{(k)} \right)^{-1} \right\|_{sp} \|\boldsymbol{\psi}_{(k)}(r)\|^2 = O_p(k).$$

As $\widehat{\boldsymbol{\Omega}}_{n,00} \xrightarrow{p} \boldsymbol{\Omega}_{00}$ implies $\widehat{\mathbb{V}}_{n(k)}(\tau_t) = \mathbb{V}_{(k)}(\tau_t) + o_p(1)$, by Slutsky's theorem and continuous mapping theorem, we obtain

$$n \left(\widehat{\mathbb{V}}_{n(k)}(\tau_t) \right)^{-1/2} \text{vec} \left(\widehat{\mathbf{A}}_n^{(k)}(\tau_t) - \mathbf{A}(\tau_t) + \boldsymbol{\Gamma}(\tau_t) \right) \xrightarrow{d} \mathcal{N}(\mathbf{0}_{dK_1 \times 1}, \mathbf{I}_{dK_1}).$$

A.3. Proof of Theorem 2

Define

$$\begin{aligned} \widehat{\mathbf{B}}_X &= \left(\sum_{t=1}^n \mathbf{y}_t \mathbf{X}_t^{(k)\top} \right) \left(\sum_{t=1}^n \mathbf{X}_t^{(k)} \mathbf{X}_t^{(k)\top} \right)^{-1}, \\ \widehat{\boldsymbol{\Gamma}}_Z &= \left(\sum_{t=1}^n \mathbf{Z}_t \mathbf{X}_t^{(k)\top} \right) \left(\sum_{t=1}^n \mathbf{X}_t^{(k)} \mathbf{X}_t^{(k)\top} \right)^{-1}. \end{aligned}$$

Then,

$$\widetilde{\mathbf{y}}_t = \mathbf{y}_t - \widehat{\mathbf{B}}_X \mathbf{X}_t^{(k)}, \quad \widetilde{\mathbf{Z}}_t = \mathbf{Z}_t - \widehat{\boldsymbol{\Gamma}}_Z \mathbf{X}_t^{(k)},$$

are just residuals from the regression $\mathbf{y}_t$ on $\mathbf{X}_t^{(k)}$ and $\mathbf{Z}_t$ on $\mathbf{X}_t^{(k)}$, respectively. In view of the Frisch–Waugh–Lovell theorem, $\boldsymbol{\Gamma}$ can be estimated from the regression

$$\widetilde{\mathbf{y}}_t = \boldsymbol{\Gamma} \widetilde{\mathbf{Z}}_t + \widetilde{\mathbf{u}}_{0t}^{(k)},$$

where $\tilde{\mathbf{u}}_{0t}^{(k)} = \mathbf{u}_{0t} + (\mathbf{A}_t - \mathbf{A}_k(t/n)) \mathbf{X}_t + (\mathbf{B}^{(k)} - \widehat{\mathbf{B}}_X + \Gamma \widehat{\Gamma}_Z) \mathbf{X}_t^{(k)}$. We then have

$$\begin{aligned} \widehat{\Gamma}_n - \Gamma &= \left(\sum_{t=1}^n \tilde{\mathbf{u}}_{0t}^{(k)} \tilde{\mathbf{Z}}_t^\top \right) \left(\sum_{t=1}^n \tilde{\mathbf{Z}}_t \tilde{\mathbf{Z}}_t^\top \right)^{-1} \\ &= \left(\sum_{t=1}^n \mathbf{u}_{0t} \tilde{\mathbf{Z}}_t^\top \right) \left(\sum_{t=1}^n \tilde{\mathbf{Z}}_t \tilde{\mathbf{Z}}_t^\top \right)^{-1} + \left(\sum_{t=1}^n (\mathbf{A}_t - \mathbf{A}_k(t/n)) \mathbf{X}_t \tilde{\mathbf{Z}}_t^\top \right) \left(\sum_{t=1}^n \tilde{\mathbf{Z}}_t \tilde{\mathbf{Z}}_t^\top \right)^{-1}, \end{aligned}$$

since, by construction, $\sum_{t=1}^n \mathbf{X}_t^{(k)} \tilde{\mathbf{Z}}_t^\top = \mathbf{0}_{K_1 \times K_2}$.

First, we note that

$$\frac{1}{n} \sum_{t=1}^n \tilde{\mathbf{Z}}_t \tilde{\mathbf{Z}}_t^\top = \frac{1}{n} \sum_{t=1}^n \mathbf{Z}_t \mathbf{Z}_t^\top - \left(\frac{1}{n} \sum_{t=1}^n \mathbf{Z}_t \mathbf{X}_t^{(k)\top} \right) \left(\frac{1}{n} \sum_{t=1}^n \mathbf{X}_t^{(k)} \mathbf{X}_t^{(k)\top} \right)^{-1} \left(\frac{1}{n} \sum_{t=1}^n \mathbf{X}_t^{(k)} \mathbf{Z}_t^\top \right).$$

By Lemma C.1, we have $\left\| \left(\frac{1}{n} \sum_{t=1}^n \mathbf{X}_t^{(k)} \mathbf{X}_t^{(k)\top} \right)^{-1} \right\|_{sp} = O_p(1/n)$. In addition, Lemma C.4(i) leads to $\left\| \frac{1}{n} \sum_{t=1}^n \mathbf{X}_t^{(k)} \mathbf{Z}_t^\top \right\|_{sp} = O_p(1)$. This implies that

$$\begin{aligned} \left\| \frac{1}{n} \sum_{t=1}^n \tilde{\mathbf{Z}}_t \tilde{\mathbf{Z}}_t^\top - \frac{1}{n} \sum_{t=1}^n \mathbf{Z}_t \mathbf{Z}_t^\top \right\|_{sp} &\leq \left\| \left(\frac{1}{n} \sum_{t=1}^n \mathbf{Z}_t \mathbf{X}_t^{(k)\top} \right) \left(\frac{1}{n} \sum_{t=1}^n \mathbf{X}_t^{(k)} \mathbf{X}_t^{(k)\top} \right)^{-1} \left(\frac{1}{n} \sum_{t=1}^n \mathbf{X}_t^{(k)} \mathbf{Z}_t^\top \right) \right\|_{sp} \\ &\leq \left\| \frac{1}{n} \sum_{t=1}^n \mathbf{Z}_t \mathbf{X}_t^{(k)\top} \right\|_{sp} \left\| \left(\frac{1}{n} \sum_{t=1}^n \mathbf{X}_t^{(k)} \mathbf{X}_t^{(k)\top} \right)^{-1} \right\|_{sp} \left\| \frac{1}{n} \sum_{t=1}^n \mathbf{X}_t^{(k)} \mathbf{Z}_t^\top \right\|_{sp} \\ &= O_p(1/n), \end{aligned}$$

so that

$$\frac{1}{n} \sum_{t=1}^n \tilde{\mathbf{Z}}_t \tilde{\mathbf{Z}}_t^\top = \frac{1}{n} \sum_{t=1}^n \mathbf{Z}_t \mathbf{Z}_t^\top + O_p(1/n). \quad (\text{A.1})$$

Similar arguments yield

$$\begin{aligned}
\frac{1}{\sqrt{n}} \sum_{t=1}^n \mathbf{u}_{0t} \tilde{\mathbf{Z}}_t^\top &= \frac{1}{\sqrt{n}} \sum_{t=1}^n \mathbf{u}_{0t} \mathbf{Z}_t^\top - \frac{1}{\sqrt{n}} \sum_{t=1}^n \mathbf{u}_{0t} \mathbf{X}_t^{\top(k)} \hat{\boldsymbol{\Gamma}}_Z^\top \\
&= \frac{1}{\sqrt{n}} \sum_{t=1}^n \mathbf{u}_{0t} \mathbf{Z}_t^\top - \left(\frac{1}{\sqrt{n}} \sum_{t=1}^n \mathbf{u}_{0t} \mathbf{X}_t^{\top(k)} \right) \left(\frac{1}{n} \sum_{t=1}^n \mathbf{X}_t^{(k)} \mathbf{X}_t^{(k)\top} \right)^{-1} \left(\frac{1}{n} \sum_{t=1}^n \mathbf{X}_t^{(k)} \mathbf{Z}_t^\top \right) \\
&= \frac{1}{\sqrt{n}} \sum_{t=1}^n \mathbf{u}_{0t} \mathbf{Z}_t^\top + O_p(1/\sqrt{n}).
\end{aligned} \tag{A.2}$$

We also have

$$\begin{aligned}
&\frac{1}{\sqrt{n}} \sum_{t=1}^n (\mathbf{A}_t - \mathbf{A}_k(t/n)) \mathbf{X}_t \tilde{\mathbf{Z}}_t^\top \\
&= \frac{1}{\sqrt{n}} \sum_{t=1}^n (\mathbf{A}_t - \mathbf{A}_k(t/n)) \mathbf{X}_t \mathbf{Z}_t^\top - \frac{1}{\sqrt{n}} \sum_{t=1}^n (\mathbf{A}_t - \mathbf{A}_k(t/n)) \mathbf{X}_t \mathbf{X}_t^{(k)\top} \hat{\boldsymbol{\Gamma}}_Z^\top \\
&= \frac{1}{\sqrt{n}} \sum_{t=1}^n (\mathbf{A}_t - \mathbf{A}_k(t/n)) \mathbf{X}_t \mathbf{Z}_t^\top - \left(\frac{1}{\sqrt{n}} \sum_{t=1}^n (\mathbf{A}_t - \mathbf{A}_k(t/n)) \mathbf{X}_t \mathbf{X}_t^{(k)\top} \right) \left(\sum_{t=1}^n \mathbf{X}_t^{(k)} \mathbf{X}_t^{(k)\top} \right)^{-1} \left(\sum_{t=1}^n \mathbf{X}_t^{(k)} \mathbf{Z}_t^\top \right) \\
&= O_p\left(\frac{n}{k^q}\right) + O_p\left(\frac{n^{3/2}}{k^q}\right) O_p\left(\frac{1}{n}\right) = O_p\left(\frac{n}{k^q}\right),
\end{aligned} \tag{A.3}$$

where the third equality follows from Lemma C.4(ii). By combining (A.1), (A.2), and (A.3), we obtain

$$\sqrt{n} \cdot (\hat{\boldsymbol{\Gamma}}_n - \boldsymbol{\Gamma}) = \left(\frac{1}{\sqrt{n}} \sum_{t=1}^n \mathbf{u}_{0t} \mathbf{Z}_t^\top \right) \left(\frac{1}{n} \sum_{t=1}^n \mathbf{Z}_t \mathbf{Z}_t^\top \right)^{-1} + o_p(1).$$

Using standard WLLN and C.L.T. for stationary linear processes (see, for example, Lemma 8.4 in Phillips 1995), we may show that

$$\frac{1}{n} \sum_{t=1}^n \mathbf{Z}_t \mathbf{Z}_t^\top \xrightarrow{p} \boldsymbol{\Sigma}_{zz},$$

and

$$\frac{1}{\sqrt{n}} \sum_{t=1}^n \text{vec}(\mathbf{u}_{0t} \mathbf{Z}_t^\top) \xrightarrow{d} \mathcal{N}(\mathbf{0}_{dK_2 \times 1}, \boldsymbol{\Omega}^{(0z)}),$$

where $\boldsymbol{\Omega}^{(0z)} = \sum_{j=-\infty}^{\infty} \mathbb{E}(\mathbf{u}_{0t} \mathbf{u}_{0t+j}^\top \otimes \mathbf{Z}_t \mathbf{Z}_{t+j}^\top)$. Theorem 2 now follows immediately from Slutsky's theorem.

A.4. Proof of Theorem 3

Similarly as in the proof of Theorem 1, define

$$\begin{aligned}\widehat{\mathbf{B}}_{YZ} &= \left(\sum_{t=1}^n \mathbf{y}_t \mathbf{Z}_t^\top \right) \left(\sum_{t=1}^n \mathbf{Z}_t \mathbf{Z}_t^\top \right)^{-1}, \\ \widehat{\mathbf{\Gamma}}_{XZ} &= \left(\sum_{t=1}^n \mathbf{X}_t^{(k)} \mathbf{Z}_t^\top \right) \left(\sum_{t=1}^n \mathbf{Z}_t \mathbf{Z}_t^\top \right)^{-1}.\end{aligned}$$

Then,

$$\check{\mathbf{y}}_t = \mathbf{y}_t - \widehat{\mathbf{B}}_{YZ} \mathbf{Z}_t, \quad \check{\mathbf{X}}_t^{(k)} = \mathbf{X}_t^{(k)} - \widehat{\mathbf{\Gamma}}_{XZ} \mathbf{Z}_t,$$

are just residuals from the regression $\mathbf{y}_t$ on $\mathbf{Z}_t$ and $\mathbf{X}_t^{(k)}$ on $\mathbf{Z}_t$, respectively. In view of the Frisch–Waugh–Lovell theorem, $\mathbf{B}^{(k)}$ can be estimated from the regression

$$\check{\mathbf{y}}_t = \mathbf{B}^{(k)} \check{\mathbf{X}}_t^{(k)} + \check{\mathbf{u}}_{0t}^{(k)},$$

where $\check{\mathbf{u}}_{0t}^{(k)} = \mathbf{u}_{0t} + (\mathbf{A}_t - \mathbf{A}_k(t/n)) \mathbf{X}_t + (\mathbf{B}^{(k)} \widehat{\mathbf{\Gamma}}_{XZ} - \widehat{\mathbf{B}}_{YZ} + \mathbf{\Gamma}) \mathbf{Z}_t$. Then, we have

$$\begin{aligned}\widehat{\mathbf{B}}_n^{(k)} - \mathbf{B}^{(k)} &= \left(\sum_{t=1}^n \check{\mathbf{u}}_{0t}^{(k)} \check{\mathbf{X}}_t^{(k)\top} \right) \left(\sum_{t=1}^n \check{\mathbf{X}}_t^{(k)} \check{\mathbf{X}}_t^{(k)\top} \right)^{-1} \\ &= \left(\sum_{t=1}^n \mathbf{u}_{0t} \check{\mathbf{X}}_t^{(k)\top} \right) \left(\sum_{t=1}^n \check{\mathbf{X}}_t^{(k)} \check{\mathbf{X}}_t^{(k)\top} \right)^{-1} + \left(\sum_{t=1}^n (\mathbf{A}_t - \mathbf{A}_k(t/n)) \mathbf{X}_t \check{\mathbf{X}}_t^{(k)\top} \right) \left(\sum_{t=1}^n \check{\mathbf{X}}_t^{(k)} \check{\mathbf{X}}_t^{(k)\top} \right)^{-1},\end{aligned}\tag{A.4}$$

since, by construction, $\sum_{t=1}^n \check{\mathbf{X}}_t^{(k)} \mathbf{Z}_t^\top = \mathbf{0}_{K_1 \times K_2}$.

Proceeding analogously as in the proof of Theorem 2, we have

$$\begin{aligned}\frac{1}{n^2} \sum_{t=1}^n \check{\mathbf{X}}_t^{(k)} \check{\mathbf{X}}_t^{(k)\top} &= \frac{1}{n^2} \sum_{t=1}^n \mathbf{X}_t^{(k)} \mathbf{X}_t^{(k)\top} + O_p\left(\frac{1}{n}\right) \\ &= \int_0^1 \{ \boldsymbol{\psi}_{(k)}(r) \boldsymbol{\psi}_{(k)}^\top(r) \otimes (\mathbf{B}_X(r) \mathbf{B}_X^\top(r)) \} dr + o_p(1).\end{aligned}\tag{A.5}$$

Second, observe that

$$\frac{1}{n} \sum_{t=1}^n \mathbf{u}_{0t} \tilde{\mathbf{X}}_t^{(k)\top} = \frac{1}{n} \sum_{t=1}^n \mathbf{u}_{0t} \mathbf{X}_t^{(k)\top} - \left(\frac{1}{n} \sum_{t=1}^n \mathbf{u}_{0t} \mathbf{Z}_t^\top \right) \left(\frac{1}{n} \sum_{t=1}^n \mathbf{Z}_t \mathbf{Z}_t^\top \right)^{-1} \left(\frac{1}{n} \sum_{t=1}^n \mathbf{Z}_t \mathbf{X}_t^{(k)\top} \right).$$

Standard WLLN for stationary linear processes gives

$$\frac{1}{n} \sum_{t=1}^n \mathbf{u}_{0t} \mathbf{Z}_t^\top \xrightarrow{p} \Sigma_{0z}, \quad \left(\frac{1}{n} \sum_{t=1}^n \mathbf{Z}_t \mathbf{Z}_t^\top \right)^{-1} \xrightarrow{p} \Sigma_{zz}^{-1}.$$

Recall that, in Lemma C.2 and Lemma C.4(i), we have shown

$$\begin{aligned} \frac{1}{n} \sum_{t=1}^n \mathbf{u}_{0t} \mathbf{X}_t^{(k)\top} &= \int_0^1 d\mathbf{B}_0(r) (\boldsymbol{\psi}_{(k)}(r) \otimes \mathbf{B}_X(r))^\top + \boldsymbol{\ell}_{1(k)}^\top \otimes (\boldsymbol{\Lambda}_{0x} + \boldsymbol{\Sigma}_{0x}) + o_p(1), \\ \frac{1}{n} \sum_{t=1}^n \mathbf{Z}_t \mathbf{X}_t^{(k)\top} &= \int_0^1 d\mathbf{B}_Z(r) (\boldsymbol{\psi}_{(k)}(r) \otimes \mathbf{B}_X(r))^\top + \boldsymbol{\ell}_{1(k)}^\top \otimes (\boldsymbol{\Lambda}_{zx} + \boldsymbol{\Sigma}_{zx}) + o_p(1). \end{aligned}$$

By combining results above, we obtain

$$\begin{aligned} \frac{1}{n} \sum_{t=1}^n \mathbf{u}_{0t} \tilde{\mathbf{X}}_t^{(k)\top} &= \int_0^1 (d\mathbf{B}_0(r) - \Sigma_{0z} \Sigma_{zz}^{-1} d\mathbf{B}_X(r)) (\boldsymbol{\psi}_{(k)}(r) \otimes \mathbf{B}_X(r))^\top \\ &\quad + \boldsymbol{\ell}_{1(k)}^\top \otimes \{ \boldsymbol{\Lambda}_{0x} + \boldsymbol{\Sigma}_{0x} - \Sigma_{0z} \Sigma_{zz}^{-1} (\boldsymbol{\Lambda}_{zx} + \boldsymbol{\Sigma}_{zx}) \} + o_p(1). \end{aligned} \tag{A.6}$$

Finally,

$$\begin{aligned} &\frac{1}{n} \sum_{t=1}^n (\mathbf{A}_t - \mathbf{A}_k(t/n)) \mathbf{X}_t \tilde{\mathbf{X}}_t^{(k)\top} \\ &= \frac{1}{n} \sum_{t=1}^n (\mathbf{A}_t - \mathbf{A}_k(t/n)) \mathbf{X}_t \mathbf{X}_t^{(k)\top} - \left(\frac{1}{n} \sum_{t=1}^n (\mathbf{A}_t - \mathbf{A}_k(t/n)) \mathbf{X}_t \mathbf{Z}_t^\top \right) \left(\frac{1}{n} \sum_{t=1}^n \mathbf{Z}_t \mathbf{Z}_t^\top \right)^{-1} \left(\frac{1}{n} \sum_{t=1}^n \mathbf{Z}_t \mathbf{X}_t^{(k)\top} \right) \\ &= O_p\left(\frac{n}{k^q}\right) + O_p\left(\frac{\sqrt{n}}{k^q}\right) = O_p\left(\frac{n}{k^q}\right) = o_p(1), \end{aligned} \tag{A.7}$$

where the second equality again follows from Lemma C.4(ii). What remains is to substitute (A.5)–(A.7) into (A.4). The conclusion then follows by applying the same arguments as those used in the proof of Theorem 1, so details are omitted.

A.5. Proof of Theorem 4

Let $\mathbf{Z}_t = (\Delta \mathbf{X}_{t-K}^\top, \dots, \Delta \mathbf{X}_{t+K}^\top)^\top$. The steps follow similarly as in the proofs of Theorem 3.

Estimation is based on the following regression

$$\check{\mathbf{y}}_t = \mathbf{B}^{(k)} \check{\mathbf{X}}_t^{(k)} + \check{\mathbf{u}}_{0t}^{(k)},$$

where $\check{\mathbf{u}}_{0t}^{(k)} = \mathbf{v}_t + \sum_{|j|>K} \mathbf{\Gamma}_j \mathbf{X}_{t-j} + (\mathbf{A}_t - \mathbf{A}_k(t/n)) \mathbf{X}_t + (\mathbf{B}^{(k)} \hat{\mathbf{\Gamma}}_{XZ} - \hat{\mathbf{B}}_{YZ} + \mathbf{\Gamma}) \mathbf{Z}_t$. Then, we have

$$\begin{aligned} \hat{\mathbf{B}}_n^{(k)} - \mathbf{B}^{(k)} &= \left(\sum_{t=K+1}^{n-2K} \check{\mathbf{u}}_{0t}^{(k)} \check{\mathbf{X}}_t^{(k)\top} \right) \left(\sum_{t=K+1}^{n-2K} \check{\mathbf{X}}_t^{(k)} \check{\mathbf{X}}_t^{(k)\top} \right)^{-1} \\ &= \left(\sum_{t=K+1}^{n-2K} \tilde{\mathbf{v}}_t \check{\mathbf{X}}_t^{(k)\top} \right) \left(\sum_{t=K+1}^{n-2K} \check{\mathbf{X}}_t^{(k)} \check{\mathbf{X}}_t^{(k)\top} \right)^{-1} \\ &\quad + \left(\sum_{t=K+1}^{n-2K} (\mathbf{A}_t - \mathbf{A}_k(t/n)) \mathbf{X}_t \check{\mathbf{X}}_t^{(k)\top} \right) \left(\sum_{t=K+1}^{n-2K} \check{\mathbf{X}}_t^{(k)} \check{\mathbf{X}}_t^{(k)\top} \right)^{-1}, \end{aligned} \quad (\text{A.8})$$

where $\tilde{\mathbf{v}}_t = \mathbf{v}_t + \sum_{|j|>K} \mathbf{\Gamma}_j \mathbf{X}_{t-j}$.

First, note that

$$\begin{aligned} &\frac{1}{(n-2K)^2} \sum_{t=K+1}^{n-2K} \check{\mathbf{X}}_t^{(k)} \check{\mathbf{X}}_t^{(k)\top} \\ &= \frac{1}{(n-2K)^2} \sum_{t=K+1}^{n-2K} \mathbf{X}_t^{(k)} \mathbf{X}_t^{(k)\top} \\ &\quad - \left(\frac{1}{(n-2K)^2} \sum_{t=K+1}^{n-2K} \mathbf{X}_t^{(k)} \mathbf{Z}_t^\top \right) \left(\frac{1}{n-2K} \sum_{t=1}^n \mathbf{Z}_t \mathbf{Z}_t^\top \right)^{-1} \left(\frac{1}{n-2K} \sum_{t=K+1}^{n-2K} \mathbf{Z}_t \mathbf{X}_t^{(k)\top} \right) \\ &= \frac{1}{(n-2K)^2} \sum_{t=K+1}^{n-2K} \mathbf{X}_t^{(k)} \mathbf{X}_t^{(k)\top} \\ &\quad - \left(\frac{1}{(n-2K)^2} \sum_{t=K+1}^{n-2K} \mathbf{X}_t^{(k)} \mathbf{Z}_t^\top \right) \left[\left(\frac{1}{n-2K} \sum_{t=1}^n \mathbf{Z}_t \mathbf{Z}_t^\top \right)^{-1} - \mathbf{\Sigma}_{zz}^{-1} \right] \left(\frac{1}{n-2K} \sum_{t=K+1}^{n-2K} \mathbf{Z}_t \mathbf{X}_t^{(k)\top} \right) \\ &\quad - \left(\frac{1}{(n-2K)^2} \sum_{t=K+1}^{n-2K} \mathbf{X}_t^{(k)} \mathbf{Z}_t^\top \right) \mathbf{\Sigma}_{zz}^{-1} \left(\frac{1}{n-2K} \sum_{t=K+1}^{n-2K} \mathbf{Z}_t \mathbf{X}_t^{(k)\top} \right) \\ &:= \frac{1}{(n-2K)^2} \sum_{t=K+1}^{n-2K} \mathbf{X}_t^{(k)} \mathbf{X}_t^{(k)\top} + \mathbf{R}_{xx,1} + \mathbf{R}_{xx,2}. \end{aligned}$$

By Lemma C.5, we have

$$\begin{aligned}\|\mathbf{R}_{xx,1}\| &\leq \left\| \frac{1}{(n-2K)^2} \sum_{t=K+1}^{n-2K} \mathbf{X}_t^{(k)} \mathbf{Z}_t^\top \right\| \left\| \left(\frac{1}{n-2K} \sum_{t=1}^n \mathbf{Z}_t \mathbf{Z}_t^\top \right)^{-1} - \Sigma_{zz}^{-1} \right\|_{sp} \left\| \frac{1}{n-2K} \sum_{t=K+1}^{n-2K} \mathbf{Z}_t \mathbf{X}_t^{(k)\top} \right\| \\ &= O_p \left(\frac{\sqrt{Kk}}{n} \right) O_p \left(\frac{K}{\sqrt{n}} \right) O_p \left(\sqrt{Kk} \right) = O_p \left(\frac{K^2 k}{n^{3/2}} \right),\end{aligned}$$

and

$$\begin{aligned}\|\mathbf{R}_{xx,2}\| &\leq \left\| \frac{1}{(n-2K)^2} \sum_{t=K+1}^{n-2K} \mathbf{X}_t^{(k)} \mathbf{Z}_t^\top \right\| \|\Sigma_{zz}^{-1}\|_{sp} \left\| \frac{1}{n-2K} \sum_{t=K+1}^{n-2K} \mathbf{Z}_t \mathbf{X}_t^{(k)\top} \right\| \\ &= O_p \left(\frac{\sqrt{Kk}}{n} \right) O_p(1) O_p \left(\sqrt{Kk} \right) = O_p \left(\frac{Kk}{n} \right).\end{aligned}$$

Given Assumption 3 and the condition $\frac{K^2}{n^{1-\gamma}} \rightarrow 0$, both terms above are just $o_p(1)$. This implies that

$$\begin{aligned}\frac{1}{(n-2K)^2} \sum_{t=K+1}^{n-2K} \check{\mathbf{X}}_t^{(k)} \check{\mathbf{X}}_t^{(k)\top} &= \frac{1}{(n-2K)^2} \sum_{t=K+1}^{n-2K} \mathbf{X}_t^{(k)} \mathbf{X}_t^{(k)\top} + o_p(1) \\ &= \frac{1}{n^2} \sum_{t=1}^n \mathbf{X}_t^{(k)} \mathbf{X}_t^{(k)\top} + o_p(1) = \mathbf{\Lambda}_{xx}^{(k)} + o_p(1),\end{aligned}$$

provided that $K = o(n)$, where $\mathbf{\Lambda}_{xx}^{(k)}$ is defined as in Theorem 1.

Second, note that

$$\begin{aligned}
& \frac{1}{n-2K} \sum_{t=K+1}^{n-K} \tilde{\mathbf{v}}_t \tilde{\mathbf{X}}_t^{(k)\top} \\
&= \frac{1}{n-2K} \sum_{t=K+1}^{n-K} \tilde{\mathbf{v}}_t \mathbf{X}_t^{(k)\top} - \left(\frac{1}{n-2K} \sum_{t=K+1}^{n-2K} \tilde{\mathbf{v}}_t \mathbf{Z}_t^\top \right) \left(\frac{1}{n-2K} \sum_{t=1}^n \mathbf{Z}_t \mathbf{Z}_t^\top \right)^{-1} \left(\frac{1}{n-2K} \sum_{t=K+1}^{n-2K} \mathbf{Z}_t \mathbf{X}_t^{(k)\top} \right) \\
&= \frac{1}{n-2K} \sum_{t=K+1}^{n-K} \tilde{\mathbf{v}}_t \mathbf{X}_t^{(k)\top} \\
&\quad - \left(\frac{1}{n-2K} \sum_{t=K+1}^{n-2K} \tilde{\mathbf{v}}_t \mathbf{Z}_t^\top \right) \left[\left(\frac{1}{n-2K} \sum_{t=1}^n \mathbf{Z}_t \mathbf{Z}_t^\top \right)^{-1} - \Sigma_{zz}^{-1} \right] \left(\frac{1}{n-2K} \sum_{t=K+1}^{n-2K} \mathbf{Z}_t \mathbf{X}_t^{(k)\top} \right) \\
&\quad - \left(\frac{1}{n-2K} \sum_{t=K+1}^{n-2K} \tilde{\mathbf{v}}_t \mathbf{Z}_t^\top \right) \Sigma_{zz}^{-1} \left(\frac{1}{n-2K} \sum_{t=K+1}^{n-2K} \mathbf{Z}_t \mathbf{X}_t^{(k)\top} \right) \\
&:= \frac{1}{n-2K} \sum_{t=K+1}^{n-K} \tilde{\mathbf{v}}_t \mathbf{X}_t^{(k)\top} + \mathbf{R}_{vx,1} + \mathbf{R}_{vx,2}.
\end{aligned}$$

It follows again from Lemma C.5 that

$$\begin{aligned}
\|\mathbf{R}_{vx,1}\| &\leq \left\| \frac{1}{n-2K} \sum_{t=K+1}^{n-2K} \tilde{\mathbf{v}}_t \mathbf{Z}_t^\top \right\| \left\| \left(\frac{1}{n-2K} \sum_{t=1}^n \mathbf{Z}_t \mathbf{Z}_t^\top \right)^{-1} - \Sigma_{zz}^{-1} \right\|_{sp} \left\| \frac{1}{n-2K} \sum_{t=K+1}^{n-2K} \mathbf{Z}_t \mathbf{X}_t^{(k)\top} \right\| \\
&= O_p \left(\sqrt{\frac{K}{n}} \right) O_p \left(\frac{K}{\sqrt{n}} \right) O_p \left(\sqrt{Kk} \right) = O_p \left(\frac{K^2 \sqrt{k}}{n} \right),
\end{aligned}$$

and

$$\begin{aligned}
\|\mathbf{R}_{vx,2}\| &\leq \left\| \frac{1}{n-2K} \sum_{t=K+1}^{n-2K} \tilde{\mathbf{v}}_t \mathbf{Z}_t^\top \right\| \left\| \Sigma_{zz}^{-1} \right\|_{sp} \left\| \frac{1}{n-2K} \sum_{t=K+1}^{n-2K} \mathbf{Z}_t \mathbf{X}_t^{(k)\top} \right\| \\
&= O_p \left(\sqrt{\frac{K}{n}} \right) O_p(1) O_p \left(\sqrt{Kk} \right) = O_p \left(\frac{K \sqrt{k}}{\sqrt{n}} \right).
\end{aligned}$$

Again, given Assumption 3 and the condition $\frac{K^2}{n^{1-\gamma}} \rightarrow 0$, both terms above are just $o_p(1)$. This

implies that

$$\begin{aligned} \frac{1}{n-2K} \sum_{t=K+1}^{n-K} \tilde{\mathbf{v}}_t \tilde{\mathbf{X}}_t^{(k)\top} &= \frac{1}{n-2K} \sum_{t=K+1}^{n-K} \tilde{\mathbf{v}}_t \mathbf{X}_t^{(k)\top} + o_p(1) \\ &= \frac{1}{n} \sum_{t=1}^n \mathbf{v}_t \mathbf{X}_t^{(k)\top} + o_p(1). \end{aligned}$$

Now, substituting all results back to (A.8) yields

$$n \cdot \left(\widehat{\mathbf{B}}_n^{(k)} - \mathbf{B}^{(k)} \right) = \left(\frac{1}{n} \sum_{t=1}^n \mathbf{v}_t \mathbf{X}_t^{(k)\top} \right) \left(\frac{1}{n^2} \sum_{t=1}^n \mathbf{X}_t^{(k)} \mathbf{X}_t^{(k)\top} \right)^{-1} + o_p(1).$$

Let $\tilde{\mathbf{\Gamma}}_{n,ux}^{(k)} := \frac{1}{n} \sum_{t=1}^n \mathbf{v}_t \mathbf{X}_t^{(k)\top}$ and

$$\tilde{\mathbb{V}}_{n(k)}(\tau_t) = \left\{ \mathbf{T}_{(k)}^\top(\tau_t) \left(\Lambda_{n,xx}^{(k)} \right)^{-1} \mathbf{T}_{(k)}(\tau_t) \right\} \otimes \widehat{\mathbf{\Omega}}_{n,0 \cdot x},$$

where $\widehat{\mathbf{\Omega}}_{n,0 \cdot x}$ is a consistent estimator of $\mathbf{\Omega}_{0 \cdot x} = \mathbf{\Omega}_{00} - \mathbf{\Omega}_{0x} \mathbf{\Omega}_{xx}^{-1} \mathbf{\Omega}_{x0}$. Then, we have

$$\begin{aligned} &n \left(\tilde{\mathbb{V}}_{n(k)}(\tau_t) \right)^{-1/2} \text{vec} \left(\widehat{\mathbf{A}}_n^{(k)}(\tau_t) - \mathbf{A}(\tau_t) + \mathbf{\Gamma}(\tau_t) \right) \\ &= \left(\left\{ \mathbf{T}_{(k)}^\top(u) \left(\Lambda_{n,xx}^{(k)} \right)^{-1} \mathbf{T}_{(k)}(u) \right\} \otimes \widehat{\mathbf{\Omega}}_{n,0 \cdot x} \right)^{-1/2} \left(\mathbf{T}_{(k)}^\top(\tau_t) \otimes \mathbf{I}_d \right) \text{vec} \left\{ \tilde{\mathbf{\Gamma}}_{n,ux}^{(k)} \left(\Lambda_{n,xx}^{(k)} \right)^{-1} \right\} + o_p(1). \end{aligned}$$

Following similar procedures as used in the proof of Corollary 1, we have the joint stochastic expansion

$$\begin{pmatrix} \Lambda_{n,xx}^{(k)} \\ \tilde{\mathbf{\Gamma}}_{n,ux}^{(k)} \end{pmatrix} = \begin{pmatrix} \Lambda_{xx}^{(k)} \\ \tilde{\mathbf{\Gamma}}_{ux}^{(k)} \end{pmatrix} + o_p(1),$$

where

$$\tilde{\mathbf{\Gamma}}_{ux}^{(k)} := \int_0^1 d\mathbf{B}_{0 \cdot x}(r) \left(\left(\boldsymbol{\psi}_{(k)}(r) \otimes \mathbf{B}_r(\mathbf{\Omega}_{xx}) \right)^\top \right)^\top,$$

where $\mathbf{B}_{0 \cdot x}(r) := \mathbf{B}_v(r) = \mathbf{B}_0(r) - \mathbf{\Omega}_{0x} \mathbf{\Omega}_{xx}^{-1} \mathbf{B}_X(r)$. By construction, $\mathbf{B}_{0 \cdot x}(r)$ and $\mathbf{B}_X(r)$ are uncorrelated and hence independently distributed. Then, as $\widehat{\mathbf{\Omega}}_{n,0 \cdot c} \xrightarrow{p} \mathbf{\Omega}_{0 \cdot c}$ implies $\tilde{\mathbb{V}}_{n(k)}(\tau_t) = \mathbb{V}_{(k)}(\tau_t) + o_p(1)$, by Slutsky's theorem and continuous mapping theorem, we obtain

$$n \left(\tilde{\mathbb{V}}_{n(k)}(\tau_t) \right)^{-1/2} \text{vec} \left(\widehat{\mathbf{A}}_n^{(k)}(\tau_t) - \mathbf{A}(\tau_t) + \mathbf{\Gamma}(\tau_t) \right) \xrightarrow{d} \mathcal{N}(\mathbf{0}_{dK_1 \times 1}, \mathbf{I}_{dK_1}).$$

B — Justification of the specification (24)

First, we show that the autocovariance matrix of $\mathbf{u}_t$: $\Gamma_u(k) := \sum_{j=0}^{\infty} \Phi_j \Sigma_\varepsilon \Phi_{j+k}^\top$, is absolute summable. Under Assumption 1, Σ_ε is positive definite and $\sum_{m=0}^{\infty} \|\Phi_m\| < \infty$, we have

$$\begin{aligned} \sum_{k=0}^{\infty} \|\Gamma_u(k)\| &\leq \sum_{k=0}^{\infty} \sum_{j=0}^{\infty} \|\Phi_j\| \|\Sigma_\varepsilon\| \|\Phi_{j+k}^\top\| \\ &= \|\Sigma_\varepsilon\| \sum_{j=0}^{\infty} \|\Phi_j\| \left(\sum_{k=0}^{\infty} \|\Phi_{j+k}^\top\| \right) \\ &\leq \|\Sigma_\varepsilon\| \left(\sum_{m=0}^{\infty} \|\Phi_m\| \right)^2 < \infty. \end{aligned}$$

Second, note that the spectral density matrix of $\mathbf{u}_t$ is given by $\mathbf{f}_{uu}(\lambda) = \frac{1}{2\pi} \Phi(e^{-i\lambda}) \Sigma_\varepsilon \Phi^\top(e^{i\lambda})$. Under Assumption 1, Σ_ε is positive definite and $\Phi(e^{-i\lambda})$ is nonsingular for all $\lambda \in [-\pi, \pi]$. Since $\mathbf{f}_{uu}(\lambda)$ is continuous in λ (implied by the absolute summability of $\Gamma_u(k)$, $\forall k$) and $[-\pi, \pi]$ is compact, we have $\inf_{\lambda \in [-\pi, \pi]} \lambda_{\min}(\mathbf{f}_{uu}(\lambda)) > 0$.

Given that $\sum_{k=0}^{\infty} \|\Gamma_u(k)\| < \infty$ and $\inf_{\lambda \in [-\pi, \pi]} \lambda_{\min}(\mathbf{f}_{uu}(\lambda)) > 0$, the conditions in Theorem 8.3.1 in Brillinger 1975 are satisfied. We may write

$$\mathbf{u}_{0t} = \sum_{j=-\infty}^{\infty} \Gamma_j \mathbf{u}_{xt} + \mathbf{v}_t, \quad (\text{B.1})$$

where $\Gamma_j = \frac{1}{2\pi} \int_{-\pi}^{\pi} \Gamma(\lambda) e^{ij\lambda} d\lambda$ satisfies $\sum_{j=-\infty}^{\infty} \|\Gamma_j\| < \infty$. $\Gamma(\lambda) = \mathbf{f}_{u_0 u_x}(\lambda) \mathbf{f}_{u_x u_x}^{-1}(\lambda)$ is the complex regression coefficient of $\mathbf{u}_{0t}$ on $\mathbf{u}_{xt}$ at frequency λ . The error series $\mathbf{v}_t$ has spectral density matrix

$$\mathbf{f}_{vv}(\lambda) = \mathbf{f}_{u_0 u_0}(\lambda) - \mathbf{f}_{u_0 u_x}(\lambda) \mathbf{f}_{u_x u_x}^{-1}(\lambda) \mathbf{f}_{u_x u_0}(\lambda).$$

This implies that (24) is achieved by truncating at $|j| = K$ so that

$$\mathbf{y}_t = \mathbf{B}^{(k)} \mathbf{X}_t^{(k)} + \sum_{j=-K}^K \Gamma_j \mathbf{X}_{t-j} + \tilde{\mathbf{v}}_t^{(k)},$$

where $\tilde{\mathbf{v}}_t^{(k)} = \mathbf{v}_t + \sum_{|j|>K} \Gamma_j \mathbf{X}_{t-j} + (\mathbf{A}_t - \mathbf{A}_k(\tau_t)) \mathbf{X}_t$.

Finally, since the cross spectral density matrix of $\mathbf{u}_{xt}$ and $\mathbf{v}_t$ is given by

$$\mathbf{f}_{vu_x}(\lambda) = \mathbf{f}_{u_0u_0}(\lambda) - \mathbf{\Gamma}(\lambda)\mathbf{f}_{u_xu_x}(\lambda) = \mathbf{0}_{d_1 \times K_1}, \quad \forall \lambda,$$

we have

$$\mathbb{E} [\mathbf{v}_t \mathbf{u}_{x,t-k}^\top] = \int_{-\pi}^{\pi} \mathbf{f}_{vu_x}(\lambda) e^{ik\lambda} d\lambda = \mathbf{0}_{d_1 \times K_1}, \quad \forall k.$$

In addition, since $\sum_{j=-\infty}^{\infty} \|\mathbf{\Gamma}_j\| < \infty$ and $(\mathbf{u}_{xt})_t$ is weakly stationary, the process $\left(\sum_{j=-\infty}^{\infty} \mathbf{\Gamma}_j \mathbf{u}_{xt}\right)_t$ is also weakly stationary, so does $(\mathbf{v}_t)_t$.

C – Useful lemmas

Lemma C.1. Under Assumptions 1-3, we have, as $n \rightarrow \infty$

$$\frac{1}{n^2} \sum_{t=1}^n \mathbf{X}_t^{(k)} \mathbf{X}_t^{\top(k)} = \mathbf{\Lambda}_{xx}^{(k)} + o_p(1),$$

where

$$\mathbf{\Lambda}_{xx}^{(k)} := \int_0^1 \left\{ \boldsymbol{\psi}_{(k)}(r) \boldsymbol{\psi}_{(k)}^\top(r) \otimes (\mathbf{B}_X(r) \mathbf{B}_X^\top(r)) \right\} dr$$

satisfies $\left\| \left(\mathbf{\Lambda}_{xx}^{(k)} \right)^{-1} \right\|_{sp} = O_p(1)$.

Proof. Let $\mathbf{V}_{n, \lfloor nr \rfloor}^x = \mathbf{\Phi}_x(1) \frac{1}{\sqrt{n}} \sum_{s=1}^{\lfloor nr \rfloor} \boldsymbol{\varepsilon}_s$. Using the BN decomposition as in Phillips and Solo 1992, we have

$$\mathbf{X}_t - \mathbf{X}_{t-1} = \mathbf{u}_{xt} = \mathbf{\Phi}_x(1) \boldsymbol{\varepsilon}_t - \Delta \tilde{\mathbf{u}}_{xt},$$

where $\tilde{\mathbf{u}}_{xt} = \sum_{j=0}^{\infty} \tilde{\mathbf{\Phi}}_{xj} \boldsymbol{\varepsilon}_{t-j}$ with $\tilde{\mathbf{\Phi}}_{xj} = \sum_{k=j+1}^{\infty} \mathbf{\Phi}_{xk}$. Then,

$$\mathbf{X}_t = \mathbf{\Phi}_x(1) \sum_{s=1}^t \boldsymbol{\varepsilon}_s + \tilde{\mathbf{u}}_{x0} - \tilde{\mathbf{u}}_{xt} + \mathbf{X}_0 = \sqrt{n} \mathbf{V}_{n,t}^x + \tilde{\mathbf{u}}_{x0} - \tilde{\mathbf{u}}_{xt} + \mathbf{X}_0.$$

We have

$$\begin{aligned}
& \frac{1}{n^2} \sum_{t=1}^n \mathbf{X}_t^{(k)} \mathbf{X}_t^{(k)\top} \\
&= \frac{1}{n^2} \sum_{t=1}^n (\boldsymbol{\psi}_{(k)}(t/n) \otimes \mathbf{X}_t) (\boldsymbol{\psi}_{(k)}(t/n) \otimes \mathbf{X}_t)^\top \\
&= \frac{1}{n^2} \sum_{t=1}^n (\boldsymbol{\psi}_{(k)}(t/n) \boldsymbol{\psi}_{(k)}^\top(t/n) \otimes \mathbf{X}_t \mathbf{X}_t^\top) \\
&= \frac{1}{n^2} \sum_{t=1}^n \left(\boldsymbol{\psi}_{(k)}(t/n) \boldsymbol{\psi}_{(k)}^\top(t/n) \otimes (\sqrt{n} \mathbf{V}_{n,t}^x) (\sqrt{n} \mathbf{V}_{n,t}^x)^\top \right) \\
&\quad + \frac{1}{n^2} \sum_{t=1}^n \left(\boldsymbol{\psi}_{(k)}(t/n) \boldsymbol{\psi}_{(k)}^\top(t/n) \otimes (\sqrt{n} \mathbf{V}_{n,t}^x) (\tilde{\mathbf{u}}_{x0} - \tilde{\mathbf{u}}_{xt} + \mathbf{X}_0)^\top \right) \\
&\quad + \frac{1}{n^2} \sum_{t=1}^n \left(\boldsymbol{\psi}_{(k)}^\top(t/n) \boldsymbol{\psi}_{(k)}^\top(t/n) \otimes (\tilde{\mathbf{u}}_{x0} - \tilde{\mathbf{u}}_{xt} + \mathbf{X}_0) (\sqrt{n} \mathbf{V}_{n,t}^x)^\top \right) \\
&\quad + \frac{1}{n^2} \sum_{t=1}^n \left(\boldsymbol{\psi}_{(k)}(t/n) \boldsymbol{\psi}_{(k)}^\top(t/n) \otimes (\tilde{\mathbf{u}}_{x0} - \tilde{\mathbf{u}}_{xt} + \mathbf{X}_0) (\tilde{\mathbf{u}}_{x0} - \tilde{\mathbf{u}}_{xt} + \mathbf{X}_0)^\top \right) \\
&:= \boldsymbol{\Lambda}_n(1) + \boldsymbol{\Lambda}_n(2) + \boldsymbol{\Lambda}_n(3) + \boldsymbol{\Lambda}_n(4),
\end{aligned}$$

so that

$$\left\| \frac{1}{n^2} \sum_{t=1}^n \mathbf{X}_t^{(k)} \mathbf{X}_t^{(k)\top} - \boldsymbol{\Lambda}_n(1) \right\| \leq \|\boldsymbol{\Lambda}_n(2)\| + \|\boldsymbol{\Lambda}_n(3)\| + \|\boldsymbol{\Lambda}_n(4)\|.$$

We first show that $\|\boldsymbol{\Lambda}_n(2)\| = O_p\left(\frac{k}{n^a}\right)$, $\|\boldsymbol{\Lambda}_n(3)\| = O_p\left(\frac{k}{n^a}\right)$, and $\|\boldsymbol{\Lambda}_n(4)\| = O_p\left(\frac{k}{n^{2a}}\right)$, where $a = \frac{p-2}{2p}$. This would imply that

$$\frac{1}{n^2} \sum_{t=1}^n \mathbf{X}_t^{(k)} \mathbf{X}_t^{(k)\top} = \boldsymbol{\Lambda}_n(1) + O_p\left(\frac{k}{n^a}\right) = \boldsymbol{\Lambda}_n(1) + o_p(1). \tag{C.1}$$

Let us start with $\|\Lambda_n(2)\|$. Notice that

$$\begin{aligned}
\|\Lambda_n(2)\| &\leq \frac{1}{n^2} \sum_{t=1}^n \left\| \psi_{(k)}(t/n) \psi_{(k)}^\top(t/n) \right\| \left\| (\sqrt{n} \mathbf{V}_{n,t}^x) (\tilde{\mathbf{u}}_{x0} - \tilde{\mathbf{u}}_{xt} + \mathbf{X}_0)^\top \right\| \\
&\leq \frac{1}{n^2} \sum_{t=1}^n \left\| \psi_{(k)}(t/n) \psi_{(k)}^\top(t/n) \right\| \sqrt{n} \left\| \mathbf{V}_{n,t}^x \right\| \left\| \tilde{\mathbf{u}}_{x0} - \tilde{\mathbf{u}}_{xt} + \mathbf{X}_0 \right\| \\
&\leq \max_{1 \leq t \leq n} \left\| \frac{1}{\sqrt{n}} (\tilde{\mathbf{u}}_{x0} - \tilde{\mathbf{u}}_{xt} + \mathbf{X}_0) \right\| \cdot \frac{1}{n^2} \sum_{t=1}^n \left\| \psi_{(k)}(t/n) \psi_{(k)}^\top(t/n) \right\| \cdot n \left\| \mathbf{V}_{n,t}^x \right\|.
\end{aligned}$$

By standard Riemann-sum approximation for the integral, we have

$$\frac{1}{n} \sum_{t=1}^n \left\| \psi_{(k)}(t/n) \psi_{(k)}^\top(t/n) \right\| = \int_0^1 \left\| \psi_{(k)}(r) \psi_{(k)}^\top(r) \right\| dr + O(1/n) = O(1)k.$$

In addition, by Lemma C.6, we have

$$\max_{1 \leq t \leq n} \left\| \frac{1}{\sqrt{n}} (\tilde{\mathbf{u}}_{x0} - \tilde{\mathbf{u}}_{xt} + \mathbf{X}_0) \right\| = O_p \left(\frac{1}{n^a} \right),$$

for $a = \frac{p-2}{2p}$. Substituting all these results above yields

$$\|\Lambda_n(2)\| = O_p \left(\frac{k}{n^a} \right).$$

Similar arguments lead to $\|\Lambda_n(3)\| = O_p \left(\frac{k}{n^a} \right)$ and $\|\Lambda_n(4)\| = O_p \left(\frac{k}{n^{2a}} \right)$.

We now move on to $\Lambda_n(1)$. By construction, $\mathbf{V}_{n,0} = \mathbf{0}$. We have

$$\begin{aligned}
&\frac{1}{n^2} \sum_{t=1}^n \left(\psi_{(k)}(t/n) \psi_{(k)}^\top(t/n) \otimes (\sqrt{n} \mathbf{V}_{n,t}^x) (\sqrt{n} \mathbf{V}_{n,t}^x)^\top \right) \\
&= \int_0^1 \left(\psi_{(k)}(\lfloor nr \rfloor / n) \psi_{(k)}^\top(\lfloor nr \rfloor / n) \otimes \mathbf{V}_{n, \lfloor nr \rfloor}^x \mathbf{V}_{n, \lfloor nr \rfloor}^{x\top} \right) dr + \frac{1}{n} \left(\psi_{(k)}(1) \psi_{(k)}^\top(1) \otimes \mathbf{V}_{n,n}^x \mathbf{V}_{n,n}^{x\top} \right) \\
&= \int_0^1 \left(\psi_{(k)}(\lfloor nr \rfloor / n) \psi_{(k)}^\top(\lfloor nr \rfloor / n) \otimes \mathbf{V}_{n, \lfloor nr \rfloor}^x \mathbf{V}_{n, \lfloor nr \rfloor}^{x\top} \right) dr + O_p(k/n),
\end{aligned}$$

where $\frac{1}{n} \left(\psi_{(k)}(1) \psi_{(k)}^\top(1) \otimes \mathbf{V}_{n,n}^x \mathbf{V}_{n,n}^{x\top} \right) = O_p(k/n) = o_p(1)$, due to the fact that $\left\| \psi_{(k)}(1) \psi_{(k)}^\top(1) \right\| =$

$O(1)k$ and (4). Notice that

$$\begin{aligned}
& \int_0^1 (\boldsymbol{\psi}_{(k)}(\lfloor nr \rfloor / n) \boldsymbol{\psi}_{(k)}^\top(\lfloor nr \rfloor / n) - \boldsymbol{\psi}_{(k)}(r) \boldsymbol{\psi}_{(k)}^\top(r)) dr \\
&= \int_0^1 (\boldsymbol{\psi}_{(k)}(\lfloor nr \rfloor / n) - \boldsymbol{\psi}_{(k)}(r)) \boldsymbol{\psi}_{(k)}^\top(r) dr + \int_0^1 \boldsymbol{\psi}_{(k)}(r) (\boldsymbol{\psi}_{(k)}(\lfloor nr \rfloor / n) - \boldsymbol{\psi}_{(k)}(r))^\top dr \\
&\quad + \int_0^1 (\boldsymbol{\psi}_{(k)}(\lfloor nr \rfloor / n) - \boldsymbol{\psi}_{(k)}(r)) (\boldsymbol{\psi}_{(k)}(\lfloor nr \rfloor / n) - \boldsymbol{\psi}_{(k)}(r))^\top dr.
\end{aligned}$$

Since

$$\begin{aligned}
\|\boldsymbol{\psi}_{(k)}(\lfloor nr \rfloor / n) - \boldsymbol{\psi}_{(k)}(r)\|^2 &= 2 \sum_{j=1}^{k-1} \left(\cos \left(j\pi \frac{\lfloor nr \rfloor}{n} \right) - \cos(j\pi r) \right)^2 \\
&\leq O(1) \left| \frac{\lfloor nr \rfloor}{n} - r \right|^2 \left(\sum_{j=1}^{k-1} j^2 \right) \leq O(1) k^3 / n^2,
\end{aligned}$$

we have

$$\|\boldsymbol{\psi}_{(k)}(\lfloor nr \rfloor / n) - \boldsymbol{\psi}_{(k)}(r)\| = O\left(\frac{k^{3/2}}{n}\right) = o(1).$$

As by Assumption 3, $k = o(n^{2/3})$, this implies that

$$\int_0^1 (\boldsymbol{\psi}_{(k)}(\lfloor nr \rfloor / n) \boldsymbol{\psi}_{(k)}^\top(\lfloor nr \rfloor / n) - \boldsymbol{\psi}_{(k)}(r) \boldsymbol{\psi}_{(k)}^\top(r)) dr = o(1). \quad (\text{C.2})$$

Thus,

$$\begin{aligned}
\boldsymbol{\Lambda}_n(1) &= \int_0^1 (\boldsymbol{\psi}_{(k)}(\lfloor nr \rfloor / n) \boldsymbol{\psi}_{(k)}^\top(\lfloor nr \rfloor / n) - \boldsymbol{\psi}_{(k)}(r) \boldsymbol{\psi}_{(k)}^\top(r)) \otimes (\mathbf{V}_{n, \lfloor nr \rfloor}^x \mathbf{V}_{n, \lfloor nr \rfloor}^{x\top}) dr \\
&\quad + \int_0^1 (\boldsymbol{\psi}_{(k)}(r) \boldsymbol{\psi}_{(k)}^\top(r) \otimes \mathbf{V}_{n, \lfloor nr \rfloor}^x \mathbf{V}_{n, \lfloor nr \rfloor}^{x\top}) dr \\
&= \int_0^1 (\boldsymbol{\psi}_{(k)}(r) \boldsymbol{\psi}_{(k)}^\top(r) \otimes \mathbf{V}_{n, \lfloor nr \rfloor}^x \mathbf{V}_{n, \lfloor nr \rfloor}^{x\top}) dr + o_p(1).
\end{aligned} \quad (\text{C.3})$$

By combining (C.1) and (C.3), we conclude that

$$\frac{1}{n^2} \sum_{t=1}^n \mathbf{X}_t^{(k)} \mathbf{X}_t^{\top(k)} = \int_0^1 (\boldsymbol{\psi}_{(k)}(r) \boldsymbol{\psi}_{(k)}^\top(r) \otimes \mathbf{V}_{n, \lfloor nr \rfloor}^x \mathbf{V}_{n, \lfloor nr \rfloor}^{x\top}) dr + O_p\left(\frac{k}{n^a}\right).$$

Finally, we show that

$$\left\| \int_0^1 (\boldsymbol{\psi}_{(k)}(r) \boldsymbol{\psi}_{(k)}^\top(r) \otimes \mathbf{V}_{n, \lfloor nr \rfloor}^x \mathbf{V}_{n, \lfloor nr \rfloor}^{x\top}) dr - \int_0^1 (\boldsymbol{\psi}_{(k)}(r) \boldsymbol{\psi}_{(k)}^\top(r) \otimes \mathbf{B}_X(r) \mathbf{B}_X^\top(r)) dr \right\|_{sp} = o_p(1). \quad (\text{C.4})$$

For any unit vector $\mathbf{z} \in \mathbb{R}^{kK_1}$, where $\mathbf{z} = \text{vec}(\mathbf{Z})$, we note that

$$\begin{aligned} & \left| \mathbf{z}^\top \left(\int_0^1 \{ \boldsymbol{\psi}_{(k)}(r) \boldsymbol{\psi}_{(k)}^\top(r) \otimes (\mathbf{V}_{n, \lfloor nr \rfloor}^x \mathbf{V}_{n, \lfloor nr \rfloor}^{x\top} - \mathbf{B}_X(r) \mathbf{B}_X^\top(r)) \} dr \right) \mathbf{z} \right| \\ &= \left\| \int_0^1 (\mathbf{Z} \boldsymbol{\psi}_{(k)}(r))^\top (\mathbf{V}_{n, \lfloor nr \rfloor}^x \mathbf{V}_{n, \lfloor nr \rfloor}^{x\top} - \mathbf{B}_X(r) \mathbf{B}_X^\top(r)) (\mathbf{Z} \boldsymbol{\psi}_{(k)}(r)) dr \right\| \\ &\leq \sup_{r \in [0,1]} \|\mathbf{V}_{n, \lfloor nr \rfloor}^x \mathbf{V}_{n, \lfloor nr \rfloor}^{x\top} - \mathbf{B}_X(r) \mathbf{B}_X^\top(r)\| \int_0^1 \|\mathbf{Z} \boldsymbol{\psi}_{(k)}(r)\|^2 dr \\ &\leq \sup_{r \in [0,1]} \|\mathbf{V}_{n, \lfloor nr \rfloor}^x \mathbf{V}_{n, \lfloor nr \rfloor}^{x\top} - \mathbf{B}_X(r) \mathbf{B}_X^\top(r)\| \text{tr} \left(\mathbf{Z}^\top \mathbf{Z} \int_0^1 \boldsymbol{\psi}_{(k)}(r) \boldsymbol{\psi}_{(k)}^\top(r) dr \right) \\ &\leq \sup_{r \in [0,1]} \|\mathbf{V}_{n, \lfloor nr \rfloor}^x \mathbf{V}_{n, \lfloor nr \rfloor}^{x\top} - \mathbf{B}_X(r) \mathbf{B}_X^\top(r)\|, \end{aligned}$$

where the last inequality follows from the fact that

$$\text{tr} \left(\mathbf{Z}^\top \mathbf{Z} \int_0^1 \boldsymbol{\psi}_{(k)}(r) \boldsymbol{\psi}_{(k)}^\top(r) dr \right) = \text{tr}(\mathbf{Z}^\top \mathbf{Z}) = \|\mathbf{Z}\|^2 = 1.$$

In addition, by (4) and the fact that $\mathbf{B}_X(\cdot)$ has continuous sample paths *a.s.*, we have

$$\sup_{r \in [0,1]} \|\mathbf{V}_{n, \lfloor nr \rfloor}^x \mathbf{V}_{n, \lfloor nr \rfloor}^{x\top} - \mathbf{B}_X(r) \mathbf{B}_X^\top(r)\| = o_p(1).$$

We now show that $\left\| \left(\boldsymbol{\Lambda}_{xx}^{(k)} \right)^{-1} \right\|_{sp} = O_p(1)$. For any fixed vector $\mathbf{z} \neq \mathbf{0}_{kK_1 \times 1}$, we need to show that

$$\mathbf{z}^\top \left(\int_0^1 (\boldsymbol{\psi}_{(k)}(r) \boldsymbol{\psi}_{(k)}^\top(r)) \otimes (\mathbf{B}_X(r) \mathbf{B}_X^\top(r)) dr \right) \mathbf{z} > 0 \quad a.s.$$

Let $\text{vec}(\mathbf{Z}) = \mathbf{z}$. Using the fact that

$$(\boldsymbol{\psi}_{(k)}(r) \boldsymbol{\psi}_{(k)}^\top(r)) \otimes (\mathbf{B}_X(r) \mathbf{B}_X^\top(r)) = (\boldsymbol{\psi}_{(k)}(r) \otimes \mathbf{B}_X(r)) (\boldsymbol{\psi}_{(k)}(r) \otimes \mathbf{B}_X(r))^\top$$

and the identity

$$(\boldsymbol{\psi}_{(k)}(r) \otimes \mathbf{B}_X(r))^\top \text{vec}(\mathbf{Z}) = \text{vec}(\mathbf{B}_X^\top(r) \mathbf{Z} \boldsymbol{\psi}_{(k)}(r)),$$

we obtain

$$\begin{aligned} & \mathbf{z}^\top \left(\int_0^1 (\boldsymbol{\psi}_{(k)}(r) \boldsymbol{\psi}_{(k)}^\top(r)) \otimes (\mathbf{B}_X(r) \mathbf{B}_X^\top(r)) dr \right) \mathbf{z} \\ &= \int_0^1 \mathbf{z}^\top \left((\boldsymbol{\psi}_{(k)}(r) \otimes \mathbf{B}_X(r)) (\boldsymbol{\psi}_{(k)}(r) \otimes \mathbf{B}_X(r))^\top \right) \mathbf{z} dr \\ &= \int_0^1 \left((\boldsymbol{\psi}_{(k)}(r) \otimes \mathbf{B}_X(r))^\top \text{vec}(\mathbf{Z}) \right)^\top \left((\boldsymbol{\psi}_{(k)}(r) \otimes \mathbf{B}_X(r))^\top \text{vec}(\mathbf{Z}) \right) dr \\ &= \int_0^1 \left(\text{vec}(\mathbf{B}_X^\top(r) \mathbf{Z} \boldsymbol{\psi}_{(k)}(r)) \right)^\top \left(\text{vec}(\mathbf{B}_X^\top(r) \mathbf{Z} \boldsymbol{\psi}_{(k)}(r)) \right) dr \\ &= \int_0^1 (\mathbf{B}_X^\top(r) \mathbf{Z} \boldsymbol{\psi}_{(k)}(r))^2 dr \geq 0 \quad a.s. \end{aligned}$$

Note that $\mathbf{Z} \boldsymbol{\psi}_{(k)}(r)$ is $K_1 \times 1$, so $\mathbf{B}_X^\top(r) \mathbf{Z} \boldsymbol{\psi}_{(k)}(r)$ is a one-dimensional Brownian motion obtained as a linear combination of the components of $\mathbf{B}_X(r)$ whenever $\mathbf{Z} \boldsymbol{\psi}_{(k)}(r) \neq 0$. Suppose, for contradiction, that

$$\int_0^1 (\mathbf{B}_X^\top(r) \mathbf{Z} \boldsymbol{\psi}_{(k)}(r))^2 dr = 0$$

holds on an event of positive probability. Then, on that event, the continuous process

$$W(r) := \mathbf{B}_X^\top(r) \mathbf{Z} \boldsymbol{\psi}_{(k)}(r)$$

satisfies $W(r) = 0$, except on a set of Lebesgue measure zero on $[0, 1]$. Since $\boldsymbol{\Omega}_{xx}$ is full and the Brownian motion has continuous sample paths, this can only happen with positive probability if $\mathbf{Z} \boldsymbol{\psi}_{(k)}(r) = 0$ for almost every $r \in [0, 1]$. If there exists $r_0 \in [0, 1]$ such that $\mathbf{v} := \mathbf{Z} \boldsymbol{\psi}_{(k)}(r_0) \neq 0$, we have $\mathbf{v}^\top \boldsymbol{\Omega}_{xx} \mathbf{v} > 0$. Then, at the fixed r_0 , $W(r_0) := \mathbf{B}_X(r_0)^\top \mathbf{v}$ has a Gaussian distribution with variance $r_0 \mathbf{v}^\top \boldsymbol{\Omega}_{xx} \mathbf{v} > 0$, so that $\mathbb{P}(W(r_0) = 0) = 0$, which implies that $\mathbb{P}(|W(r_0)| > 0) = 1$. By continuity of $r \mapsto W(r)$, there exists a neighbourhood of r_0 such that $|W(r)| > 0$ for all $r \in \mathcal{N}_{r_0}$, which implies that $\int_0^1 W(r)^2 dr \geq \int_{\mathcal{N}_{r_0}} W(r)^2 dr > 0$, *a.s.*, a contradiction. Hence

$$\int_0^1 (\mathbf{Z} \boldsymbol{\psi}_{(k)}(r)) (\mathbf{Z} \boldsymbol{\psi}_{(k)}(r))^\top dr = \mathbf{0}_{K_1 \times K_1}.$$

Since the basis is orthonormal, we have

$$\int_0^1 \boldsymbol{\psi}_{(k)}(r) \boldsymbol{\psi}_{(k)}^\top(r) dr = \mathbf{I}_k.$$

Therefore,

$$\int_0^1 (\mathbf{Z} \boldsymbol{\psi}_{(k)}(r)) (\mathbf{Z} \boldsymbol{\psi}_{(k)}(r))^\top dr = \mathbf{Z} \left(\int_0^1 \boldsymbol{\psi}_{(k)}(r) \boldsymbol{\psi}_{(k)}^\top(r) dr \right) \mathbf{Z}^\top = \mathbf{Z} \mathbf{Z}^\top.$$

If the left-hand side equals $\mathbf{0}_{K_1 \times K_1}$, then $\mathbf{Z} \mathbf{Z}^\top = \mathbf{0}_{K_1 \times K_1}$, which implies $\mathbf{Z} = \mathbf{0}_{K_1 \times k}$ and hence $\text{vec}(\mathbf{Z}) = \mathbf{z} = \mathbf{0}_{kK_1 \times 1}$, a contradiction. $\square$

Lemma C.2. Under Assumptions 1-3, we have, as $n \rightarrow \infty$

$$\frac{1}{n} \sum_{t=1}^n \mathbf{u}_{0t} \mathbf{X}_t^\top = \boldsymbol{\Gamma}_{ux}^{(k)} + o_p(1),$$

where

$$\boldsymbol{\Gamma}_{ux}^{(k)} := \int_0^1 d\mathbf{B}_0(r) (\boldsymbol{\psi}_{(k)}(r) \otimes \mathbf{B}_X(r))^\top + \boldsymbol{\ell}_{1(k)} \otimes (\boldsymbol{\Lambda}_{0x} + \boldsymbol{\Sigma}_{0x})$$

satisfies $\left\| \boldsymbol{\Gamma}_{ux}^{(k)} \right\|_{sp} = O_p(1)$. $\boldsymbol{\ell}_{1(k)}$ denotes an $k \times 1$ vector with the first element equal to one and all remaining elements equal to zero.

Proof. Let $\mathbf{U}_n(r) = \boldsymbol{\Phi}_0(1) \frac{1}{\sqrt{n}} \sum_{s=1}^{\lfloor nr \rfloor} \boldsymbol{\varepsilon}_s$. We note that

$$\begin{aligned} \frac{1}{n} \sum_{t=1}^n \mathbf{u}_{0t} \mathbf{X}_t^\top &= \frac{1}{n} \sum_{t=1}^n \mathbf{u}_{0t} (\boldsymbol{\psi}_{(k)}(t/n) \otimes \mathbf{X}_{t-1})^\top + \frac{1}{n} \sum_{t=1}^n \mathbf{u}_{0t} (\boldsymbol{\psi}_{(k)}(t/n) \otimes \mathbf{u}_{xt})^\top \\ &:= \Gamma_n(1) + \Gamma_n(2). \end{aligned}$$

We will show that

$$\Gamma_n(1) = \int_0^1 d\mathbf{U}_n(r) (\boldsymbol{\psi}_{(k)}(r) \otimes \mathbf{V}_{n, \lfloor nr \rfloor}^x)^\top + \boldsymbol{\ell}_{1(k)}^\top \otimes \boldsymbol{\Lambda}_{0x} + o_p(1), \quad (\text{C.5})$$

$$\Gamma_n(2) = \boldsymbol{\ell}_{1(k)}^\top \otimes \boldsymbol{\Sigma}_{0x} + o_p(1). \quad (\text{C.6})$$

These together with triangular inequality implies

$$\left\| \frac{1}{n} \sum_{t=1}^n \mathbf{u}_{0t} \mathbf{X}_t^{\top(k)} - \int_0^1 d\mathbf{U}_n(r) (\boldsymbol{\psi}_{(k)}(r) \otimes \mathbf{V}_{n, \lfloor nr \rfloor}^x)^{\top} - \boldsymbol{\ell}_{1(k)}^{\top} \otimes (\boldsymbol{\Lambda}_{0x} + \boldsymbol{\Sigma}_{0x}) \right\| = o_p(1).$$

Let us start with (C.5). BN decomposition gives

$$\begin{aligned} \Gamma_n(1) &= \Phi_0(1) \frac{1}{n} \sum_{t=1}^n \boldsymbol{\varepsilon}_t (\boldsymbol{\psi}_{(k)}(t/n) \otimes \mathbf{X}_{t-1})^{\top} - \frac{1}{n} \sum_{t=1}^n \Delta \tilde{\mathbf{u}}_{0t} (\boldsymbol{\psi}_{(k)}(t/n) \otimes \mathbf{X}_{t-1})^{\top} \\ &= \Phi_0(1) \frac{1}{\sqrt{n}} \sum_{t=1}^n \boldsymbol{\varepsilon}_t (\boldsymbol{\psi}_{(k)}(t/n) \otimes \mathbf{V}_{n,t-1}^x)^{\top} + \Phi_0(1) \frac{1}{n} \sum_{t=1}^n \boldsymbol{\varepsilon}_t (\boldsymbol{\psi}_{(k)}(t/n) \otimes (\tilde{\mathbf{u}}_{x,0} - \tilde{\mathbf{u}}_{x,t-1} + \mathbf{X}_0))^{\top} \\ &\quad - \frac{1}{n} \sum_{t=1}^n \Delta \tilde{\mathbf{u}}_{0t} (\boldsymbol{\psi}_{(k)}(t/n) \otimes \mathbf{X}_{t-1})^{\top} \\ &:= \Gamma_n(1, 1) + \Gamma_n(1, 2) + \Gamma_n(1, 3). \end{aligned}$$

Now let us consider $\Gamma_n(1, 1)$. We have

$$\begin{aligned} \Gamma_n(1, 1) &= \sum_{t=1}^n \left(\Phi_0(1) \frac{1}{\sqrt{n}} \boldsymbol{\varepsilon}_t \right) (\boldsymbol{\psi}_{(k)}(t/n) \otimes \mathbf{V}_{n,t-1}^x)^{\top} \\ &= \int_0^1 d\mathbf{U}_n(r) (\boldsymbol{\psi}_{(k)}(\lfloor nr \rfloor/n) \otimes \mathbf{V}_{n, \lfloor nr \rfloor - 1}^x)^{\top} \\ &= \int_0^1 d\mathbf{U}_n(r) (\boldsymbol{\psi}_{(k)}(r) \otimes \mathbf{V}_{n, \lfloor nr \rfloor}^x)^{\top} + \int_0^1 d\mathbf{U}_n(r) (\boldsymbol{\psi}_{(k)}(\lfloor nr \rfloor/n) \otimes \mathbf{V}_{n, \lfloor nr \rfloor - 1}^x - \boldsymbol{\psi}_{(k)}(r) \otimes \mathbf{V}_{n, \lfloor nr \rfloor}^x)^{\top} \\ &= \int_0^1 d\mathbf{U}_n(r) (\boldsymbol{\psi}_{(k)}(r) \otimes \mathbf{V}_{n, \lfloor nr \rfloor}^x)^{\top} + o_p(1), \end{aligned}$$

if we could show that

$$\int_0^1 d\mathbf{U}_n(r) (\boldsymbol{\psi}_{(k)}(\lfloor nr \rfloor/n) \otimes \mathbf{V}_{n, \lfloor nr \rfloor - 1}^x - \boldsymbol{\psi}_{(k)}(r) \otimes \mathbf{V}_{n, \lfloor nr \rfloor}^x)^{\top} = o_p(1). \quad (\text{C.7})$$

Note that

$$\begin{aligned}
& \int_0^1 d\mathbf{U}_n(r) \left(\boldsymbol{\psi}_{(k)}(\lfloor nr \rfloor/n) \otimes \mathbf{V}_{n, \lfloor nr \rfloor - 1}^x - \boldsymbol{\psi}_{(k)}(r) \otimes \mathbf{V}_{n, \lfloor nr \rfloor}^x \right)^\top \\
&= \int_0^1 d\mathbf{U}_n(r) \left\{ \left(\boldsymbol{\psi}_{(k)}(\lfloor nr \rfloor/n) - \boldsymbol{\psi}_{(k)}(r) \right) \otimes \left(\mathbf{V}_{n, \lfloor nr \rfloor - 1}^x - \mathbf{V}_{n, \lfloor nr \rfloor}^x \right) \right\}^\top \\
&\quad + \int_0^1 d\mathbf{U}_n(r) \left\{ \left(\boldsymbol{\psi}_{(k)}(\lfloor nr \rfloor/n) - \boldsymbol{\psi}_{(k)}(r) \right) \otimes \mathbf{V}_{n, \lfloor nr \rfloor}^x \right\}^\top \\
&\quad + \int_0^1 d\mathbf{U}_n(r) \left\{ \boldsymbol{\psi}_{(k)}(r) \otimes \left(\mathbf{V}_{n, \lfloor nr \rfloor - 1}^x - \mathbf{V}_{n, \lfloor nr \rfloor}^x \right) \right\}^\top.
\end{aligned}$$

We have shown in (C.2) that $\|\boldsymbol{\psi}_{(k)}(\lfloor nr \rfloor/n) - \boldsymbol{\psi}_{(k)}(r)\| = O\left(\frac{k^{3/2}}{n}\right) = o(1)$ for all $r \in [0, 1]$. In addition, by the definition of $\mathbf{V}_{n, \lfloor nr \rfloor}$, $\|\mathbf{V}_{n, \lfloor nr \rfloor - 1}^x - \mathbf{V}_{n, \lfloor nr \rfloor}^x\| = O_p(1/\sqrt{n})$. Then, each of the integrals above is $o_p(1)$ and hence (C.7) holds.

Moving on to $\Gamma_n(1, 2)$. We note that

$$\begin{aligned}
& \left\| \frac{1}{n} \sum_{t=1}^n \boldsymbol{\varepsilon}_t \left(\boldsymbol{\psi}_{(k)}(t/n) \otimes (\tilde{\mathbf{u}}_{x,0} - \tilde{\mathbf{u}}_{x,t-1} + \mathbf{X}_0) \right)^\top \right\| \\
&\leq \frac{1}{n} \sum_{t=1}^n \left\| \boldsymbol{\varepsilon}_t \left(\boldsymbol{\psi}_{(k)}(t/n) \otimes (\tilde{\mathbf{u}}_{x,0} - \tilde{\mathbf{u}}_{x,t-1} + \mathbf{X}_0) \right)^\top \right\| \\
&\leq \max_{1 \leq t \leq n} \left\| \frac{1}{\sqrt{n}} (\tilde{\mathbf{u}}_{x,0} - \tilde{\mathbf{u}}_{x,t} + \mathbf{X}_0) \right\| \frac{1}{\sqrt{n}} \sum_{t=1}^n \|\boldsymbol{\varepsilon}_t\| \|\boldsymbol{\psi}_{(k)}(t/n)\| = O_p\left(\frac{\sqrt{k}}{n^a}\right),
\end{aligned}$$

which follows from Lemma C.6 and the fact that $\|\boldsymbol{\psi}_{(k)}(t/n)\| = O(1)\sqrt{k}$, which holds uniformly over $t = 1, 2, \dots, n$. This implies that $\|\Gamma_n(1, 2)\| = O_p\left(\frac{\sqrt{k}}{n^a}\right) = o_p(1)$.

Finally, we consider $\Gamma_n(1, 3)$. Summation by parts leads to

$$\begin{aligned}
\Gamma_n(1, 3) &= -\frac{1}{n} \sum_{t=1}^n \Delta \tilde{\mathbf{u}}_{0t} \left(\boldsymbol{\psi}_{(k)}(t/n) \otimes \mathbf{X}_{t-1} \right)^\top \\
&= \frac{1}{n} \sum_{t=1}^n \tilde{\mathbf{u}}_{0t} \left\{ \left(\boldsymbol{\psi}_{(k)}((t+1)/n) \otimes \mathbf{X}_t \right) - \left(\boldsymbol{\psi}_{(k)}(t/n) \otimes \mathbf{X}_{t-1} \right) \right\}^\top + o_p(1) \\
&= \frac{1}{n} \sum_{t=1}^n \tilde{\mathbf{u}}_{0t} \left\{ \left(\boldsymbol{\psi}_{(k)}((t+1)/n) - \boldsymbol{\psi}_{(k)}(t/n) \right) \otimes \mathbf{X}_t + \boldsymbol{\psi}_{(k)}(t/n) \otimes (\mathbf{X}_t - \mathbf{X}_{t-1}) \right\}^\top + o_p(1) \\
&= \frac{1}{n} \sum_{t=1}^n \tilde{\mathbf{u}}_{0t} \left(\boldsymbol{\psi}_{(k)}(t/n) \otimes \mathbf{u}_{xt} \right)^\top + o_p(1),
\end{aligned}$$

where the final equality follows from the Lipschitz-continuity of the base function, which implies that $\|\boldsymbol{\psi}_{(k)}((t+1)/n) - \boldsymbol{\psi}_{(k)}(t/n)\| = O(k^{3/2}/n)$, together with Cauchy-Schwarz inequality and the fact that $\frac{1}{n} \sum_{t=1}^n \|\mathbf{X}_t\|^2 = O_p(n)$, so that $\left\| \frac{1}{n} \sum_{t=1}^n \tilde{\mathbf{u}}_{0t} \{(\boldsymbol{\psi}_{(k)}((t+1)/n) - \boldsymbol{\psi}_{(k)}(t/n)) \otimes \mathbf{X}_t\}^\top \right\| = O_p\left(\frac{k^{3/2}}{\sqrt{n}}\right) = o_p(1)$. In addition,

$$\begin{aligned} & \frac{1}{n} \sum_{t=1}^n \tilde{\mathbf{u}}_{0t} (\boldsymbol{\psi}_{(k)}(t/n) \otimes \mathbf{u}_{xt})^\top \\ &= \frac{1}{n} \sum_{t=1}^n (\boldsymbol{\psi}_{(k)}^\top(t/n) \otimes (\tilde{\mathbf{u}}_{0t} \mathbf{u}_{x,t}^\top - \boldsymbol{\Lambda}_{0x})) + \left(\frac{1}{n} \sum_{t=1}^n \boldsymbol{\psi}_{(k)}^\top(t/n) \right)^\top \otimes \boldsymbol{\Lambda}_{0x} \\ &= \left(\int_0^1 \boldsymbol{\psi}_{(k)}(r) dr \right)^\top \otimes \boldsymbol{\Lambda}_{0x} + O_p\left(\sqrt{\frac{k}{n}}\right) \\ &= \boldsymbol{\ell}_{1(k)}^\top \otimes \boldsymbol{\Lambda}_{0x} + o_p(1). \end{aligned}$$

The $O_p\left(\sqrt{k/n}\right)$ term follows by applying C.L.T. for the stationary and ergodic process $(\tilde{\mathbf{u}}_{0t} \mathbf{u}_{x,t}^\top - \boldsymbol{\Lambda}_{0x})_t$, together with $\|\boldsymbol{\psi}_{(k)}(t/n)\| = O(1)\sqrt{k}$. Moreover, by the Riemann-sum approximation, $\frac{1}{n} \sum_{t=1}^n \boldsymbol{\psi}_{(k)}(t/n) \rightarrow \int_0^1 \boldsymbol{\psi}_{(k)}(r) dr = \boldsymbol{\ell}_{1(k)}$ as $n \rightarrow \infty$, where $\boldsymbol{\ell}_{1(k)}$ is a unit vector with 1 in the first coordinate and 0 in all other coordinates.

By combining all results obtained above, we get

$$\left\| \Gamma_n(1) - \int_0^1 d\mathbf{U}_n(r) (\boldsymbol{\psi}_{(k)}(r) \otimes \mathbf{V}_{n, \lfloor nr \rfloor}^x)^\top - \boldsymbol{\ell}_{1(k)}^\top \otimes \boldsymbol{\Lambda}_{0x} \right\| = o_p(1).$$

Using Theorem 2.2 in Kurtz and Protter 1991, the joint convergence result of

$$\begin{pmatrix} \frac{1}{\sqrt{n}} \sum_{t=1}^{\lfloor nr \rfloor} \mathbf{u}_{0t} \\ \frac{1}{\sqrt{n}} \sum_{t=1}^{\lfloor nr \rfloor} \mathbf{u}_{xt} \end{pmatrix} \Rightarrow \begin{pmatrix} \mathbf{B}_0(r) \\ \mathbf{B}_X(r) \end{pmatrix},$$

on $D([0, 1], \mathbb{R}^{d+K_1})$, together with similar arguments as used to establish (C.4), we obtain

$$\left\| \int_0^1 d\mathbf{U}_n(r) (\boldsymbol{\psi}_{(k)}(r) \otimes \mathbf{V}_{n, \lfloor nr \rfloor}^x)^\top - \int_0^1 d\mathbf{B}_0(r) (\boldsymbol{\psi}_{(k)}(r) \otimes \mathbf{B}_X(r))^\top \right\|_{sp} = o_p(1).$$

Turning to $\Gamma_n(2)$. This is straightforward, since same reasoning for $\frac{1}{n} \sum_{t=1}^n \tilde{\mathbf{u}}_{0t} (\boldsymbol{\psi}_{(k)}(t/n) \otimes \mathbf{u}_{xt})^\top =$

$\ell_{1(k)}^\top \otimes \Lambda_{0x} + o_p(1)$ gives $\Gamma_n(2) = \ell_{1(k)}^\top \otimes \Sigma_{0x} + o_p(1)$. Thus, we have

$$\frac{1}{n} \sum_{t=1}^n \mathbf{u}_{0t} \mathbf{X}_t^{\top(k)} = \int_0^1 d\mathbf{B}_0(r) (\boldsymbol{\psi}_{(k)}(r) \otimes \mathbf{B}_X(r))^\top + \ell_{1(k)} \otimes (\Lambda_{0x} + \Sigma_{0x}) + o_p(1).$$

To show $\left\| \Gamma_{ux}^{(k)} \right\|_{sp} = O_p(1)$, it is suffice to show that $\left\| \int_0^1 d\mathbf{B}_0(r) (\boldsymbol{\psi}_{(k)}(r) \otimes \mathbf{B}_X(r))^\top \right\|_{sp} = O_p(1)$. Let $\mathbf{u} \in \mathbb{R}^d$ and $\mathbf{v} = (\mathbf{v}_0^\top, \dots, \mathbf{v}_{k-1}^\top)^\top \in \mathbb{R}^{K_1 k}$ be unit vectors such that $\|\mathbf{u}\| = 1$ and $\|\mathbf{v}\| = 1$. Furthermore, define

$$q_{(k)} := \mathbf{u}^\top \left(\int_0^1 d\mathbf{B}_0(r) (\boldsymbol{\psi}_{(k)}(r) \otimes \mathbf{B}_x(r))^\top \right) \mathbf{v} := \int_0^1 g_{(k)}(r) \mathbf{u}^\top d\mathbf{B}_0(r).$$

By Itô isometry, we have

$$\mathbb{E} [q_{(k)}^2] = (\mathbf{u}^\top \Omega_{00} \mathbf{u}) \mathbb{E} \left[\int_0^1 g_{(k)}^2(r) dr \right].$$

Recall that

$$g_{(k)}(r) = (\boldsymbol{\psi}_{(k)}(r) \otimes \mathbf{B}_x(r))^\top \mathbf{v} = \sum_{i=0}^{k-1} \psi_i(r) \mathbf{B}_x^\top(r) \mathbf{v}_i.$$

We then have

$$\begin{aligned} \mathbb{E} \left[\int_0^1 g_{(k)}^2(r) dr \right] &= \int_0^1 \mathbb{E} \left[\left(\sum_{i=0}^{k-1} \psi_i(r) \mathbf{B}_x^\top(r) \mathbf{v}_i \right)^2 \right] dr \\ &= \sum_{i,\ell} \mathbf{v}_i^\top \Omega_{xx} \mathbf{v}_\ell \int_0^1 r \psi_i(r) \psi_\ell(r) dr. \end{aligned}$$

Let $\mathbf{G}_{(k)}$ be the $k \times k$ matrix with (i, ℓ) th element equal to $\int_0^1 r \psi_i(r) \psi_\ell(r) dr$. Cauchy-Schwartz inequality gives

$$\left| \int_0^1 r \psi_i(r) \psi_\ell(r) dr \right| \leq \left(\int_0^1 r^2 \psi_i^2(r) dr \right)^{1/2} \left(\int_0^1 r^2 \psi_\ell^2(r) dr \right)^{1/2} \leq \int_0^1 \psi_i^2(r) dr = 1,$$

which follows from the fact that $\{\psi_i(r)\}_i$ is orthonormal. This implies that $\|\mathbf{G}_{(k)}\|_{sp} \leq 1$.

Substituting this above yields

$$\begin{aligned}\mathbb{E} \left[\int_0^1 g_{(k)}^2(r) dr \right] &= \sum_{i,\ell} \mathbf{v}_i^\top \boldsymbol{\Omega}_{xx} \mathbf{v}_\ell \int_0^1 r \psi_i(r) \psi_\ell(r) dr \\ &\leq \|\mathbf{G}_{(k)}\|_{sp} \sum_{i=0}^{k-1} \|\boldsymbol{\Omega}_{xx}^{1/2} \mathbf{v}_i\|^2 \leq \|\boldsymbol{\Omega}_{xx}^{1/2}\|_{sp}^2 \sum_{i=0}^{k-1} \|\mathbf{v}_i\|^2 < \infty,\end{aligned}$$

which implies that $q_{(k)} = O_p(1)$. This completes the proof. $\square$

Lemma C.3. Under Assumptions 1-3, we have, as $n \rightarrow \infty$

$$\frac{1}{n^2} \sum_{t=1}^n (\mathbf{A}_t - \mathbf{A}_k(t/n)) \mathbf{X}_t \mathbf{X}_t^\top = \int_0^1 (\mathbf{A}(r) - \mathbf{A}_k(r)) (\mathbf{B}_X(r)) (\boldsymbol{\psi}_{(k)}(r) \otimes \mathbf{B}_X(r))^\top dr + O_p\left(\frac{1}{n^a k^{q-1/2}}\right),$$

where $a = \frac{p-2}{2p}$. Furthermore, $\left\| \int_0^1 (\mathbf{A}(r) - \mathbf{A}_k(r)) (\mathbf{B}_X(r)) (\boldsymbol{\psi}_{(k)}(r) \otimes \mathbf{B}_X(r))^\top dr \right\|_{sp} = O_p\left(\frac{1}{k^q}\right)$.

Proof. Note that

$$\begin{aligned}& \frac{1}{n^2} \sum_{t=1}^n (\mathbf{A}_t - \mathbf{A}_k(t/n)) \mathbf{X}_t \mathbf{X}_t^\top \\ &= \frac{1}{n^2} \sum_{t=1}^n (\mathbf{A}_t - \mathbf{A}_k(t/n)) (\sqrt{n} \mathbf{V}_{n,t} + \tilde{\mathbf{u}}_{x0} - \tilde{\mathbf{u}}_{xt} + \mathbf{X}_0) \\ & \quad \times (\boldsymbol{\psi}_{(k)}(t/n) \otimes (\sqrt{n} \mathbf{V}_{n,t} + \tilde{\mathbf{u}}_{x0} - \tilde{\mathbf{u}}_{xt} + \mathbf{X}_0))^\top \\ &= \frac{1}{n^2} \sum_{t=1}^n (\mathbf{A}_t - \mathbf{A}_k(t/n)) (\sqrt{n} \mathbf{V}_{n,t}) (\boldsymbol{\psi}_{(k)}(t/n) \otimes (\sqrt{n} \mathbf{V}_{n,t}))^\top \\ & \quad + \frac{1}{n^2} \sum_{t=1}^n (\mathbf{A}_t - \mathbf{A}_k(t/n)) (\sqrt{n} \mathbf{V}_{n,t}) (\boldsymbol{\psi}_{(k)}(t/n) \otimes (\tilde{\mathbf{u}}_{x0} - \tilde{\mathbf{u}}_{xt} + \mathbf{X}_0))^\top \\ & \quad + \frac{1}{n^2} \sum_{t=1}^n (\mathbf{A}_t - \mathbf{A}_k(t/n)) (\tilde{\mathbf{u}}_{x0} - \tilde{\mathbf{u}}_{xt} + \mathbf{X}_0) (\boldsymbol{\psi}_{(k)}(t/n) \otimes \sqrt{n} \mathbf{V}_{n,t})^\top \\ & \quad + \frac{1}{n^2} \sum_{t=1}^n (\mathbf{A}_t - \mathbf{A}_k(t/n)) (\tilde{\mathbf{u}}_{x0} - \tilde{\mathbf{u}}_{xt} + \mathbf{X}_0) (\boldsymbol{\psi}_{(k)}(t/n) \otimes (\tilde{\mathbf{u}}_{x0} - \tilde{\mathbf{u}}_{xt} + \mathbf{X}_0))^\top \\ &:= \boldsymbol{\Delta}_n(1) + \boldsymbol{\Delta}_n(2) + \boldsymbol{\Delta}_n(3) + \boldsymbol{\Delta}_n(4),\end{aligned}$$

so that

$$\left\| \frac{1}{n^2} \sum_{t=1}^n (\mathbf{A}_t - \mathbf{A}_k(t/n)) \mathbf{X}_t \mathbf{X}_t^{\top(k)} - \boldsymbol{\Delta}_n(1) \right\| \leq \|\boldsymbol{\Delta}_n(2)\| + \|\boldsymbol{\Delta}_n(3)\| + \|\boldsymbol{\Delta}_n(4)\|.$$

We first show that $\|\boldsymbol{\Delta}_n(2)\| = O_p\left(\frac{1}{k^{q-1/2}n^a}\right)$, $\|\boldsymbol{\Delta}_n(3)\| = O_p\left(\frac{1}{k^{q-1/2}n^a}\right)$, and $\|\boldsymbol{\Delta}_n(4)\| = O_p\left(\frac{1}{n^{2a}k^{q-1/2}}\right)$.

This would imply that

$$\frac{1}{n^2} \sum_{t=1}^n (\mathbf{A}_t - \mathbf{A}_k(t/n)) \mathbf{X}_t \mathbf{X}_t^{\top(k)} = \boldsymbol{\Delta}_n(1) + O_p\left(\frac{1}{k^{q-1/2}n^a}\right). \quad (\text{C.8})$$

Let us start with $\boldsymbol{\Delta}_n(2)$. We note that

$$\begin{aligned} \|\boldsymbol{\Delta}_n(2)\| &= \left\| \frac{1}{n} \sum_{t=1}^n (\mathbf{A}_t - \mathbf{A}_k(t/n)) (\mathbf{V}_{n,t}) \left(\boldsymbol{\psi}_{(k)}(t/n) \otimes \frac{1}{\sqrt{n}} (\tilde{\mathbf{u}}_{x0} - \tilde{\mathbf{u}}_{xt} + \mathbf{X}_0) \right)^{\top} \right\| \\ &\leq \max_{1 \leq t \leq n} \left\| \frac{1}{\sqrt{n}} (\tilde{\mathbf{u}}_{x0} - \tilde{\mathbf{u}}_{xt} + \mathbf{X}_0) \right\| \frac{1}{n} \sum_{t=1}^n \|(\mathbf{A}_t - \mathbf{A}_k(t/n)) (\mathbf{V}_{n,t})\| \|\boldsymbol{\psi}_{(k)}(t/n)\|. \end{aligned}$$

By Lemma C.6 we have $\max_{1 \leq t \leq n} \left\| \frac{1}{\sqrt{n}} (\tilde{\mathbf{u}}_{x0} - \tilde{\mathbf{u}}_{xt} + \mathbf{X}_0) \right\| = O_p\left(\frac{1}{n^a}\right)$. In addition, since

$$\begin{aligned} &\frac{1}{n} \sum_{t=1}^n \|(\mathbf{A}_t - \mathbf{A}_k(t/n)) \mathbf{V}_{n,t}\| \|\boldsymbol{\psi}_{(k)}(t/n)\| \\ &= \int_0^1 \|(\mathbf{A}(\lfloor nr \rfloor/n) - \mathbf{A}_k(\lfloor nr \rfloor/n)) \mathbf{V}_{n,\lfloor nr \rfloor}\| \|\boldsymbol{\psi}_{(k)}(\lfloor nr \rfloor/n)\| dr + O_p(1/n) \\ &= O_p\left(\frac{1}{k^{q-1/2}}\right) + o_p(1), \end{aligned}$$

which follows from Cauchy-Schwarz inequality and (11):

$$\begin{aligned} &\int_0^1 \|(\mathbf{A}(r) - \mathbf{A}_k(r)) \mathbf{V}_{n,\lfloor nr \rfloor}\| \|\boldsymbol{\psi}_{(k)}(r)\| dr \\ &\leq \left(\int_0^1 \|\mathbf{A}(r) - \mathbf{A}_k(r)\|^2 dr \right)^{1/2} \left(\int_0^1 \|\mathbf{V}_{n,\lfloor nr \rfloor}\|^2 dr \right)^{1/2} \left(\int_0^1 \|\boldsymbol{\psi}_{(k)}(r)\|^2 dr \right)^{1/2} \\ &\leq \left(\int_0^1 \|\mathbf{A}(r) - \mathbf{A}_k(r)\|^2 dr \right)^{1/2} \left(\int_0^1 \|\mathbf{V}_{n,\lfloor nr \rfloor}\|^2 dr \right)^{1/2} \left(\int_0^1 \|\boldsymbol{\psi}_{(k)}(r)\|^2 dr \right)^{1/2} \\ &= O_p\left(\frac{1}{k^{q-1/2}}\right). \end{aligned}$$

We thus conclude that $\|\Delta_n(2)\| = O_p\left(\frac{1}{k^{q-1/2}n^a}\right)$. Following similar argument, we can show that $\|\Delta_n(3)\| = O_p\left(\frac{1}{k^{q-1/2}n^a}\right)$. For $\Delta_n(4)$, we have

$$\begin{aligned}
& \|\Delta_n(4)\| \\
&= \left\| \frac{1}{n} \sum_{t=1}^n (\mathbf{A}_t - \mathbf{A}_k(t/n)) \left(\frac{1}{\sqrt{n}} (\tilde{\mathbf{u}}_{x0} - \tilde{\mathbf{u}}_{xt} + \mathbf{X}_0) \right) \left(\boldsymbol{\psi}_{(k)}(t/n) \otimes \frac{1}{\sqrt{n}} (\tilde{\mathbf{u}}_{x0} - \tilde{\mathbf{u}}_{xt} + \mathbf{X}_0) \right)^\top \right\| \\
&\leq \max_{1 \leq t \leq n} \left\| \frac{1}{\sqrt{n}} (\tilde{\mathbf{u}}_{x0} - \tilde{\mathbf{u}}_{xt} + \mathbf{X}_0) \right\| \max_{1 \leq t \leq n} \left\| \frac{1}{\sqrt{n}} (\tilde{\mathbf{u}}_{x0} - \tilde{\mathbf{u}}_{xt} + \mathbf{X}_0) \right\| \frac{1}{n} \sum_{t=1}^n \|\mathbf{A}_t - \mathbf{A}_k(t/n)\| \|\boldsymbol{\psi}_{(k)}(t/n)\| \\
&= O_p\left(\frac{1}{n^a}\right) O_p\left(\frac{1}{n^a}\right) O\left(\frac{1}{k^{q-1/2}}\right) = O_p\left(\frac{1}{n^{2a}k^{q-1/2}}\right) = o_p(1).
\end{aligned}$$

Finally, we consider $\Delta_n(1)$. We have

$$\begin{aligned}
& \Delta_n(1) \\
&= \frac{1}{n} \sum_{t=1}^n (\mathbf{A}_t - \mathbf{A}_k(t/n)) \mathbf{V}_{n,t} (\boldsymbol{\psi}_{(k)}(t/n) \otimes \mathbf{V}_{n,t})^\top \\
&= \int_0^1 (\mathbf{A}(\lfloor nr \rfloor/n) - \mathbf{A}_k(\lfloor nr \rfloor/n)) (\mathbf{V}_{n, \lfloor nr \rfloor}) (\boldsymbol{\psi}_{(k)}(\lfloor nr \rfloor/n) \otimes \mathbf{V}_{n, \lfloor nr \rfloor})^\top dr + O_p(1/n) \\
&= \int_0^1 (\mathbf{A}(r) - \mathbf{A}_k(r)) (\mathbf{V}_{n, \lfloor nr \rfloor}) (\boldsymbol{\psi}_{(k)}(r) \otimes \mathbf{V}_{n, \lfloor nr \rfloor})^\top dr + o_p(1),
\end{aligned}$$

where the second inequality follows from a Riemann-sum approximation of integral, the final equality follows from the continuity of $\mathbf{A}(\cdot)$ and the fact that $\|\boldsymbol{\psi}_{(k)}(\lfloor nr \rfloor/n) - \boldsymbol{\psi}_{(k)}(r)\| = O\left(\frac{k^{3/2}}{n}\right) = o(1)$ for all $r \in [0, 1]$. This establishes (C.8). Finally, by Cauchy-Schwarz inequality, $\int_0^1 \|\boldsymbol{\psi}_{(k)}(r)\|^2 dr = kO(1)$, and $\sup_{r \in [0, 1]} \|\mathbf{V}_{n, \lfloor nr \rfloor} - \mathbf{B}_X(r)\| = o_p(1)$, we could show that $\|\mathbf{R}_{\Delta_n(1)}\| = O_p(k^{-(q-1/2)}) = o_p(1)$, where

$$\mathbf{R}_{\Delta_n(1)} = \int_0^1 (\mathbf{A}(r) - \mathbf{A}_k(r)) \left\{ (\mathbf{V}_{n, \lfloor nr \rfloor}) (\boldsymbol{\psi}_{(k)}(r) \otimes \mathbf{V}_{n, \lfloor nr \rfloor})^\top - (\mathbf{B}_X(r)) (\boldsymbol{\psi}_{(k)}(r) \otimes \mathbf{B}_X(r))^\top \right\} dr.$$

Therefore,

$$\begin{aligned} & \frac{1}{n^2} \sum_{t=1}^n (\mathbf{A}_t - \mathbf{A}_k(t/n)) \mathbf{X}_t \mathbf{X}_t^\top{}^{(k)} \\ &= \int_0^1 (\mathbf{A}(r) - \mathbf{A}_k(r)) (\mathbf{B}_X(r)) (\boldsymbol{\psi}_{(k)}(r) \otimes \mathbf{B}_X(r))^\top dr + O_p \left(\frac{1}{k^{q-1/2} n^a} \right). \end{aligned}$$

Now we show that $\left\| \int_0^1 (\mathbf{A}(r) - \mathbf{A}_k(r)) (\mathbf{B}_X(r)) (\boldsymbol{\psi}_{(k)}(r) \otimes \mathbf{B}_X(r))^\top dr \right\|_{sp} = O_p \left(\frac{1}{k^q} \right)$. Let $\mathbf{u} \in \mathbb{R}^d$ and $\mathbf{v} = (\mathbf{v}_0^\top, \dots, \mathbf{v}_{k-1}^\top)^\top \in \mathbb{R}^{K_1 k}$ be unit vectors such that $\|\mathbf{u}\| = 1$ and $\|\mathbf{v}\| = 1$. Furthermore, define

$$z_{(k)} := \mathbf{u}^\top \int_0^1 (\mathbf{A}(r) - \mathbf{A}_k(r)) (\mathbf{B}_X(r)) (\boldsymbol{\psi}_{(k)}(r) \otimes \mathbf{B}_X(r))^\top dr \mathbf{v},$$

and

$$\mathbf{H}_{(k)}(r) := \mathbf{A}(r) - \mathbf{A}_k(r), \quad \mathbf{G}_{(k)}(r) := \mathbf{u} \mathbf{B}_X^\top(r) (\boldsymbol{\psi}_{(k)}(r) \otimes \mathbf{B}_X(r))^\top \mathbf{v}.$$

We then have

$$\begin{aligned} |z_{(k)}| &= \left| \int_0^1 \text{tr} (\mathbf{H}_{(k)}^\top(r) \mathbf{G}_{(k)}(r)) dr \right| \\ &\leq \left(\int_0^1 \|\mathbf{H}_{(k)}(r)\|^2 dr \right)^{1/2} \left(\int_0^1 \|\mathbf{G}_{(k)}(r)\|^2 dr \right)^{1/2}. \end{aligned}$$

By (11), we have $\left(\int_0^1 \|\mathbf{H}_{(k)}(r)\|^2 dr \right)^{1/2} = O(1/k^q)$. We also have

$$\begin{aligned} \int_0^1 \|\mathbf{G}_{(k)}(r)\|^2 dr &\leq \int_0^1 \|\mathbf{B}_X(r)\|^4 \left\| \sum_{j=0}^{k-1} \psi_j(r) \mathbf{v}_j \right\|^2 dr \\ &\leq \sup_{r \in [0,1]} \|\mathbf{B}_X(r)\|^4 \int_0^1 \left\| \sum_{j=0}^{k-1} \psi_j(r) \mathbf{v}_j \right\|^2 dr. \end{aligned}$$

Observe that

$$\begin{aligned} \int_0^1 \left\| \sum_{j=0}^{k-1} \psi_j(r) \mathbf{v}_j \right\|^2 dr &= \int_0^1 \sum_{j,\ell} \psi_j(r) \psi_\ell(r) \mathbf{v}_j^\top \mathbf{v}_\ell dr \\ &= \sum_{j=0}^{k-1} \|\mathbf{v}_j\|^2 = \|\mathbf{v}\|^2 = 1, \end{aligned}$$

which follows from the orthonormality of $\{\psi_j(r)\}_j$. This implies that $\int_0^1 \|\mathbf{G}_{(k)}(r)\|^2 dr = O_p(1)$, since $\sup_{r \in [0,1]} \|\mathbf{B}_X(r)\|^4 = O_p(1)$. $\square$

Lemma C.4. Under Assumptions 1-3 and (19), we have, as $n \rightarrow \infty$

(i)

$$\frac{1}{n} \sum_{t=1}^n \mathbf{Z}_t \mathbf{X}_t^\top = \Gamma_{zx}^{(k)} + o_p(1),$$

where

$$\Gamma_{zx}^{(k)} := \int_0^1 d\mathbf{B}_Z(r) (\psi_{(k)}(r) \otimes \mathbf{B}_X(r))^\top + \boldsymbol{\ell}_{1(k)} \otimes (\boldsymbol{\Lambda}_{zx} + \boldsymbol{\Sigma}_{zx})$$

satisfies $\left\| \Gamma_{zx}^{(k)} \right\|_{sp} = O_p(1)$. $\boldsymbol{\ell}_{1(k)}$ denotes an $k \times 1$ vector with the first element equal to one and all remaining elements equal to zero.

(ii)

$$\left\| \frac{1}{n} \sum_{t=1}^n (\mathbf{A}_t - \mathbf{A}_k(t/n)) \mathbf{X}_t \mathbf{Z}_t^\top \right\| = O_p\left(\frac{\sqrt{n}}{k^q}\right)$$

Proof. We only show (ii), as (i) is identical to Lemma C.2. Again, by BN decomposition, we have

$$\begin{aligned} & \frac{1}{n} \sum_{t=1}^n (\mathbf{A}_t - \mathbf{A}_k(t/n)) \mathbf{X}_t \mathbf{Z}_t^\top \\ &= \frac{1}{n} \sum_{t=1}^n (\mathbf{A}_t - \mathbf{A}_k(t/n)) (\sqrt{n} \mathbf{V}_{n,t}^x + \tilde{\mathbf{u}}_{x0} - \tilde{\mathbf{u}}_{xt} + \mathbf{X}_0) (\Phi_z(1) \boldsymbol{\varepsilon}_t - \Delta \tilde{\mathbf{u}}_{zt})^\top \\ &= \frac{1}{n} \sum_{t=1}^n (\mathbf{A}_t - \mathbf{A}_k(t/n)) (\sqrt{n} \mathbf{V}_{n,t}^x) (\Phi_z(1) \boldsymbol{\varepsilon}_t)^\top - \frac{1}{n} \sum_{t=1}^n (\mathbf{A}_t - \mathbf{A}_k(t/n)) (\sqrt{n} \mathbf{V}_{n,t}^x) (\Delta \tilde{\mathbf{u}}_{zt})^\top \\ & \quad + \frac{1}{n} \sum_{t=1}^n (\mathbf{A}_t - \mathbf{A}_k(t/n)) (\tilde{\mathbf{u}}_{x0} - \tilde{\mathbf{u}}_{xt} + \mathbf{X}_0) (\Phi_z(1) \boldsymbol{\varepsilon}_t)^\top - \frac{1}{n} \sum_{t=1}^n (\mathbf{A}_t - \mathbf{A}_k(t/n)) (\tilde{\mathbf{u}}_{x0} - \tilde{\mathbf{u}}_{xt} + \mathbf{X}_0) (\Delta \tilde{\mathbf{u}}_{zt})^\top \\ &:= \boldsymbol{\Delta}_n^z(1) + \boldsymbol{\Delta}_n^z(2) + \boldsymbol{\Delta}_n^z(3) + \boldsymbol{\Delta}_n^z(4). \end{aligned}$$

Let us start with $\Delta_n^z(1)$. By F.C.L.T. and C.M.T., we have $\sup_{1 \leq t \leq n} \|\mathbf{V}_{n,t}^x\| = O_p(1)$. In addition, $\frac{1}{n} \sum_{t=1}^n \|\mathbf{A}_t - \mathbf{A}_k(t/n)\|^2 = \int_0^1 \|\mathbf{A}(r) - \mathbf{A}_k(r)\|^2 dr + O(1/n) = O(1/k^q)$, which follows from a Riemann-sum approximation of integral and (11). Under Assumption 1, we have $\frac{1}{n} \sum_{t=1}^n \|\Phi_z(1)\varepsilon_t\|^2 = O_p(1)$. Combining all these yields

$$\begin{aligned} \|\Delta_n^z(1)\| &\leq \frac{1}{n} \sum_{t=1}^n \|\mathbf{A}_t - \mathbf{A}_k(t/n)\| \|\sqrt{n}\mathbf{V}_{n,t}^x\| \|\Phi_z(1)\varepsilon_t\| \\ &\leq \sup_{1 \leq t \leq n} \|\sqrt{n}\mathbf{V}_{n,t}^x\| \left(\frac{1}{n} \sum_{t=1}^n \|\mathbf{A}_t - \mathbf{A}_k(t/n)\|^2 \right)^{1/2} \left(\frac{1}{n} \sum_{t=1}^n \|\Phi_z(1)\varepsilon_t\|^2 \right)^{1/2} \\ &= O_p(\sqrt{n})O_p(1/k^q) = O_p\left(\frac{\sqrt{n}}{k^q}\right), \end{aligned}$$

where the second equality follows from Cauchy-Schwarz inequality. Proceeding with similar reasoning as above, we could show that

$$\begin{aligned} \|\Delta_n^z(2)\| &\leq \frac{1}{n} \sum_{t=1}^n \|\mathbf{A}_t - \mathbf{A}_k(t/n)\| \|\sqrt{n}\mathbf{V}_{n,t}^x\| \|\Delta\tilde{\mathbf{u}}_{zt}\| \\ &\leq \sup_{1 \leq t \leq n} \|\sqrt{n}\mathbf{V}_{n,t}^x\| \left(\frac{1}{n} \sum_{t=1}^n \|\mathbf{A}_t - \mathbf{A}_k(t/n)\|^2 \right)^{1/2} \left(\frac{1}{n} \sum_{t=1}^n \|\Delta\tilde{\mathbf{u}}_{zt}\|^2 \right)^{1/2} \\ &= O_p(\sqrt{n})O_p(1/k^q) = O_p\left(\frac{\sqrt{n}}{k^q}\right), \end{aligned}$$

and

$$\begin{aligned} \|\Delta_n^z(3)\| &\leq \frac{1}{n} \sum_{t=1}^n \|\mathbf{A}_t - \mathbf{A}_k(t/n)\| \|\tilde{\mathbf{u}}_{x0} - \tilde{\mathbf{u}}_{xt} + \mathbf{X}_0\| \|\Delta\tilde{\mathbf{u}}_{zt}\| \\ &\leq \max_{1 \leq t \leq n} \left\| \frac{1}{\sqrt{n}} (\tilde{\mathbf{u}}_{x0} - \tilde{\mathbf{u}}_{xt} + \mathbf{X}_0) \right\| \left(\frac{1}{n} \sum_{t=1}^n \|\mathbf{A}_t - \mathbf{A}_k(t/n)\|^2 \right)^{1/2} \left(\frac{1}{n} \sum_{t=1}^n \|\Delta\tilde{\mathbf{u}}_{zt}\|^2 \right)^{1/2} \\ &= O_p(\sqrt{n})O_p(\sqrt{n}/k^q) = O_p\left(\frac{n^{\frac{1}{2}-a}}{k^q}\right) = o_p(1), \end{aligned}$$

where the final equality follows from Lemma C.6. The case for $\|\Delta_n^z(4)\| = o_p(1)$ is also similar. $\square$

Lemma C.5. For the model specification (24), let $\mathbf{Z}_t = (\Delta\mathbf{X}_{t-K}^\top, \dots, \Delta\mathbf{X}_{t+K}^\top)^\top$. Under As-

sumptions 1-3, we have, as $n \rightarrow \infty$

- (i) $\left\| \frac{1}{n-2K} \sum_{t=K+1}^{n-K} \tilde{\mathbf{v}}_t \mathbf{Z}_t^\top \right\| = O_p \left(\sqrt{K/n} \right);$
- (ii) $\left\| \left(\frac{1}{n-2K} \sum_{t=K+1}^{n-K} \mathbf{Z}_t \mathbf{Z}_t^\top \right)^{-1} - \Sigma_{zz}^{-1} \right\|_{sp} = O_p \left(K/\sqrt{n} \right);$
- (iii) $\left\| \frac{1}{n-2K} \sum_{t=K+1}^{n-K} \mathbf{Z}_t \mathbf{X}_t^{(k)\top} \right\| = O_p \left(\sqrt{Kk} \right);$
- (iv)

$$\frac{1}{n} \sum_{t=1}^n \mathbf{v}_t \mathbf{X}_t^{\top(k)} = \int_0^1 d\mathbf{B}_v(r) \left((\psi_{(k)}(r) \otimes \mathbf{B}_r(\boldsymbol{\Omega}_{xx}))^\top \right)^\top + o_p(1),$$

$$\text{where } \left\| \int_0^1 d\mathbf{B}_v(r) \left((\psi_{(k)}(r) \otimes \mathbf{B}_r(\boldsymbol{\Omega}_{xx}))^\top \right)^\top \right\|_{sp} = O_p(1).$$

Proof. Proof of (i). Recall that $\tilde{\mathbf{v}}_t = \mathbf{v}_t + \sum_{|j|>K} \boldsymbol{\Gamma}_j \mathbf{X}_{t-j}$. Then,

$$\frac{1}{n-2K} \sum_{t=K+1}^{n-K} \tilde{\mathbf{v}}_t \mathbf{Z}_t^\top = \frac{1}{n-2K} \sum_{t=K+1}^{n-K} \mathbf{v}_t \mathbf{Z}_t^\top + \frac{1}{n-2K} \sum_{t=K+1}^{n-K} \left(\sum_{|j|>K} \boldsymbol{\Gamma}_j \mathbf{X}_{t-j} \right) \mathbf{Z}_t^\top.$$

By Lemma A5 in Saikkonen 1991, we have $\left\| \frac{1}{n-2K} \sum_{t=K+1}^{n-K} \left(\sum_{|j|>K} \boldsymbol{\Gamma}_j \mathbf{X}_{t-j} \right) \mathbf{Z}_t^\top \right\| = o_p \left(\sqrt{K/n} \right)$. In addition, Lemma A6 in Saikkonen 1991 gives $\left\| \frac{1}{n-2K} \sum_{t=K+1}^{n-K} \mathbf{v}_t \mathbf{Z}_t^\top \right\| = O_p \left(\sqrt{K/n} \right)$. (i) follows accordingly.

Proof of (ii). This is just Lemma A4 in Saikkonen 1991 so details are omitted.

Proof of (iii). Since K_1 is fixed and $\mathbf{X}_0 = O_p(1)$, for notation simplicity we shall prove for the scalar case $K_1 = 1$ and assume that $X_0 = 0$. Note that

$$\begin{aligned} \frac{1}{n-2K} \sum_{t=K+1}^{n-K} \mathbf{Z}_t \mathbf{X}_t^{(k)\top} &= \frac{1}{n-2K} \sum_{t=K+1}^{n-K} \mathbf{Z}_t (\psi_{(k)}(t/n) \otimes X_t)^\top \\ &= \frac{1}{n-2K} \sum_{t=K+1}^{n-K} \mathbf{Z}_t X_t \psi_{(k)}^\top(t/n), \end{aligned}$$

which is of dimension $(2K+1) \times k$. Consider a generic element $\frac{1}{n-2K} \sum_{t=K+1}^{n-K} \psi_r(t/n) X_t u_{x,t+j}$,

where $j = 1, 2, \dots, 2K + 1$, $r = 1, 2, \dots, k$. Observe that

$$\begin{aligned}
\mathbb{E}(X_t u_{x,t+j} X_s u_{x,s+j}) &= \sum_{\ell_1=1}^t \sum_{\ell_2=1}^s \mathbb{E}(u_{x,\ell_1} u_{x,t+j} u_{x,\ell_2} u_{x,s+j}) \\
&= \sum_{\ell_1=1}^t \sum_{\ell_2=1}^s \kappa_x(t+j-\ell_1, \ell_2-\ell_1, s+j-\ell_1) + \sum_{\ell_1=1}^t \sum_{\ell_2=1}^s \gamma_x(t+j-\ell_1) \gamma_x(s+j-\ell_2) \\
&\quad + \sum_{\ell_1=1}^t \sum_{\ell_2=1}^s \gamma_x(\ell_2-\ell_1) \gamma_x(s-t) + \sum_{\ell_1=1}^t \sum_{\ell_2=1}^s \gamma_x(s+j-\ell_1) \gamma_x(t+j-\ell_2),
\end{aligned}$$

where $\kappa_x(m_1, m_2, m_3) := \text{cum}(u_{x,t}, u_{x,t+m_1}, u_{x,t+m_2}, u_{x,t+m_3})$ is the fourth order cumulant of $u_{x,t}$ and $\gamma_x(j)$ is the j th autocovariance of $u_{x,t}$. Then,

$$\begin{aligned}
&\mathbb{E} \left(\frac{1}{n-2K} \sum_{t=K+1}^{n-K} \psi_r(t/n) X_t u_{x,t+j} \right)^2 \\
&= \frac{1}{(n-2K)^2} \sum_t \sum_s \psi_r(t/n) \psi_r(s/n) \mathbb{E}(X_t u_{x,t+j} X_s u_{x,s+j}) \\
&= \frac{1}{(n-2K)^2} \sum_t \sum_s \psi_r(t/n) \psi_r(s/n) \sum_{\ell_1} \sum_{\ell_2} \kappa_x(t+j-\ell_1, \ell_2-\ell_1, s+j-\ell_1) \\
&\quad + \frac{1}{(n-2K)^2} \sum_t \sum_s \psi_r(t/n) \psi_r(s/n) \sum_{\ell_1} \sum_{\ell_2} \gamma_x(t+j-\ell_1) \gamma_x(s+j-\ell_2) \\
&\quad + \frac{1}{(n-2K)^2} \sum_t \sum_s \psi_r(t/n) \psi_r(s/n) \sum_{\ell_1} \sum_{\ell_2} \gamma_x(\ell_2-\ell_1) \gamma_x(s-t) \\
&\quad + \frac{1}{(n-2K)^2} \sum_t \sum_s \psi_r(t/n) \psi_r(s/n) \sum_{\ell_1} \sum_{\ell_2} \gamma_x(s+j-\ell_1) \gamma_x(t+j-\ell_2).
\end{aligned}$$

Now consider the first term above. Note that $|\psi_r(t/n)| \leq \sqrt{2}$ uniformly over any r and t . We

have

$$\begin{aligned}
& \left| \frac{1}{(n-2K)^2} \sum_t \sum_s \psi_r(t/n) \psi_r(s/n) \sum_{\ell_1} \sum_{\ell_2} \kappa_x(t+j-\ell_1, \ell_2-\ell_1, s+j-\ell_1) \right| \\
& \leq \frac{O(1)}{(n-2K)^2} \sum_t \sum_s \sum_{\ell_1} \sum_{m_2 \in \mathbb{Z}} |\kappa_x(t+j-\ell_1, m_2, s+j-\ell_1)| \\
& \leq \frac{O(1)}{(n-2K)^2} \sum_t \sum_s \sum_{m_1 \in \mathbb{Z}} \sum_{m_2 \in \mathbb{Z}} |\kappa_x(m_1, m_2, m_1+s-t)| \\
& \leq \frac{O(1)}{(n-2K)^2} \sum_{m_1 \in \mathbb{Z}} \sum_{m_2 \in \mathbb{Z}} \sum_{m_3 \in \mathbb{Z}} |\kappa_x(m_1, m_2, m_3)| = O\left(\frac{1}{n-2K}\right),
\end{aligned}$$

since Assumption 1 guarantees that $\sum_{m_1 \in \mathbb{Z}} \sum_{m_2 \in \mathbb{Z}} \sum_{m_3 \in \mathbb{Z}} |\kappa_x(m_1, m_2, m_3)| < \infty$. Moving on to the second term:

$$\begin{aligned}
& \left| \frac{1}{(n-2K)^2} \sum_t \sum_s \psi_r(t/n) \psi_r(s/n) \sum_{\ell_1} \sum_{\ell_2} \gamma_x(t+j-\ell_1) \gamma_x(s+j-\ell_2) \right| \\
& \leq \frac{O(1)}{(n-2K)^2} \left(\sum_t \sum_{\ell_1} |\gamma_x(t+j-\ell_1)| \right) \left(\sum_s \sum_{\ell_2} |\gamma_x(s+j-\ell_2)| \right) \\
& \leq \frac{O(1)}{(n-2K)^2} \left\{ \sum_t \left(\sum_{h_1 \in \mathbb{Z}} |\gamma_x(h_1)| \right) \right\} \left\{ \sum_s \left(\sum_{h_2 \in \mathbb{Z}} |\gamma_x(h_2)| \right) \right\} = O(1),
\end{aligned}$$

since Assumption 1(ii) guarantees that $\sum_{h \in \mathbb{Z}} |\gamma_x(h)| < \infty$. Using similar arguments, we could show that the third and fourth terms above are also $O(1)$. This implies that $\frac{1}{n-2K} \sum_{t=K+1}^{n-K} \psi_r(t/n) X_t u_{x,t+j} = O_p(1)$, so that $\left\| \frac{1}{n-2K} \sum_{t=K+1}^{n-K} \mathbf{Z}_t \mathbf{X}_t^{(k)\top} \right\| = O_p\left(\sqrt{Kk}\right)$. This completes the proof of (iii).

Proof of (iv). This is again similar to Lemma C.2 so details are omitted. □

Lemma C.6. $\max_{1 \leq t \leq n} \|\tilde{\mathbf{u}}_{xt}\| = O_p(n^{1/p})$.

Proof. For the linear process $\mathbf{u}_{xt}$ defined in (3), we have the following Beveridge-Nelson (BN) decomposition

$$\mathbf{u}_{xt} = \Phi_x(1) \varepsilon_t - \Delta \tilde{\mathbf{u}}_{xt},$$

where $\Phi_x(1) = \sum_{j=0}^{\infty} \Phi_{xj}$, $\tilde{\mathbf{u}}_{xt} = \sum_{j=0}^{\infty} \tilde{\Phi}_{xj} \varepsilon_{t-j}$, and $\tilde{\Phi}_{xj} = \sum_{k=j+1}^{\infty} \Phi_{xk}$. By Minkowski inequality

ity and the definition of L_p norm, we have

$$\|\tilde{\mathbf{u}}_{xt}\|_p = \left\| \sum_{j=0}^{\infty} \tilde{\Phi}_{xj} \boldsymbol{\varepsilon}_{t-j} \right\|_p \leq \sum_{j=0}^{\infty} \left\| \tilde{\Phi}_{xj} \boldsymbol{\varepsilon}_{t-j} \right\|_p = \sum_{j=0}^{\infty} \left(\mathbb{E} \left\| \tilde{\Phi}_{xj} \boldsymbol{\varepsilon}_{t-j} \right\|^p \right)^{1/p} = \|\boldsymbol{\varepsilon}_1\|_p \sum_{j=0}^{\infty} \left\| \tilde{\Phi}_{xj} \right\| < \infty,$$

where $\sum_{j=0}^{\infty} \left\| \tilde{\Phi}_{xj} \right\| < \infty$ follows from Lemma 2.1 in Phillips and Solo 1992. Then, by Markov inequality, we have, for any $M > 0$,

$$\mathbb{P} \left(n^{-1/p} \max_{1 \leq t \leq n} |\tilde{\mathbf{u}}_{xt}| > M \right) \leq \sum_{t=1}^n \mathbb{P} (n^{-1/p} |\tilde{\mathbf{u}}_{xt}| > M) \leq \frac{n \mathbb{E} |\tilde{\mathbf{u}}_{xt}|^p}{(n^{1/p} M)^p} < \infty,$$

which implies that $\max_{1 \leq t \leq n} \|\tilde{\mathbf{u}}_{xt}\| = O_p(n^{1/p})$. □

D – Chebyshev time polynomials

Recall that Chebyshev time polynomials are defined by

$$\psi_0(t/n) = 1, \quad \psi_i(t/n) = \sqrt{2} \cos(i\pi(t/n)). \quad (\text{D.1})$$

For a particular time coordinate, say $t = \lfloor nr \rfloor$ for $r \in [0, 1]$, we have

$$\psi_i(\lfloor nr \rfloor / n) \sim \sqrt{2} \cos(i\pi r),$$

so that we can redefine the Chebyshev time polynomials for any $r \in [0, 1]$:

$$\psi_0(r) = 1, \quad \psi_i(r) = \sqrt{2} \cos(i\pi r). \quad (\text{D.2})$$

The Chebyshev polynomials form an orthonormal basis in $L^2[0, 1] = \left\{ a(r) : \int_0^1 a^2(r) dr < \infty \right\}$, in which the inner product is given by $\langle a_1, a_2 \rangle = \int_0^1 a_1(r) a_2(r) dr$ and the induced norm $\|a\| =$

$\langle a, a \rangle$. To see this, notice that, for $i \neq j$, we have

$$\begin{aligned}
\langle \psi_i, \psi_j \rangle &= \int_0^1 \psi_i(r) \psi_j(r) dr \\
&= 2 \int_0^1 \cos(i\pi r) \cos(j\pi r) dr \\
&= \int_0^1 [\cos((i+j)\pi r) + \cos((i-j)\pi r)] dr \\
&= \left[\frac{\sin((i+j)\pi r)}{(i+j)\pi} + \frac{\sin((i-j)\pi r)}{(i-j)\pi} \right] \Big|_0^1 = 0.
\end{aligned}$$

If $i = j = 0$, $\|\psi_i\| = 1$ holds trivially. For $i = j \neq 0$, we have

$$\begin{aligned}
\|\psi_i\| &= 2 \int_0^1 \cos^2(i\pi r) dr \\
&= \int_0^1 [1 + \cos(2i\pi r)] dr \\
&= 1 + \left[\frac{1}{2i\pi} \sin(2i\pi r) \right] \Big|_0^1 = 1.
\end{aligned}$$

The Chebyshev series expansion for a generic function $a(r) : [0, 1] \rightarrow \mathbb{R}$ is defined as

$$a(r) \sim \sum_{i=0}^{\infty} b_i \psi_i(r). \quad (\text{D.3})$$

Now taking inner product on (D.3) gives

$$\langle a, \psi_k \rangle = \sum_{i=0}^{\infty} b_i \langle \psi_i, \psi_k \rangle = b_k \langle \psi_k, \psi_k \rangle.$$

Since $\langle \psi_k, \psi_k \rangle = 2 \int_0^\pi \cos^2 i\theta d\theta = \int_0^\pi (1 + \cos 2i\theta) = \pi + \frac{1}{2i \sin 2i\theta} \Big|_0^\pi = \pi$, we have

$$b_k = \frac{1}{\pi} \int_0^1 \frac{a(r) \psi_k(r)}{\sqrt{1-r^2}} dr,$$

for $k = 0, 1, 2, \dots$.

We write $(S_m^\top a)(r)$ as the partial sum of the Chebyshev series of degree m :

$$(S_m^\top a)(r) = \sum_{i=0}^m b_i \psi_i(r).$$

Then, by Theorem 5.2 in Mason and Handscomb 2002, we know that the Chebyshev series expansion converges in $\mathcal{L}_2$ in the sense that

$$\int_0^1 \frac{[a(r) - (S_m^\top a)(r)]^2}{\sqrt{1-r^2}} dr \rightarrow 0, \quad (\text{D.4})$$

as $m \rightarrow \infty$. However, (D.4) is not enough for our purpose, as we need the explicit convergence rate. This is provided in the following theorem.

Theorem D.1. If $a(r) : [0, 1] \rightarrow \mathbb{R}$ has $q + 1$ continuous derivatives, then $|(S_m^\top a)(r) - a(r)| = O\left(\frac{1}{m^q}\right)$ for any $r \in [0, 1]$.

Proof. We follow closely the steps as the proofs of Theorem 5.14 in Mason and Handscomb 2002. We first introduce the operator L_m , which is defined by the relationship

$$L_m a := (S_m^\top a)(r) - a(r).$$

Let Π_n be the linear space of polynomials of degree n . For any p_q in Π_q , based on the definition of the operator L_m and $S_m^\top$, we have $S_m^\top p_q \equiv p_q$ so that $L_m p_q = S_m^\top p_q - p_q = 0$.

Using Peano's theorem (Davis 1975), we obtain

$$(S_m^\top a)(r) - a(r) = \int_0^1 a^{(q+1)}(r) K_m(r, t) dt,$$

where $K_m(r, t) = \frac{1}{q!} \{S_m(r-t)_+^q - (r-t)_+^q\}$, $(r-t)_+^q = (r-t)^q$ if $r \geq t$ and zero otherwise.

We note that

$$S_m(r-t)_+^q = \sum_{i=0}^m b_{iq} \psi_i(r),$$

where $b_{iq} = \frac{1}{\pi} \int_t^1 \frac{(r-t)^m \psi_i(r)}{\sqrt{1-r^2}} dr$. By change of variables $r = \cos \theta$, $t = \cos \phi$, we obtain

$$b_{iq} = \frac{1}{\pi} \int_0^\phi (\cos \theta - \cos \phi)^q \cos i\theta d\theta.$$

We then show that $b_{iq} = O\left(\frac{1}{i^{q+1}}\right)$. To see this, notice that applying integration by parts once gives

$$\begin{aligned}
\left| \int_0^\phi (\cos \theta - \cos \phi)^q \cos i\theta \, d\theta \right| &= \frac{1}{i} \left| \int_0^\phi (\cos \theta - \cos \phi)^q \, d \sin i\theta \right| \\
&= \frac{1}{i} \left| \int_0^\phi \sin i\theta \, d(\cos \theta - \cos \phi)^q \right| \\
&= \frac{q}{i} \left| \int_0^\phi (\cos \theta - \cos \phi)^{q-1} \sin \theta \sin i\theta \, d\theta \right| \\
&= \frac{q}{i} \left| \int_0^\phi (\cos \theta - \cos \phi)^{q-1} \sin \theta \sin i\theta \, d\theta \right| \\
&= \frac{q}{2i} \left| \int_0^\phi (\cos \theta - \cos \phi)^{q-1} [\cos((i-1)\theta) - \cos((i+1)\theta)] \, d\theta \right| \\
&\leq \frac{q}{2i} |(\cos \theta - \cos \phi)^{q-1}| |\cos((i-1)\theta) - \cos((i+1)\theta)| \leq \frac{O(1)}{i}.
\end{aligned} \tag{D.5}$$

If we continue from (D.5), we see that there are two terms:

$$\frac{q}{2i} \int_0^\phi (\cos \theta - \cos \phi)^{q-1} \cos((i-1)\theta) \, d\theta, \quad \frac{q}{2i} \int_0^\phi (\cos \theta - \cos \phi)^{q-1} \cos((i+1)\theta) \, d\theta.$$

Applying integration by parts on $\int_0^\phi (\cos \theta - \cos \phi)^{q-1} \cos((i-1)\theta) \, d\theta$ gives

$$\begin{aligned}
\left| \int_0^\phi (\cos \theta - \cos \phi)^{q-1} \cos((i-1)\theta) \, d\theta \right| &= \frac{(q-1)}{(i-1)} \left| \int_0^\phi (\cos \theta - \cos \phi)^{q-2} \sin \theta \sin((i-1)\theta) \, d\theta \right| \\
&= \frac{(q-1)}{2(i-1)} \left| \int_0^\phi (\cos \theta - \cos \phi)^{q-2} [\cos((i-2)\theta) - \cos(i\theta)] \, d\theta \right| \\
&\leq \frac{O(1)}{i-1}.
\end{aligned}$$

Applying integration by parts on $\int_0^\phi (\cos \theta - \cos \phi)^{q-1} \cos((i+1)\theta) \, d\theta$ gives

$$\begin{aligned}
\left| \int_0^\phi (\cos \theta - \cos \phi)^{q-1} \cos((i+1)\theta) \, d\theta \right| &= \frac{(q-1)}{(i+1)} \left| \int_0^\phi (\cos \theta - \cos \phi)^{q-2} \sin \theta \sin((i+1)\theta) \, d\theta \right| \\
&= \frac{(q-1)}{2(i+1)} \left| \int_0^\phi (\cos \theta - \cos \phi)^{q-2} [\cos(i\theta) - \cos((i+2)\theta)] \, d\theta \right| \\
&\leq \frac{O(1)}{i+1}.
\end{aligned}$$

This indicates that $\int_0^\phi (\cos \theta - \cos \phi)^{q-1} \cos((i+1)\theta) d\theta$ is just smaller order term. We continue from (D.5) to obtain

$$\begin{aligned} \left| \int_0^\phi (\cos \theta - \cos \phi)^q \cos i\theta d\theta \right| &\leq \frac{q}{2i} \left| \int_0^\phi (\cos \theta - \cos \phi)^{q-1} [\cos((i-1)\theta) - \cos((i+1)\theta)] d\theta \right| \\ &\leq \frac{q}{2i} \left| \int_0^\phi (\cos \theta - \cos \phi)^{q-1} \cos((i-1)\theta) d\theta \right| \\ &\leq \frac{O(1)}{i(i-1)} \end{aligned}$$

As $a(r)$ is assumed to have $q+1$ continuous derivative, we can apply integration by parts q times, so that

$$\left| \int_0^\phi (\cos \theta - \cos \phi)^q \cos i\theta d\theta \right| \leq \frac{O(1)}{i(i-1) \cdots (i-q)} = O\left(\frac{1}{i^{q+1}}\right),$$

which implies that $b_{iq} = O\left(\frac{1}{i^{q+1}}\right)$.

Finally, since

$$\begin{aligned} |S_m(r-t)_+^q - (r-t)_+^q| &= \left| \sum_{i=m+1}^{\infty} b_{iq} \psi_i(r) \right| \leq \sum_{i=m+1}^{\infty} |b_{iq}| = \frac{1}{m^q} \sum_{i=m+1}^{\infty} \frac{1}{m} \frac{1}{\left(\frac{i}{m}\right)^{q+1}} \\ &= \frac{1}{m^q} \int_1^{\infty} \frac{1}{x^{q+1}} dx + o(1) = O\left(\frac{1}{m^q}\right), \end{aligned}$$

so that

$$|(S_m^\top a)(r) - a(r)| \leq |a^{(q+1)}(r)| |K_m(r, t)| = O\left(\frac{1}{m^q}\right).$$

□

E – Additional empirical results

This section reports additional empirical results corresponding to the analysis in section 5 using alternative measures of consumption. More specifically, following a change of methodology outlined in Martin Lettau's website (<https://sites.google.com/view/martinlettau/data>), we consider using total personal consumption expenditures (PCE) instead of nondurables and services consumption excluding clothing and footwear (NDS) as a measure of unobserved total consumption.

As in Section 5, we first report estimation results based on the sample period ending in 2019Q4.

Figure E1 presents the empirical estimates of $\beta_{a,t}$ and $\beta_{y,t}$ together with their 95% bootstrap confidence intervals. The estimates exhibit substantial time variation, broadly consistent with the patterns documented in Figure 1, although the magnitude of $\hat{\beta}_{y,t}$ appears larger than that obtained using the NDS consumption measure. When post-COVID observations are included, as shown in Figure E2, the estimates change noticeably, and in particular $\hat{\beta}_{y,t}$ remains statistically insignificant from around 1990 onwards.

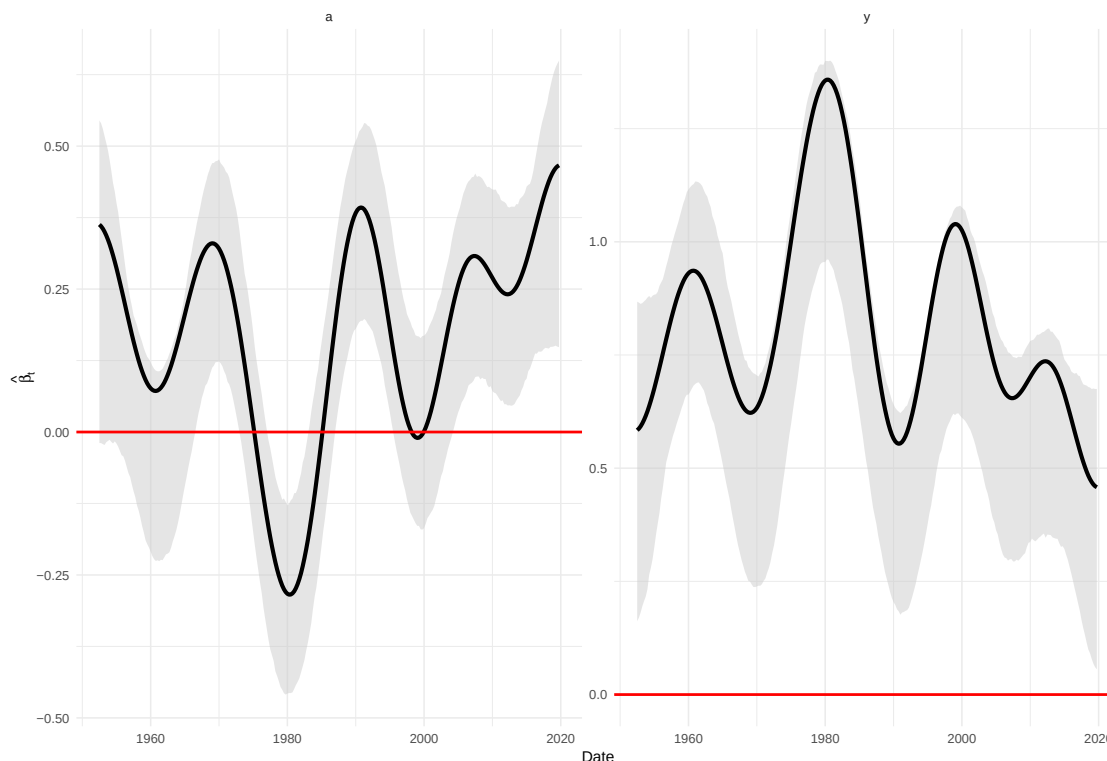


Figure E1 – Empirical estimates for $\beta_{a,t}$ and $\beta_{y,t}$ based on PCE consumption measure.

Notes. See Note in Figure 1.

Table E1 report out-of-sample forecasting results when *cay* is constructed using the PCE consumption measure. Overall, predictive performance is weaker than that reported in section 5. In the pre-pandemic evaluation sample, however, the time-varying *cay* measure still outperforms its fixed-coefficient counterpart, with most of the gains arising during expansion periods where it is able to outperform the historical-average benchmark. Once post-COVID observations are included, forecasting performance based on the time-varying specification deteriorates, a pattern that is clearly illustrated in Figure E3.

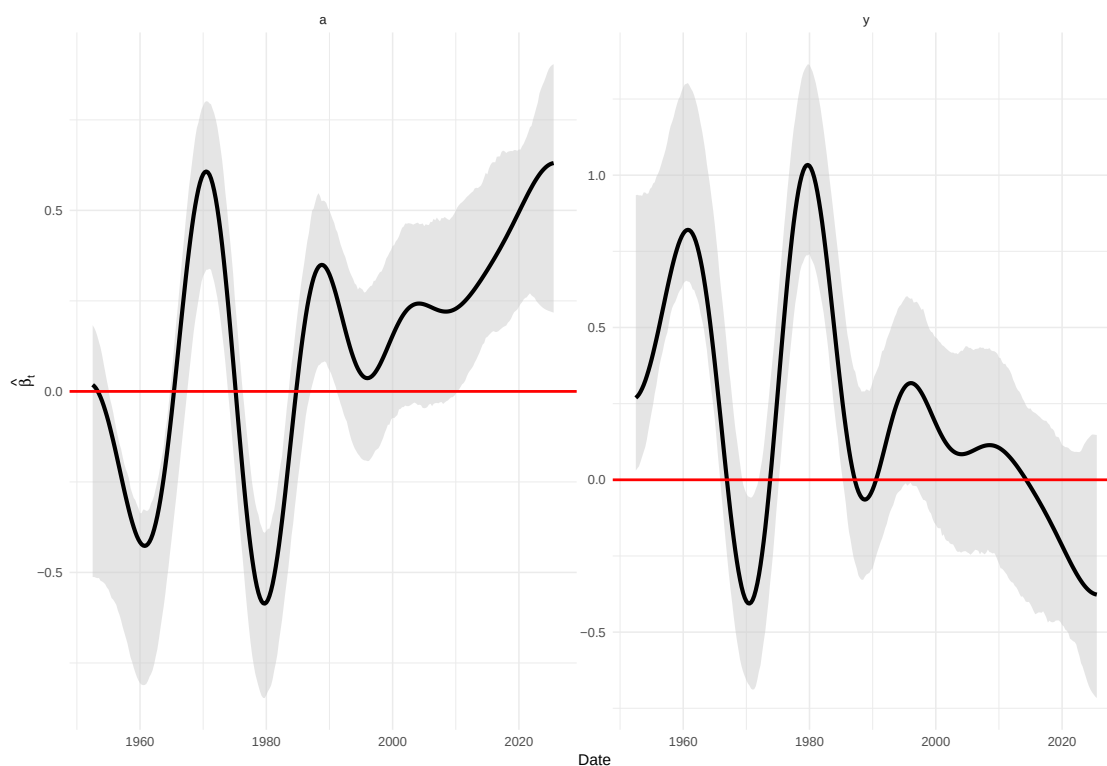


Figure E2 — Empirical estimates for $\beta_{a,t}$ and $\beta_{y,t}$ based on PCE consumption measure.

Notes. See Note in Figure 1.

Table E1 — Quarterly U.S. equity premium out-of-sample forecasting results based on the PCE consumption measure

Predictor	Overall		Recession		Expansion	
	$R^2\%$	p -value	$R^2\%$	p -value	$R^2\%$	p -value
Panel A: Evaluation sample 1995Q1–2019Q4						
$\widehat{\text{cay}}_f$	-3.171	0.321	-0.319	0.483	-4.134	0.326
$\widehat{\text{cay}}_{\text{tvp}}$	-0.871	0.611	-3.510	0.939	0.020	0.333
Panel B: Evaluation sample 1995Q1–2024Q4						
$\widehat{\text{cay}}_f$	-7.675	0.673	-1.831	0.580	-10.148	0.659
$\widehat{\text{cay}}_{\text{tvp}}$	-8.767	0.915	-1.772	0.801	-11.726	0.895

Notes. See Note in Table 3.

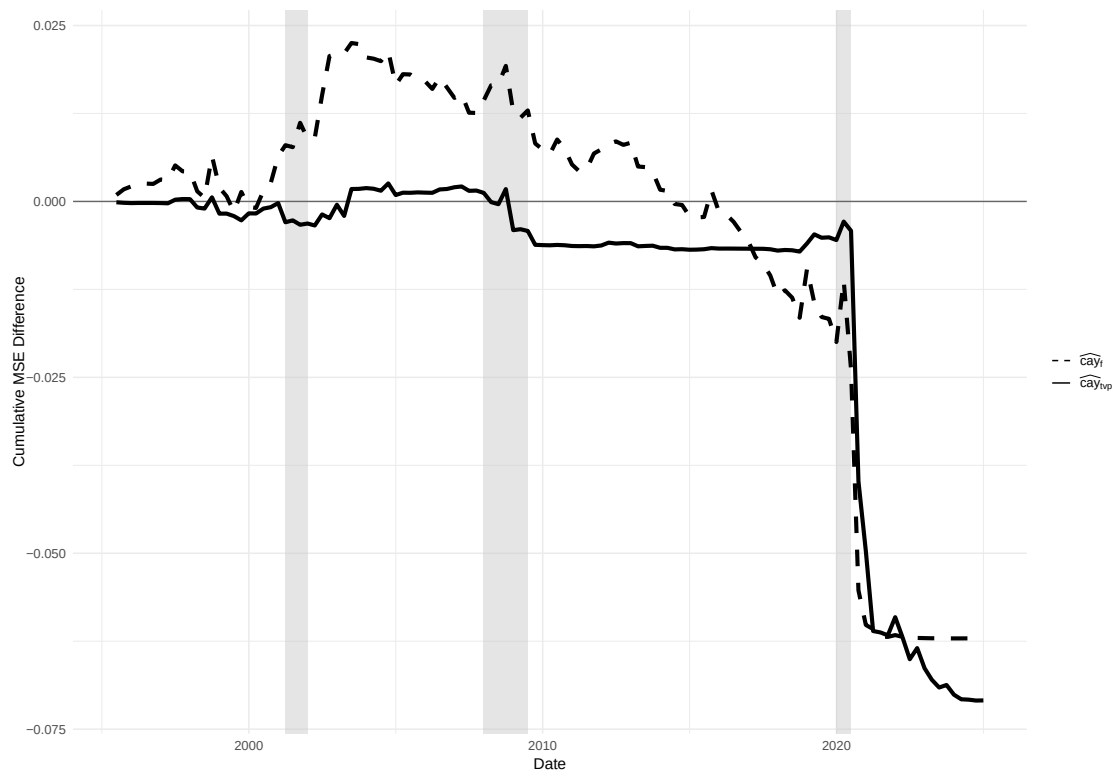


Figure E3 – Cumulative difference in MSE difference relative to the historical mean benchmark based on the PCE consumption measure

Notes. See Note in Figure 3

F — Data Appendix

The data construction broadly follows the empirical implementation in Dauber and Lawrenz 2025. We provide a detailed description of the data sources and variable definitions here for completeness.

Consumption and labor income data are obtained from the National Income and Product Accounts (NIPA) published by the Bureau of Economic Analysis (BEA). Two alternative measures of consumption are considered. The first is the conventional nondurables and services consumption measure excluding clothing and footwear (NDS). The second is total personal consumption expenditures (PCE). taxes. Each series is separately deflated to 2017-dollars with its respective deflator. Labor income is defined as the sum of wages and salaries, personal current transfer receipts, and employer contributions for employee pension and insurance funds minus employee contributions for government social insurance, and

Aggregate wealth is defined as the net worth of households and nonprofit organizations. The source is the Board of Governors of the Federal Reserve System, Financial Accounts. The aggregate wealth and labor income series are all deflated to 2017-dollars using the PCE chain-type price deflator.

Consumption, aggregate wealth and labor income series are all transformed into per capita terms and put in the logarithmic form. The population data are retrieved from FRED, Federal Reserve Economic Data.

Stock returns are measured as the CRSP Value-Weighted index return. The equity premium is defined as the index return in excess of the lagged three-month Treasury-bill rate.